

Australian/New Zealand Standard

Guidelines for scaffolding

AS/NZS 4576:1995

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Guidelines for scaffolding

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PREFACE

This Standard was prepared by the Joint Standards Australia/Standards New Zealand Committee BD/36 on Scaffolding.

Every attempt has been made to ensure that the information in this Standard is compatible with Worksafe Australia's publications *National Standard for Plant* and *National Occupational Health and Safety Certification Standard for Users and Operators of Industrial Equipment*, and with *Health and Safety in Employment Act 1992* in New Zealand.

Persons using this Standard are advised to familiarize themselves with the specific statutory requirements relating to scaffolding and the certification of scaffolders in New Zealand and in the States and Territories of Australia.

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STANDARDS AUSTRALIA/STANDARDS NEW ZEALAND

Australian/New Zealand Standard**Guidelines for scaffolding**

SECTION 1 SCOPE AND GENERAL

1.1 SCOPE

This Standard gives practical guidance for the training and certification of scaffolders, the preparation of sites for scaffolding, and the safe selection, supply, erection, alteration, dismantling, maintenance, inspection and use of scaffolding and scaffolding equipment.

1.2 INCLUSIONS

This Standard gives guidance for the general use of scaffolding, including:

- Portable ladders used as access to platforms.
- Portable trestle ladders supporting scaffold planks.
- Ladder bracket scaffolds.
- Temporary stairways, landings, ramps and other access ways.
- Temporary catch platforms including covered ways, safety nets and footpath hoardings (gantries) comprised of scaffolding equipment.
- Temporarily installed working platforms, including swing stages and boatswain's chairs.
- Temporary loading platforms and cantilevered crane loading bays.
- Temporary seating stands, stages, lighting towers and camera towers for concerts, sporting events, and so on, substantially comprised of scaffolding equipment.
- Mast climbing work platforms.

1.3 EXCLUSIONS

This Standard does not give guidance for:

- Temporarily installed guardrails.
- Portable ladders not used in connection with scaffolding.
- Permanently installed walkways, stairways, ladders, catwalks and the like.
- Building maintenance units.
- Crane boxes, scissor hoists, boom-type elevating work platforms (i.e. cherry pickers), lifts, and (men and materials) hoists.
- Falsework, shoring, back-propping and any other frame-work for the support of floors, walls, roofs and structural members of buildings, structures, ships, boats or mines.

1.4 REFERENCED DOCUMENTS

The documents in Appendix A are referred to in this Standard.

1.5 DEFINITIONS

For the purpose of this Standard and the scaffolding industry, the definitions given in Appendix B apply.

SECTION 2 TRAINING AND COMPETENCY

2.1 GENERAL

Australian and New Zealand occupational health and safety laws require employers to provide scaffolders with the information, instruction, training and supervision necessary to allow them to work safely.

In order for scaffolding to be erected, altered and dismantled correctly, safely and efficiently, scaffolders must:

- Know the basic relevant rules of physics and mechanics.
- Be able to read and understand the supplier's information, general site plans, design drawings and specifications for scaffolds. (An ability to make simple calculations of dead load and live load is often needed.)
- Have a thorough knowledge of the scaffolding equipment being used.
- Have thorough knowledge of the construction methods and design requirements associated with that equipment.
- Be able to recognize common hazards at the worksite and be capable of taking effective precautions, to control risks to health and safety arising from these hazards.
- Visually inspect scaffolding equipment for faults.
- Have the physical skills needed for scaffolding construction.
- Be competent in manual lifting techniques.
- Work safely and confidently at heights.
- Correctly use the various tools, ropes and gin wheels.
- Erect and dismantle scaffolding in the correct sequence.

2.2 FURTHER INFORMATION FROM REGULATORY AUTHORITIES

Contact your local regulatory authority for further information about:

- Training of scaffolders.
- Assessment and application procedures for certificates of competency.
- Recognition of prior learning and overseas qualifications.
- Suspension and cancellation of certificates of competency.

2.3 SUPERVISION OF TRAINEES

Trainees engaged in the erection, alteration or dismantling of scaffolding should be directly supervised by a competent scaffolder.

The supervising scaffolder should monitor the work, to ensure compliance with regulatory requirements and recommended practice and be in a position to take immediate charge in the event of an emergency.

Trainees should not be required to carry out scaffolding work in an isolated location out of sight and earshot of the supervising scaffolder.

The appropriate ratio between certificated scaffolders and trainee scaffolders in a gang depends on the level of experience and competence gained by each trainee, the complexity of the scaffolding work being undertaken and the risks associated with any mistakes that may be made by trainees. The ratio should be determined by the employer through consultation with the supervising scaffolder, taking into account any requirements of the regulatory authority. The supervising scaffolder will be carrying out the employer's legal obligation to supervise trainee scaffolders to perform the work safely and without risk to health.

2.4 TRAINING PROGRAMS

Employers should provide trainee scaffolders with a structured and comprehensive training program.

Training programs should include both formal instruction and supervised practical work experience. Ideally, the formal instruction will be delivered as a series of separate modules or topics with periods of continuing work experience between each module.

Instructors should be competent industrial trainers who have gained extensive experience as practising scaffolders. A thorough knowledge and understanding of the regulatory requirements for scaffolding and the recommended work practices for scaffolders are essential.

The information in this Standard should be supplemented with examples of suppliers' documented information and other relevant information from the regulatory authority, industry groups and training establishments.

The work experience of trainees should occur in a logical sequence, starting from the least demanding tasks and skills, with minimum risks to health and safety, and progressing through to the most demanding tasks and skills, with the greatest risks to health and safety.

The person in charge of a gang that includes one or more trainees should be a scaffolder with extensive experience, a consistent record of using recommended work practices and a willingness and ability to pass on knowledge and skills.

2.5 GAINING EXPERIENCE

Trainees with no scaffolding experience should each gain work experience on the ground as a 'groundie' with a gang of experienced scaffolders. In this way, each trainee learns to recognize scaffolding components and gains an understanding of erection techniques while remaining safely on the ground.

Ground work exposes trainees to plans and specifications and the fundamentals of estimating and preparing materials. It also provides experience in the preparation of ropes and gin wheels, the fixing of bends and hitches to haul materials, and the basic manual handling techniques needed to minimize strain when shifting and passing up large quantities of gear.

A trainee who has become a competent groundie should be introduced to working aloft by gradually increasing the degree of complexity of the tasks.

A trainee who has gained the confidence and ability to work without supervision is ready to be assessed (or to complete a staged assessment) for a certificate of competency. Some sample examination questions set in Australia are given in Appendix C.

2.6 RECORD OF TRAINING

There should be a formal record of training for each trainee scaffolder. This can take the form of a logbook held by the trainee.

The record of training should include details of the formal instruction modules and the scaffolding-related work practices that have been experienced. Entries should be progressively recorded and verified by a signature of the person overseeing that item of training or experience.

Training records should be taken into account when assessing the trainee for a certificate of competency and should influence the nature and extent of the assessment. They will also provide a useful reference for a prospective employer.

2.7 AUSTRALIAN CERTIFICATES OF COMPETENCY

There are nationally uniform certification requirements in force in all Commonwealth, State and Territory jurisdictions in Australia.

Under these requirements, certification applies where a person or object could fall more than 4 m from a scaffold or from an adjacent open floor.

The types of national certificate classes are:

- *Basic scaffolding* Basic scaffolding certificates include prefabricated scaffolds, bracket scaffolds, work associated with ropes and gin wheels and the installation of safety nets, static lines and cantilever materials hoists with a working load limit of up to 500 kg.
- *Intermediate scaffolding* Intermediate scaffolding certificates cover basic scaffolding and also include tube-and-coupler scaffolds, cantilevered crane loading platforms, cantilevered and spurred scaffolds, barrow ramps and sloping platforms, scaffolding associated with perimeter screens and shutters and mast-climbing work platforms.
- *Advanced scaffolding* Advanced scaffolding certificates cover basic and intermediate scaffolding and also include hung scaffolds, suspended scaffolds and the installation of all types of cantilevered hoists, including personnel and materials hoists.

National rigging certificates also cover some of this work, such as the erection, alteration and dismantling of mast-climbers, cantilevered crane loading bays, fully-fabricated hung scaffolds, suspended scaffolds and cantilevered builders' hoists.

Scaffolders who hold old State and Territory certificates can continue to operate under those certificates in accordance with State and Territory laws.

2.8 NEW ZEALAND CERTIFICATES OF COMPETENCY

Under New Zealand regulations, certification applies where a person or object could fall more than 5 m from a scaffold.

The types of New Zealand certificate classes are:

- *Basic scaffolding* The equipment range includes free standing modular system scaffolds, ropes, gin wheels, static lines and fall arrest systems.
- *Suspended scaffolding* The equipment range includes hand haul and mechanical boatswain's chairs, building maintenance units and swinging stages (hand haul and mechanical types).
- *Advanced scaffolding* The equipment range includes free standing modular system scaffolds, tube-and-coupler scaffolds including tube-and-coupler covered ways and gantries, scaffolding associated with perimeter safety screens and shutters, cantilevered hoists with a working load limit not exceeding 250 kg (for materials only), ropes, gin wheels, safety nets for public protection, catch nets, static lines, fall arrest systems, bracket scaffolds (tank and formwork types), cantilevered load platforms supported by a scaffold, cantilevered scaffolds, spurred scaffolds, barrow ramps, sloping platforms, mast climbers, hung scaffolds including scaffolds hanging from tubes, wire ropes and chains.

The equipment range excludes hand-haul and mechanical types of boatswain's chairs, building maintenance units and swinging stages.

SECTION 3 TOOLS AND EQUIPMENT

3.1 INTRODUCTION

There are a range of tools and work practices that scaffolders need to be familiar with.

The tools and equipment that are commonly used to erect, maintain and dismantle scaffolding are described in this Section (see also Figure 3.1).

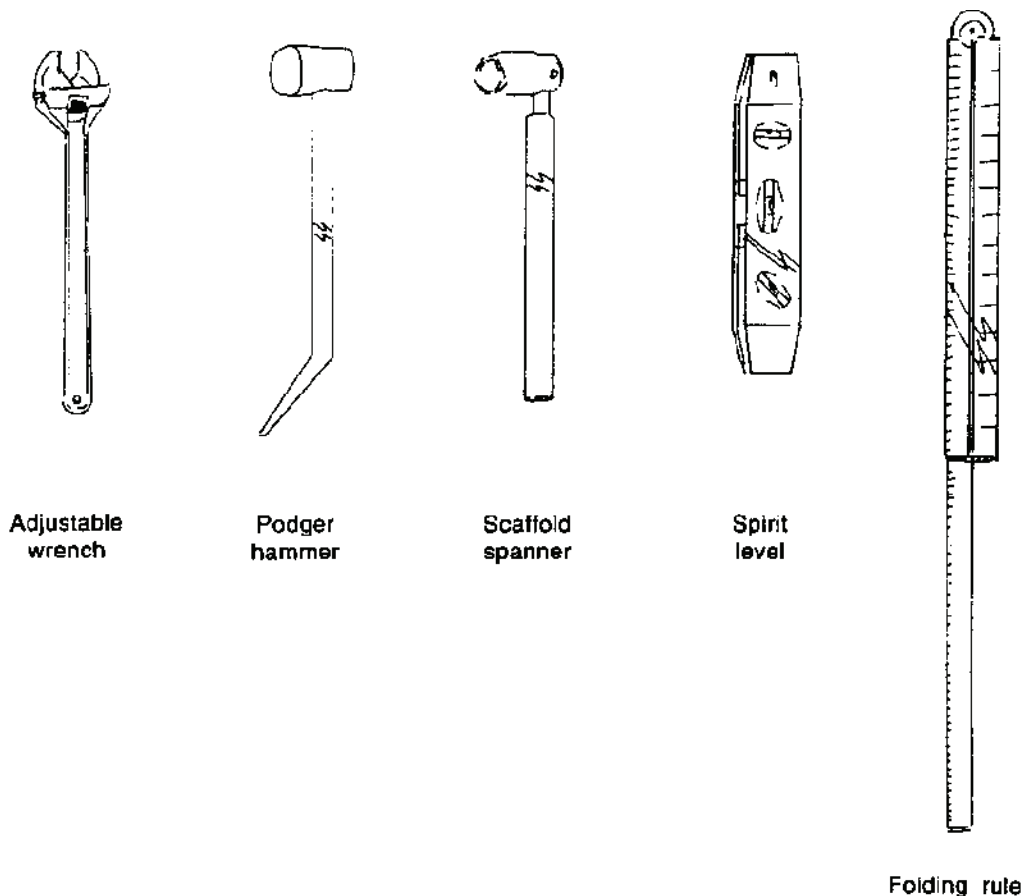


FIGURE 3.1 EXAMPLES OF COMMONLY USED SCAFFOLDING TOOLS

3.2 SCAFFOLD SPANNERS

‘Scaffold spanner’ and ‘scaffold key’ are defined in Appendix B.

The handle of scaffold spanners should be not less than 200 mm nor more than 250 mm long. Scaffold spanners with a worn rivet or a sloppy head should be repaired or replaced.

3.3 PODGER HAMMERS

‘Podger hammer’ is defined in Appendix B.

Podger hammers with a loose head or a hairline crack at the juncture between the head and the shaft should be replaced.

3.4 ADJUSTABLE WRENCHES

An adjustable wrench (shifter or shifting spanner) is normally used where an obstruction makes the use of a scaffold spanner impossible.

Care should be taken to avoid burring nuts and bolt heads when using an adjustable wrench.

The arm of an adjustable wrench used for couplers should be not less than 200 mm nor more than 250 mm long.

3.5 SPIRIT LEVELS

Spirit levels are used to check that scaffolding members are horizontal or plumb.

When working aloft, use spirit levels that are compact enough for use with one hand and for safe stowing on a scaffolder's belt when not in use.

3.6 RULES AND TAPE MEASURES

Rules and tape measures should be carried in a purpose-designed pouch on the scaffolder's belt or clipped to the inside of the belt, to minimize the risk of dislodgment while the scaffolder is working aloft.

3.7 SCAFFOLD BELTS

A scaffolder's belt (see Figure 3.7) should be used to safely carry scaffold tools while they are not used.

The belt should be made of sturdy leather, canvas webbing or material of equivalent strength and characteristics.

Leather or canvas frogs (bayonet holders) should be used to secure scaffold spanners, podger hammers and adjustable wrenches. Folding rules and small spirit levels should be secured in carpenter's chisel pouches or similar.

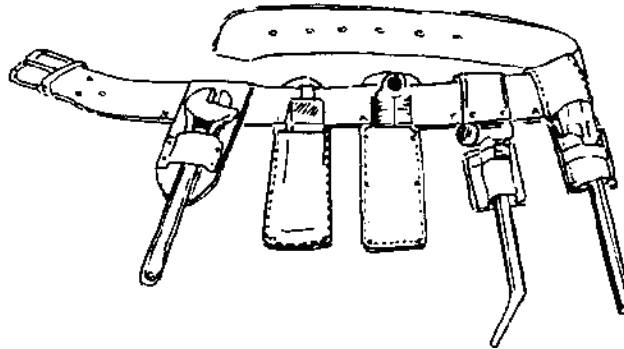


FIGURE 3.7 A TYPICAL SCAFFOLDER'S BELT

3.8 INDUSTRIAL SAFETY HELMETS

Industrial safety helmets complying with AS 1801 or NZS 5806 should be worn, wherever there is a risk of objects falling from above and on any work site where a hard hat sign is displayed.

3.9 GLOVES

Gloves should be close fitting and non-slip.

3.10 FOOTWEAR

Footwear should be comfortable, provide maximum grip and give protection from pinching, jamming and crushing. A range of lightweight flexible footwear with steel or plastic protective caps is available (see AS/NZS 2210.1 and AS/NZS 2210.2).

3.11 FIBRE ROPE

Fibre rope used as a handline for the raising or lowering of scaffolding gear should be at least 12 mm in diameter. Rope used for haulage with a gin wheel should be at least 16 mm in diameter.

To avoid high-stranding and excessive kinking, a new coil of rope should be laid on a flat surface and unwound through the coil in an anti-clockwise direction, carefully removing any kinks and loops as they form.

Where natural fibre rope is not used immediately, it should be laid out in the sun until it has dried and softened.

Prior to being cut, rope should be whipped with twine on either side of the proposed cut (see Figure 3.11), to prevent the rope from unlaying or fraying.

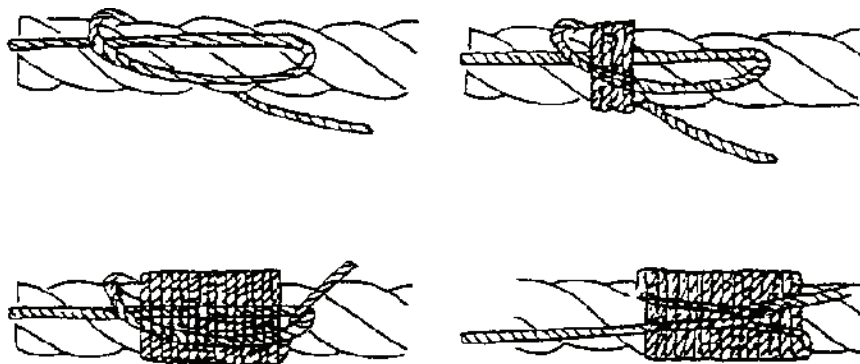


FIGURE 3.11 COMMON WHIPPING

The length of each whipping should be at least equal to the rope diameter.

Fibre rope should be stored:

- Away from exhaust gases and boilers, radiators, steampipes and other sources of heat.
- In a dry cool room with good air circulation.
- In loose coils hanging on large wooden pegs well above the floor.

Fibre rope should not be stored on the ground or floor or in boxes or cupboards with restricted air circulation.

Wet rope should be naturally dried and cleaned before being stored. Do not disturb a frozen rope until it has completely thawed, because frozen fibres may be damaged when handled.

Rope in use should be thoroughly inspected along its entire length on a daily basis. Look for:

- External wear.
- Cuts and abrasions.
- Burns.
- Powdered or hairy fibres.
- High-stranding or kinking.
- Grit and dirt lodged in strands.
- Internal wear between strands.

If there is any reasonable doubt about the soundness of a rope, it should be replaced.

When a competent person has condemned a rope, it should be destroyed at once or cut up into short lengths so that it cannot be used for lifting purposes.

Common bends and hitches for fibre rope are illustrated in Appendix D.

Further advice on the care and use of fibre rope is given in AS 4142.1.

3.12 WEBBING SLINGS

Flat webbing slings should comply with AS 1353.1 and be used in accordance with AS 1353.2.

SECTION 4 SITE FOR ERECTION

4.1 INTRODUCTION

There are a range of hazards, such as powerlines, traffic, plant and explosive substances that must be identified and assessed prior to erecting a scaffold. Specific requirements for hazards are given in Section 5.

4.2 PUBLIC PLACES

Protective measures must be taken to separate the public from the dangers of scaffolding while erecting, altering or dismantling scaffolding, particularly on, over or adjacent to any public thoroughfare or adjoining property.

Check with the local authority to see if a permit is required.

4.3 WORK AREAS FOR SCAFFOLDERS

The preparation of a work area for scaffolders must allow for the safe erection, alteration or dismantling of scaffolding, including the safety of all site personnel.

There must be a firm supporting surface for storing equipment and erecting the scaffold.

Debris and unnecessary materials and equipment should be removed and space provided for the delivery, unloading and stacking of scaffolding equipment.

Isolate the work area from other site personnel.

An example of well-stacked scaffolding equipment is given in Figure 4.3.

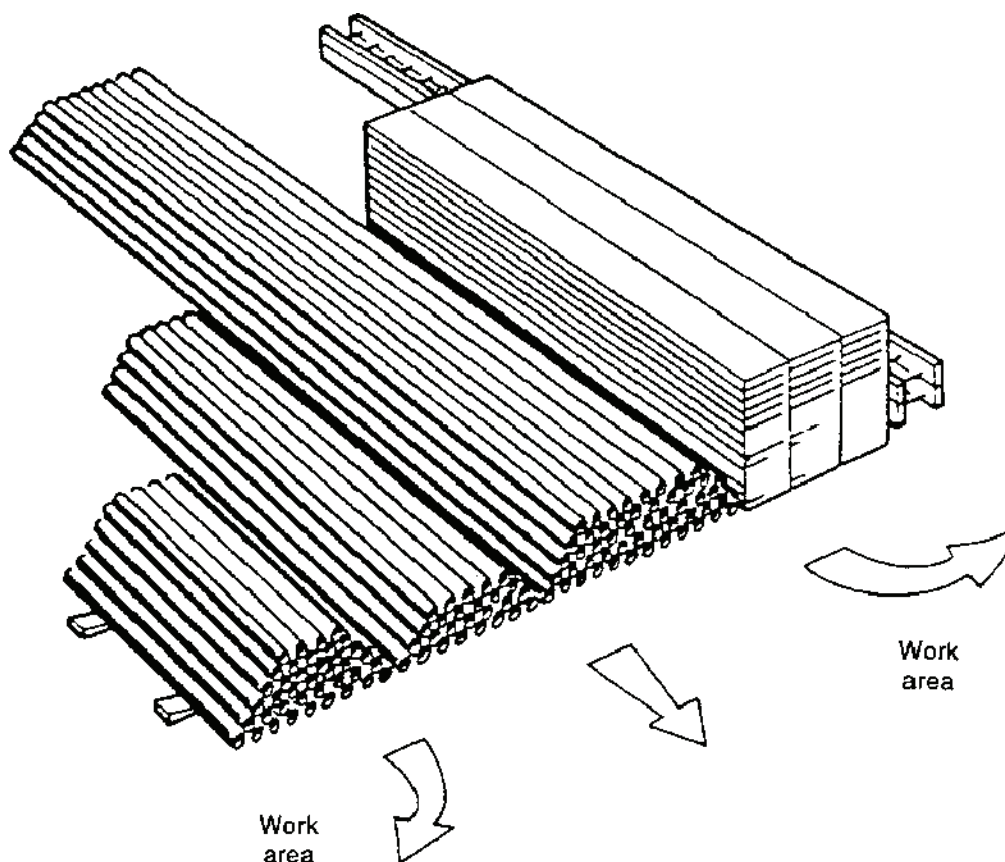


FIGURE 4.3 TYPICAL EXAMPLE OF WELL-STACKED SCAFFOLDING EQUIPMENT

SECTION 5 HAZARDS

5.1 OCCUPATIONAL HEALTH AND SAFETY

Under occupational health and safety laws in Australia and New Zealand:

- Plant and associated systems of work must not cause unnecessary risks to the health of workers and others, including members of the public.
- Plant and associated systems of work must be maintained in a safe condition.
- Employees must cooperate with their employer.
- Employees must take care of the health and safety of themselves, fellow workers and others.

The essential elements of minimizing risk are:

- Use expert advice.
- Identify hazards.
- Assess risk.
- Control risk.

5.2 EXPERT ADVICE

Wherever possible, expert advice should be sought by the main employer during the planning stage of the project and when unforeseen circumstances or changes to the project schedule occur, to identify potential hazards and to assess and control risks.

Expert advice is available from a number of sources, including equipment suppliers, health and safety professionals, industry organizations, occupational health and safety authorities, scaffolding and access design consultants, subcontractors and workplace health and safety committees.

5.3 IDENTIFICATION OF HAZARDS

Hazards that have the potential to cause injury or illness are commonly associated with elevated work, scaffolding and other temporary access equipment, and include:

- Engaging in the erection and dismantling of a scaffold or access equipment.
- Using a scaffold or equipment.
- Being in the vicinity of elevated work, a scaffold or equipment (including the general public).
- Working at heights.
- Falls from heights.
- Falling objects.
- Manual handling.
- Electricity.
- Corrosive substances.
- Volatile atmospheres.
- Movement of cranes, vehicles and machinery.
- Weak or unstable supporting structures.
- High winds and storms.

5.4 PROXIMITY TO POWERLINES

5.4.1 General Powerlines are a potential hazard to persons erecting, working from or in the vicinity of a scaffold.

Isolate or shield any electrical wires. Electrical wires or apparatus that pass through a scaffold should be de-energized or protected by the supply authority and should be fully enclosed by a non-conductive material such as moisture-resistant flooring-grade particle board complying with AS 1859, dry timber, plywood or similar non-conductive material.

The possibility of movement of powerlines caused by strong wind should be considered.

5.4.2 Australian requirements The clearance between scaffolds and any transmission line, main apparatus or transmission apparatus should be not less than (see Figure 5.4.2):

- 4.0 m where any metal member is used.
- 1.5 m where only non-conductive materials such as dry timber or plywood are used.

Advice should be sought from the local electricity supply authority for any reduction to the above clearances.

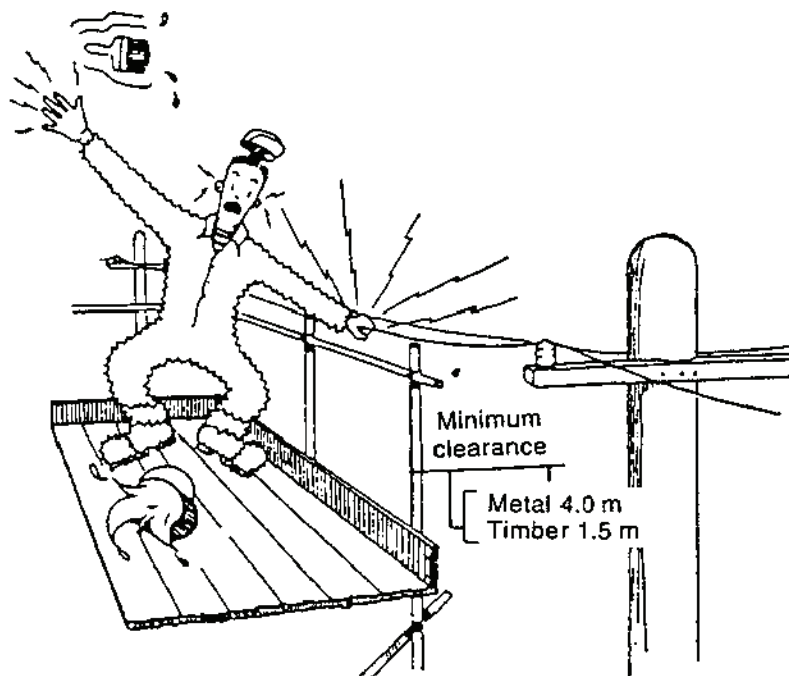


FIGURE 5.4.2 CLEARANCE TO ELECTRICAL CONDUCTOR WIRES

Do not erect scaffolding until the necessary measures have been taken to minimize risk and a written authorization has been received from the electricity supply authority.

High voltage mains (i.e. more than 600 V) near scaffolding should be de-energized, short-circuited and earthed, or re-routed prior to erection of the scaffolding.

Low voltage mains (i.e. not more than 250 V) and medium voltage mains (i.e. in the range of 250 V to 600 V) should be de-energized, short-circuited and earthed, or re-routed where practicable. Make sure that inadvertent re-energizing of mains cannot occur while work is in progress.

Low and medium voltage mains that cannot be de-energized should be insulated by the supply authority for the full length of the scaffolding plus a minimum distance beyond each end of the scaffolding of 5.0 m. Although this insulation (e.g. tiger tails) is a safeguard against contact with live wires under dry conditions, a combination of small gaps in the insulation and wet weather conditions can cause an unsafe situation, which can result in severe shocks.

The scaffolding may only be erected where no part is allowed to touch or fall across the insulated wires.

5.4.3 New Zealand requirements The minimum distances from electrical conductors in any direction under normal conditions for the erection of scaffolding are given in Table 5.4.3. Scaffolding components must not be located within 4.0 m of any conductors of an overhead electrical power line without written permission of the owner of the line.

TABLE 5.4.3
MINIMUM DISTANCES FROM ELECTRICAL CONDUCTORS
FOR THE ERECTION OF SCAFFOLDING

Line voltage (and span)	Minimum distance m
Not exceeding 66 kV (maximum span 125 m)	4.0
Exceeding 66 kV (maximum span 125 m)	5.0
Any voltage (span greater than 125 m but less than 250 m)	6.0
Any voltage (span greater than 250 m but less than 500 m)	8.0
Any voltage (span exceeding 500 m)	As agreed with the owner of the line, but not less than 8.0

5.5 PROXIMITY TO HAZARDOUS PLANT

5.5.1 Identification Operational plant, vehicular traffic and corrosive or explosive substances pose hazards affecting the safe use of a scaffold or the structural integrity of the scaffold. The risks from these hazards should be identified, evaluated and controlled.

5.5.2 Vehicular traffic Take precautions to prevent scaffold being endangered by the movement of vehicles and other plant. Traffic damage to scaffolds is a common problem, which can be solved by ensuring that motor vehicles and mobile plant are re-routed away from the location of the scaffold. Where this is not practicable, guards or fenders should be installed to shield the scaffold from traffic damage. Where both of the above measures are not possible, a person should be assigned to direct traffic in the vicinity of the scaffold.

5.5.3 Cranes Take precautions to prevent scaffold being endangered by the movement of cranes.

A scaffold in the operational radius of a crane is in danger of damage from suspended loads. It may be necessary to evacuate the scaffold and its vicinity during lifting operations. Make sure that the scaffold does not contain any unnecessary obstructions, such as over-length transoms, putlogs, tie tubes or over-height standards.

Where it is not practicable to suspend crane operations for the life of the scaffold, the operational radius of the crane should be limited, so that loads cannot come into contact with the scaffold or its supporting structure, and clear instructions should be given to crane personnel to take particular care while moving loads near to scaffolding.

5.5.4 Plant with moving parts Fail-safe lock-out procedures must be used to prevent plant from being activated where scaffolds are constructed on, over or inside items of plant, such as overhead gantry cranes, conveyors, turbines and crushers.

5.5.5 Boilers Clearances between power generation boilers and scaffolding should allow for the expansion of the boiler during firing and its contraction following shutdown.

5.5.6 Corrosive substances Heavy concentrations of acids, alkalis and salts can corrode scaffolding components, leading to structural failure of the scaffold. Where any corrosive substance is to be stored on or near a scaffold, compatible corrosion-resistant scaffolding materials should be used. Where this is not practicable, the frequency of scaffold inspections should be increased to detect early indications of structural deterioration.

5.5.7 Explosive atmospheres Petrochemical plants, powdered milk factories and flour mills are common examples of workplaces with high explosion risks. The use of scaffolding can increase the risk of an explosion if the scaffolding equipment can spark upon impact. It may be necessary to:

- Remove the hazard before constructing the scaffold.
- Construct the scaffold from non-conductive material such as timber.
- Provide non-electrically powered scaffolding hoists for suspended scaffolds, such as manual- or pneumatic-powered scaffolding hoists.

5.6 RISK ASSESSMENT

Risks associated with hazards can be assessed by considering the following three key factors:

- *Frequency* The frequency of a particular type of injury occurring can be gauged from incident recall, accident records, industry statistics and a consideration of possibilities. One injury per 1000 person hours is generally regarded as low, and one injury per 100 person hours is generally regarded as high.
- *Duration* The duration of a person's exposure to a hazard is often expressed as a percentage of each work day. Less than 20 percent is generally regarded as low and more than 60 percent is generally regarded as high.
- *Severity* The potential severity of injuries is generally classified as being low for scratches and bruises and high for death and permanent incapacity.

5.7 RISK CONTROL

Where a risk assessment reveals a significant risk to health or safety, controls are necessary to minimize the probability of the risk occurring.

Wherever possible, the identified hazard should be eliminated. For example, prefabricating work on the ground can eliminate the hazard of persons falling.

Where it is not possible to eliminate hazards, risks can be controlled by:

- Selecting a less hazardous form of scaffolding or access system.
- Modifying the design of the system.
- Isolating the scaffold.

Where these measures do not adequately control risk, the use of appropriate personal protective equipment may also be necessary. Personal protective equipment should only be used where other solutions are not practicable. For example, individual fall arrest systems should not be the primary means of protecting people working at heights.

SECTION 6 SELECTION

6.1 OCCUPATIONAL HEALTH AND SAFETY

Equipment suppliers, scaffolding designers and scaffolding contractors must consider occupational health and safety laws, taking into account the appropriate risks for the projects that should have been established by the principal employer.

Scaffolds should:

- Be constructed in accordance with a verified design or the supplier's documented information.
- Incorporate adequate and appropriate means of access and egress.
- Have safeguards to protect persons within their vicinity from hazards arising from their presence and use.
- Be used safely to carry out the necessary work.

6.2 CRITERIA

While selecting a scaffold for a particular job, consider:

- The proximity of public space or adjoining property.
- The proximity and operating radius of vehicles, cranes and other moving machinery.
- The proximity of powerlines or hazardous substances.
- The strength and condition of the ground or other supporting structure.
- The profile of the work face and any adjacent structure.
- The levels at which the job will be carried out.
- The possibility of significant impact forces being applied to the scaffold, such as demolition work and crane loading of materials.
- The exposure of the scaffold to significant environmental loads such as wind-loading, snow loads, ice build-up and heavy rain on shade cloth.
- The entry and exit for workers, materials and plant.
- The systems required to promptly rescue persons from the scaffold in the event of an accident or other emergency.
- The type of work to be undertaken.
- The materials and equipment to be used for the job and their dimensions and weight.
- The number of people required to carry out the job.
- The expected duration of the job.
- The work that will precede and follow the job.

6.3 SUPPORTING STRUCTURE

Where doubt exists about the adequacy of the supporting structure, it should be assessed by a qualified person. The assessment should take into account:

- The most adverse combination of dead loads, environmental loads and live loads.
- Any other loads on the supporting structures.
- Any equipment or other plant likely to be used on or from supporting structures.
- Any alterations to be made to supporting structures.
- Likely deterioration of supporting structures.

Where an assessment indicates the need for a supporting structure to be strengthened, the measures adopted should be as specified by a qualified person.

6.4 FITNESS FOR PURPOSE

The scaffold designer should be provided with the following information, so that the scaffold is suitable and safe for its intended use:

- The desired location of working platforms.
- Specific requirements for public protection or protection of adjoining property.
- The location of powerlines, services, public thoroughfares and operational plant.
- The nature and location of any hazardous substances.
- Clearances required at the work face and in relation to other structures or plant.
- Specific requirements for access to and egress from working platforms.
- The range of acceptable dimensions for working platforms.
- Specific edge-protection requirements.
- The maximum number of platforms to be worked on, or loaded at any one time.
- The required duty loadings for working platforms.
- Details of expected significant environmental loads.
- The maximum allowable dead load of the scaffold, which may be relevant where the scaffold is to be lifted by crane, built inside a vessel or used in a situation where its weight needs to be known.
- Any limitations and critical features of the supporting structure, such as point load limitations or counterweights on roofs.
- Any other relevant information.

SECTION 7 MATERIALS

7.1 COMPLIANCE WITH STANDARDS

The first step in the process of safely erecting and using scaffolding is to carefully select the correct scaffolding equipment for the particular job.

When ordering equipment for scaffolding, the buyer should specify that the equipment is intended to be used for the construction of a scaffold. New or unused scaffolding equipment should comply with any relevant Standards that are listed in Table 7.1.

Upon request, the supplier of any equipment should provide a statement that the equipment has been designed, manufactured and, where relevant, tested and marked in compliance with the requirements of any relevant Standards that are listed in Table 7.1. If the equipment is not new, the supplier should also provide a statement that the equipment, when new, complied with the relevant Standards that are listed in Table 7.1 at the time of its manufacture and also that, as supplied, it is in a fit and serviceable condition.

Suppliers of scaffolding equipment should provide adequate information that will enable the equipment to be used according to design specifications. The information should include guidance for the servicing and inspection of the equipment and the rejection of faulty equipment.

TABLE 7.1
STANDARDS FOR SCAFFOLDING EQUIPMENT

Type of equipment	Standard
Boatswain's chair seats	AS/NZS 1576.1
Chain	AS 2321, NZS/ISO 1835,
Cradles	AS 1576.4, AS/NZS 1576.1
Eyebolts	AS 2317, NZS/BS 4278
Fibre ropes	AS 4142.2, NZS/BS 2052
Plywood	AS/NZS 2269
Portable ladders—metal	AS 1892.1, NZS 5233
Portable ladders—timber	AS 1892.2, NZS 3609
Prefabricated platform units	AS/NZS 1576.3
Prefabricated scaffold components	AS/NZS 1576.3
Rigging screws and turnbuckles	AS 2319, NZS/BS 4429
Scaffold accessories	AS 1576.2
Scaffold couplers	AS 1576.2
Scaffold planks	AS 1577, NZS 3620
Scaffold tubes	AS/NZS 1576.3
Scaffolding hoists	AS 1418.2
Scaffolding hoist protective devices	AS 1576.4
Shackles	AS 2741, NZS/BS 3551
Sheave blocks	AS 2089
Slings—chain	AS 3775
Slings—fibre rope	AS 1380
Slings—flat synthetic-webbing	AS 1353.1
Slings—wire coil flat	AS 1438
Slings—wire rope	AS 1666, NZS/ISO 7531
Splitheads	AS/NZS 1576.5
Stairways—temporary	AS/NZS 1576.1
Steel wire ropes	AS 3569, NZS/BS 302.2
Thimbles for fibre rope	NZS 1583
Thimbles for wire rope	AS 1138
Trestles (other than trestle ladders)	AS/NZS 1576.5
Wire rope grips (bulldog grips)	AS 2076

7.2 INSPECTION OF USED EQUIPMENT

Used scaffolding equipment should be inspected to identify items that are unsuitable or that fail to comply with relevant Standards. Use the supplier's guidelines to determine the suitability of scaffolding equipment for further use. Table 7.2 and Section 13 give criteria for general inspection.

Rejected scaffolding equipment should be isolated by clear marking or tagging, or placing in containers or areas that are clearly designated for the storage of rejected equipment. Each item of rejected scaffolding equipment will need to be treated in one of the following ways:

- Repaired (e.g. replacing bolts on couplers, replacing missing wedges on modular scaffolding, re-binding ends of timber planks, appropriate re-welding).
- Reduced in length (e.g. shortening of tube, planks or wire rope to remove defective ends).
- Downgraded (e.g. downgrading a scaffold plank for use as a soleplate, provided the fault does not adversely affect the performance of the soleplate).
- Scrapped.

Records of tests, maintenance, inspections and alterations should be kept and made available upon request.

TABLE 7.2
UNSUITABLE OR DEFECTIVE SCAFFOLDING EQUIPMENT

Item	Defects
1 Scaffold tubes (see Clause 13.8)	Outside diameter less than 47.5 mm Steel wall thickness less than 3.6 mm In New Zealand, hot-dipped galvanized steel wall thickness less than 2.9 mm Aluminium wall thickness less than 4.2 mm End cut not square to axis End flame cut Unduly pitted Heavily corroded Distorted, twisted, bent or split
2 Couplers	No manufacturer's or supplier's mark Distorted, stretched or cracked Bent or stretched rivets or pins Threaded blind holes Stripped threads on bolts or nuts Seized bolts or nuts
3 Baseplates	Bent or distorted so as to prevent an even bearing Thickness less than 6 mm Surface area less than 225 cm ² (e.g. 150 mm × 150 mm) Spigot length less than 50 mm Spigot diameter less than 16 mm (or equivalent cross-sectional area)

(continued)

TABLE 7.2 (continued)

Item	Defects
4 Adjustable leg/base plate/castor	Extension exceeds 600 mm Missing fixed stop enabling less than 150 mm length of spigot to remain in tube A clearance between the inside diameter of the tube and the top of the spigot of more than 5 mm Spigot distortion causing it to jam in the tube Seized nuts
5 Castors	Manufacturer's or supplier's mark absent Working load limit not marked Fitted with pneumatic tyres Wheel diameter less than 125 mm Pintle length (either internal spigot or external socket) less than 150 mm Diametrical clearance between the pintle and tube more than 3 mm Eccentricity of the axle relative to the pintle more than 65 mm Wheel brake ineffective or missing No means to positively fix castor to the standard (see Appendix B)
6 Prefabricated structural components (see Clause 13.10)	Ends flame cut Butt-welded in the length Unduly pitted Heavily corroded Distorted, twisted, bent or split Welds cracked, broken or missing Locking devices damaged, inoperative, unrestrained or missing
7 Timber scaffold planks (see Figure 13.9(A))	Marking not in compliance with AS 1577 or NZS 3620 Width of less than 220 mm Nominal thickness reduced by more than 10% Laminations in a laminated plank are separating Warped, twisted, broken, split or worn End hoop iron broken or damaged End fixing missing Painted or treated in any way that may conceal defects Deep burns Deep oil stains that render the surface slippery Nails projecting Rot of any kind
8 Metal scaffold planks (see Figure 13.9(B))	Marking not in compliance with AS 1577 or NZS 3620 Width of less than 220 mm Distorted, twisted, bent, split or crushed Heavily corroded Welds cracked, broken or missing Rivets broken or missing End piece crushed or missing

(continued)

TABLE 7.2 (continued)

Item	Defects
9 Steel wire ropes (see Figure 13.15)	For suspended scaffolding hoists, any visible broken wire For hung scaffolds, more than 10 percent of the visible wires broken in a length equal to eight rope diameters One or more broken wires immediately below a metallised socket Outer wires worn more than 33 percent of their nominal diameter Birdcaging or birdnesting Strands high standing Core popped Severe kinking Strands crushed Wire-rope grips used on suspension ropes or secondary ropes (see Clause 11.5.8) Thimbles distorted Corrosion evident
10 Chains	Links deformed, chipped, nicked, cracked or stretched Nominal diameter or dimension reduced by more than 10 percent through wear Not a genuine lifting chain Corrosion evident
11 Shackles	Working load limit not marked Crown or pin diameter reduced by more than 10 percent through wear Bolt used in place of a shackle pin Distortion or over-straining evident Corrosion evident

SECTION 8 GENERAL DESIGN

8.1 INTRODUCTION

Section 8 provides general advice for the safe construction of the basic types of scaffolds.

8.2 GENERAL

The construction of a scaffold must comply with the relevant Australian or New Zealand occupational health and safety requirements.

Those responsible for the erection, alteration and dismantling of scaffolds should ensure that an erected scaffold complies with the relevant requirements of AS/NZS 1576.1, AS 1576.2, AS/NZS 1576.3, AS 1576.4 and AS/NZS 1576.5. The ways to achieve this are:

- Build the scaffold to the supplier's information about the system.
- Build the scaffold in accordance with AS 1576.3 Supplement 1.
- Build the scaffold to a design that has been verified by a competent person as complying with the relevant requirements of AS/NZS 1576.1, AS 1576.2, AS/NZS 1576.3, AS 1576.4 and AS/NZS 1576.5.

The design specifications of a scaffold can be either a complex set of drawings or held in the mind of the responsible scaffolder. It is the verification that the design complies with the relevant requirements of AS/NZS 1576.1, AS 1576.2, AS/NZS 1576.3, AS 1576.4 and AS/NZS 1576.5 that is important, not the form the design takes.

General loading information is given in Appendix E.

8.3 FOUNDATIONS

The foundations for a scaffold must be adequate to carry and distribute the loads imposed at each standard and of the whole loaded scaffold.

The loads likely to be carried by the scaffold's standards may need to be calculated, to enable a determination as to whether the supporting surface can hold them safely. Calculations must include the dead load and the live load. The dead load includes the self-weight of the standard plus the connected components, including ledgers, transoms, putlogs, braces, ties, planks, guardrails and attachments such as sheeting. The live load is the duty live load for persons plus materials for each platform in each bay, which should be based on the values given in Table 8.7.

To calculate the live load on a standard, assume that each standard in that bay supports one third of the duty live load on each platform in each adjoining bay. This is to allow for off-centre loading of platforms and concentrated loads placed closer to one standard.

NOTE: As an example, consider a heavy duty scaffold.

The live load on any one standard (i.e. between two bays, each of 675 kg) based on the worst possible case of loading in adjacent bays = $675 \text{ kg} \times 2 \div 3 = 450 \text{ kg}$.

Assume that the dead load of the scaffold supported by this standard is 250 kg. Then, the design load for this standard = $450 \text{ kg} + 250 \text{ kg} = 700 \text{ kg}$.

Now that the design load on the standard is known, the minimum length of the soleplate can be determined for a known ground or floor load capacity.

Assume the ground load capacity is 2000 kg/m^2 and a typical solid timber scaffold plank that is 220 mm wide will be used.

Therefore, the minimum acceptable length for the soleplate = $700 \text{ kg} \div 2000 \text{ kg/m}^2 \div 0.220 \text{ m} = 1.6 \text{ m}$.

The duty of the scaffold, the height of the scaffold and the ground conditions need to be considered when the foundation is designed. Where an excavation is planned near a scaffold, the supporting structure should be able to support the scaffold without any subsidence. The foundation must be maintained in good condition for the life of the scaffold.

On hard level surfaces such as steel or concrete, standards should be supported on base plates; however, sole plates may not be necessary if the surface does not require protection.

On timber flooring, pedestrian pavements and any other supporting structure that is likely to be adversely affected by point loads from standards, the load from the standards should be distributed by baseplates and soleplates. Soleplates should be at least equivalent in size and strength to scaffold planks and be long enough to prevent visual subsidence or damage to any supporting structure. As a general rule, each soleplate should be long enough so that it supports at least two standards. On surfaces such as soft asphalt and compacted gravel, soleplates with a size of at least 500 mm × 220 mm must be used under the baseplates. Scaffold plank soleplates may not be sufficient under some heavy duty scaffolds, or very high scaffolds. (See Figures 8.3(A) and 8.3(B)).

Where scaffolds require special design by a competent person, the soleplates should also be subject to special design. Advice must be obtained from a competent person, such as an engineer experienced in structural design, before scaffolds are erected on verandahs, suspended flooring systems or compacted soil. Where the scaffold is founded on sloping ground (see Figure 8.3(C)), each standard must have a stable foundation. Slopes steeper than 1 in 10 should be assessed by a competent person. On any other surfaces, the advice of a competent person is needed (see Figure 8.3(D)).

8.4 STABILITY

A scaffold must have the stability to prevent it from overturning.

Stability may be achieved by:

- Tying the scaffold to a supporting structure.
- Guying to a supporting structure.
- Increasing the dead load by securely attaching counterweights near the base.
- Adding back-up bays to increase the base dimension.

Where dead load alone is used to provide stability, the stability must comply with the requirements of AS/NZS 1576.1 (Clause 2.7 in the 1995 edition). Where the stability of a scaffold is in doubt, assessment should be carried out by a competent person.

8.5 TYING

Suppliers' information about tie methods and spacings should be followed. Where no specific information is supplied, use the relevant requirements of AS 1576.3 Supplement 1 for tying of tube-and-coupler scaffolds.

Drilled-in anchors, such as friction and chemical anchors, may only be used to secure ties where it is not practicable to use other methods. Where drilled-in anchors are used, they must have a safety factor of 3 and a competent person must assess the suitability of the supporting material.

Ties must not obstruct clear access along the full length of the working and access platforms.

Figure 8.5 shows some tie assemblies.

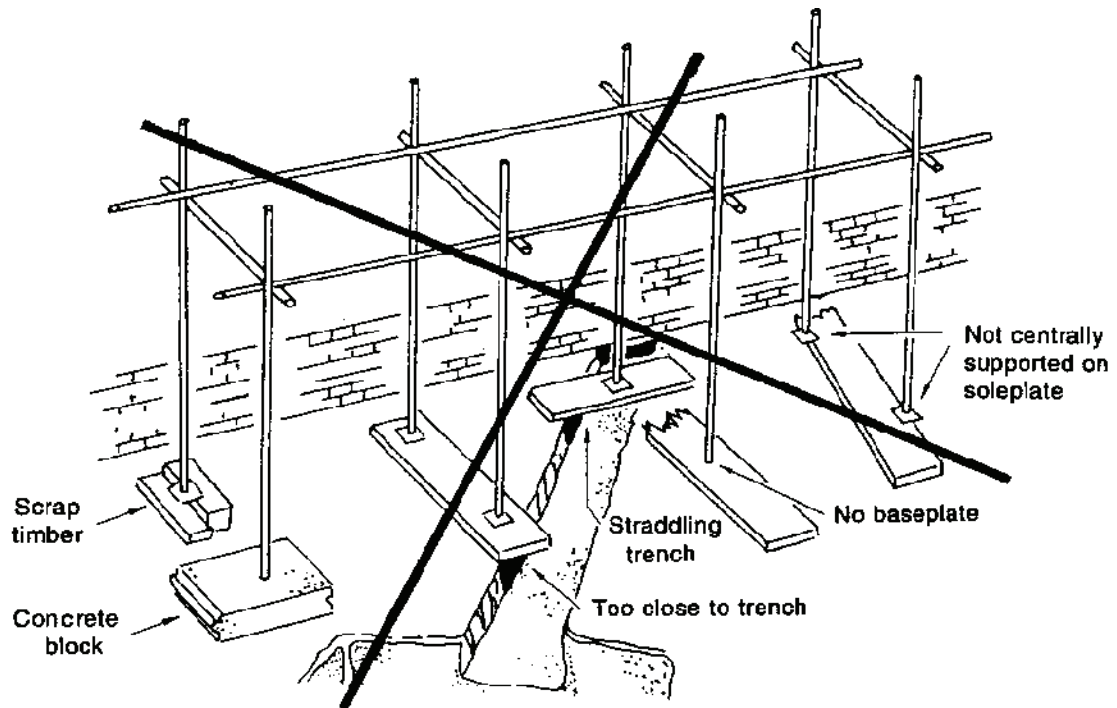


FIGURE 8.3(A) UNSAFE FOUNDATION

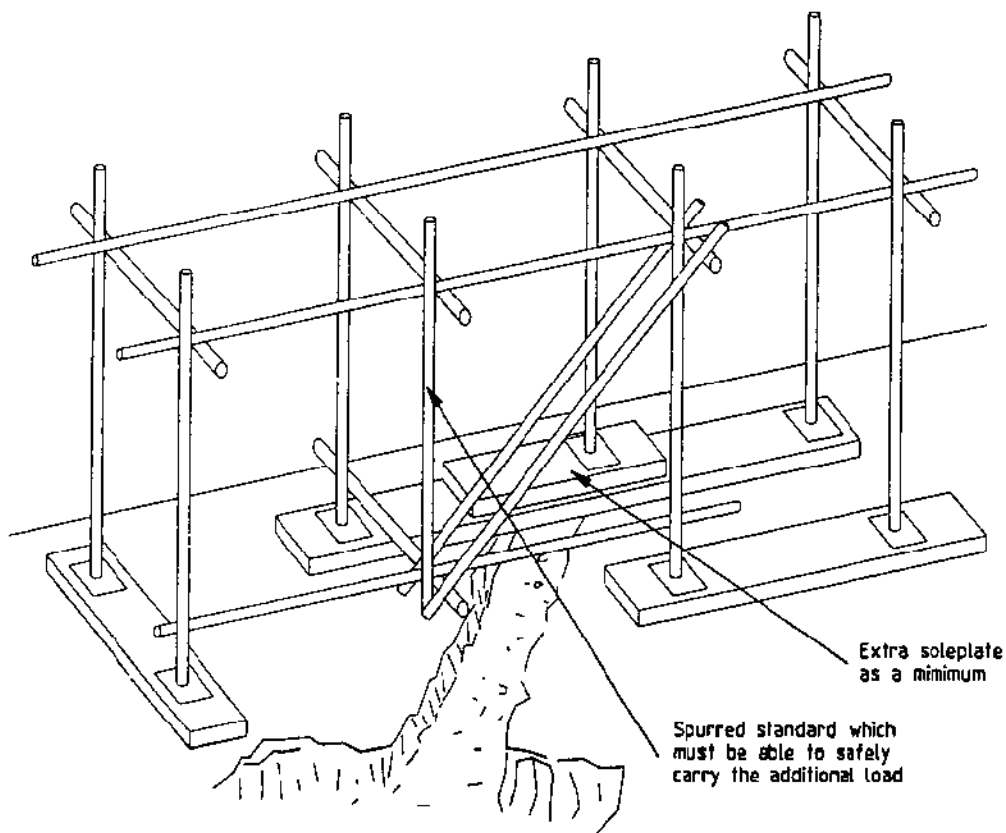
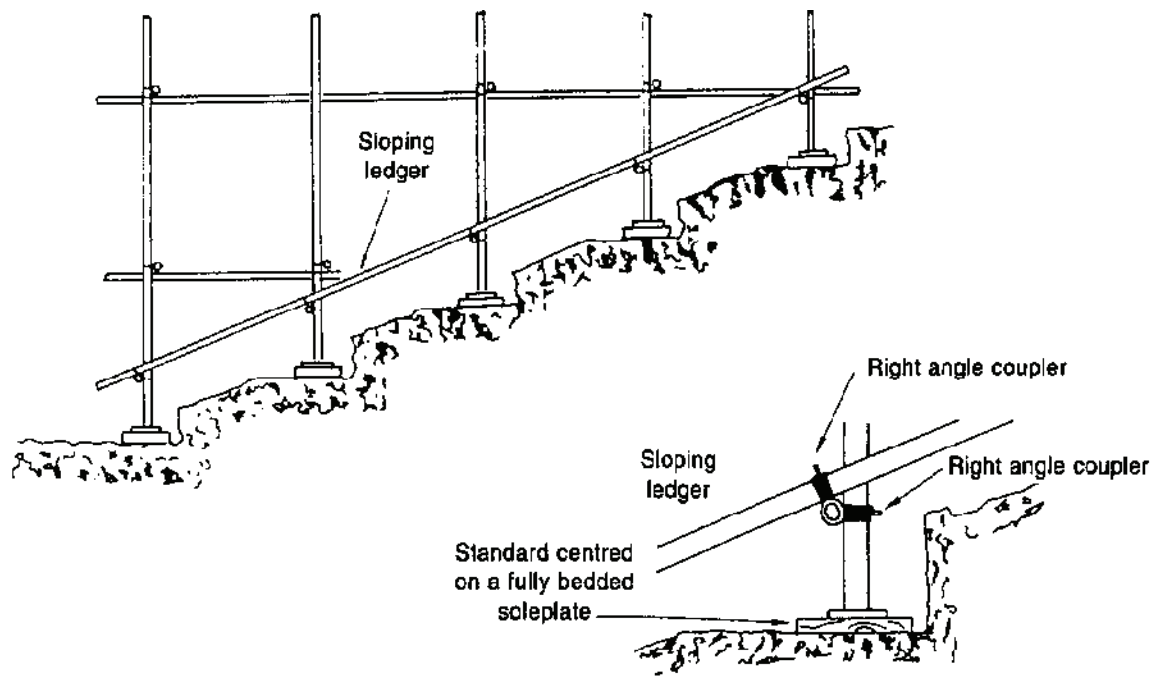
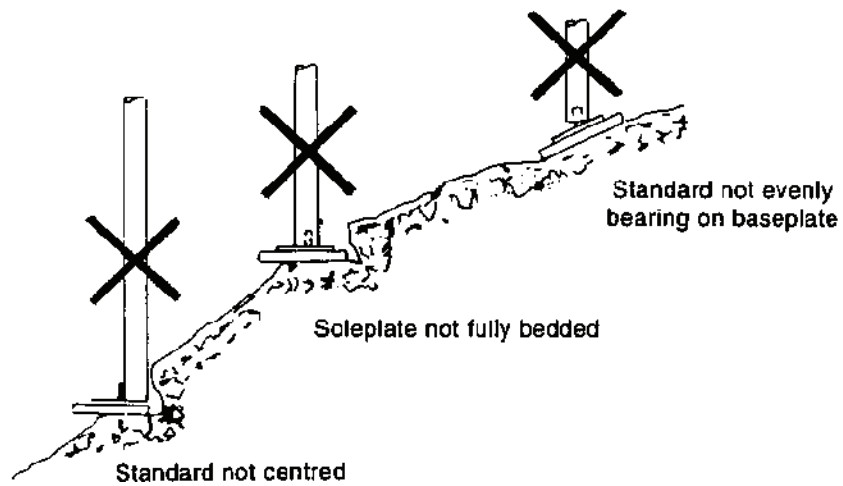


FIGURE 8.3(B) ADEQUATE FOUNDATION



(a) Good practice



(b) Unsafe practice

FIGURE 8.3(C) FOUNDATIONS ON SLOPING GROUND

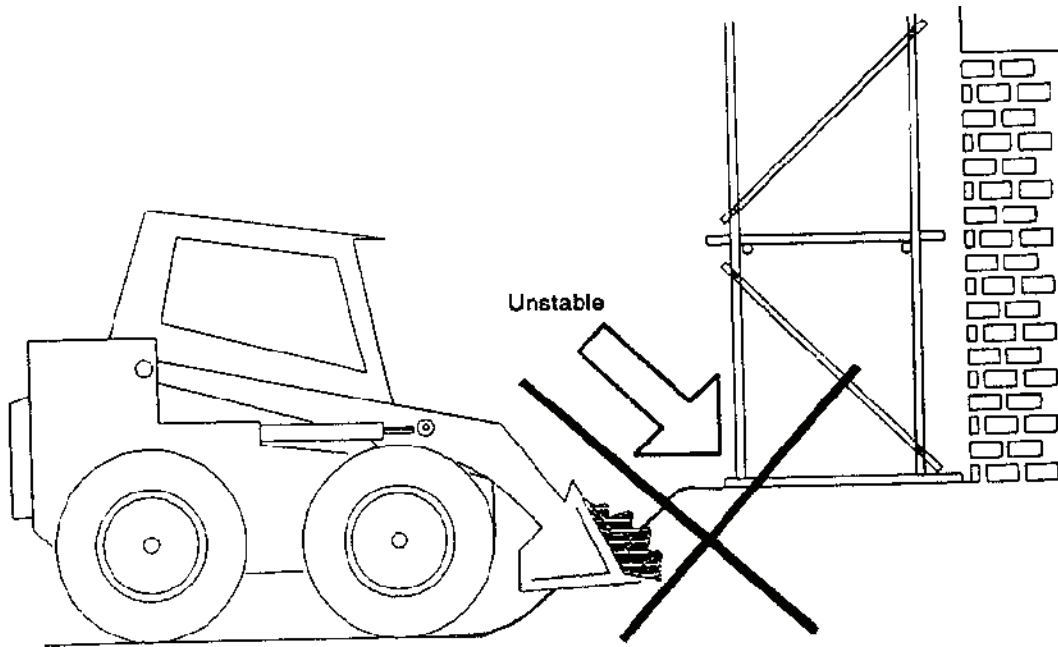


FIGURE 8.3(D) UNDERMINING OF SUPPORTING FOUNDATION

8.6 RIGIDITY

Rigidity is achieved through the use of fabricated frame units, fixed mesh panels or diagonal bracing systems. Where diagonal bracing systems are used, they should extend from the base of the scaffold to the top lift. Bracing does not have to extend to the height of the top guardrail. Typical transverse bracing is illustrated in Figure 8.6.

8.7 WORKING PLATFORMS

Each scaffold should be designed to carry the required number of working platforms and to support its live loads.

Working platforms should be constructed from either prefabricated platform units or scaffold planks.

The working platform should be wide enough to accommodate materials and plant, and allow clear and unobstructed access along its entire length.

Clear and unobstructed access should be not less than 450 mm wide, where passage is required by persons and hand tools only.

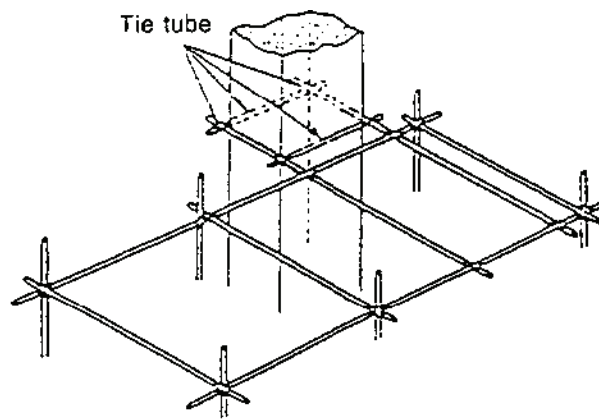
Working platforms, except suspended scaffolds should have a duty classification and dimensions complying with Table 8.7. Scaffolds designed for working platforms that support loads in excess of heavy duty should be provided with prominent signs that display the rated working load limit per platform per bay.

Working platforms should not be pitched at an angle steeper than 7° (slope of 1 to 8) to the horizontal and should have a slip-resistant surface.

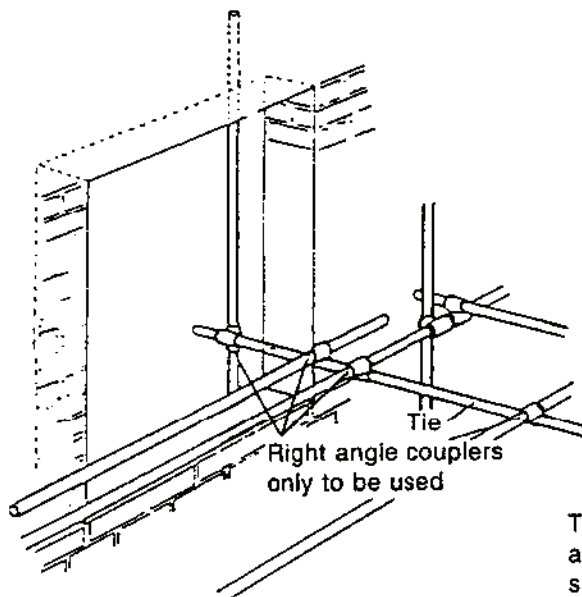
Planks or decking forming the surface of a working platform should be of uniform thickness, fixed to prevent uplift or displacement in normal use and positioned to avoid significant gaps and tripping hazards.

Planks should be butted not lapped, except at returns, curved faces or unusual profiles.

Examples of unsafe working platforms are given in Figure 8.7.



Box tie



Through ties should be placed as close as possible to the window reveal and secured with right angle couplers.

Through tie

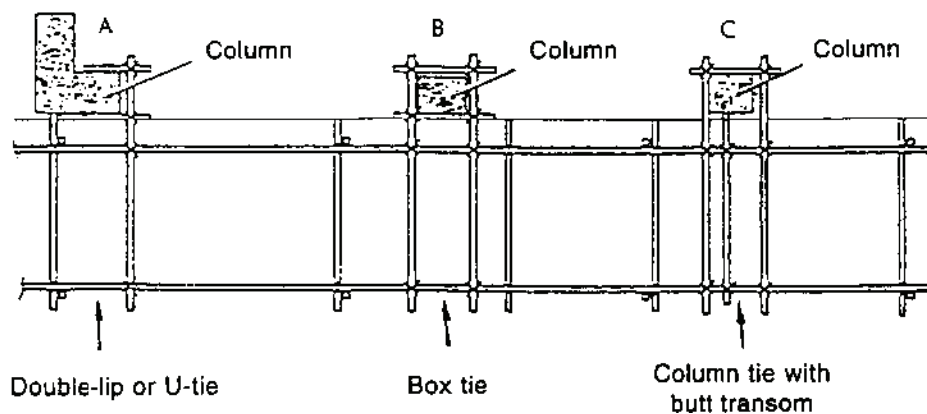


FIGURE 8.5 SOME TIE ASSEMBLIES

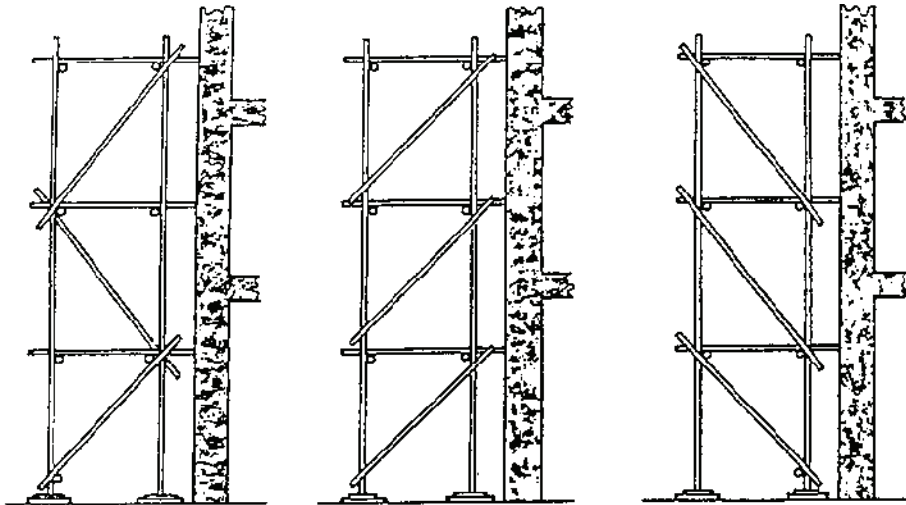


FIGURE 8.6 TYPICAL TRANSVERSE BRACING FOR A SCAFFOLD

TABLE 8.7
REQUIREMENTS FOR WORKING PLATFORMS

Duty classification as specified in AS/NZS 1576.1	Approximate maximum total load for persons and materials kg per platform per bay	Approximate maximum mass of any single concentrated load of materials or equipment (as part of total load) kg	Minimum length and width of platform mm
Light duty*	225	100	450
Medium duty	450	150	900
Heavy duty	675	200	1 000

* Materials must not be stored on light duty working platforms that have the minimum allowable width.

8.8 EDGE PROTECTION

Edge protection should be provided to the open sides and ends of any platform from which a person or object could fall 2 m or more.

Where the nature of the work makes it difficult for a person to be fully aware of the proximity of the platform edge (e.g. overhead work or welding), edge protection should be provided, regardless of the height of the platform.

Suitable forms of edge protection are:

- Guardrails, midrails and toeboards.
- Guardrails and toeboards, together with suitable infill such as brickguards or 17 mm plywood.
- Mesh panels performing the function of a guardrail and incorporating kickplates.

Do not use fibre rope or steel wire rope as guardrailing. Only chain should be used to close off access openings in guardrailing.

Guardrails and midrails should be supported at intervals not exceeding 3 m, parallel with the platform and not further than 100 mm outside the platform edge. The guardrail should be between 900 mm and 1100 mm above the platform. The midrails should be positioned approximately midway between the guardrail and the toeboard.

Guardrails, midrails and stanchions may be constructed from one of the following:

- Scaffold tube.
- Purpose designed components.
- Oregon, hardwood or other timber of equivalent strength and characteristics with nominal cross-sectional dimensions of 100 mm × 50 mm.

Toeboards and kickplates should be as least as strong and as rigid as timber scaffold planks and should be securely fixed. They should extend not less than 150 mm above the platform surface. The gap between the platform and the toeboard or kickplate should not exceed 10 mm.

Brickguards should be made of mesh having steel wire with a minimum diameter of 4 mm with an aperture not exceeding 50 mm × 50 mm (see Figure 8.8(A)). They should be securely fixed. Brickguards may incorporate a kickplate.

Fabricated mesh panels that are used instead of guardrails should be at least as strong as a guardrail. They should be made of mesh consisting of steel wire with a diameter of not less than 4 mm and apertures not exceeding 50 mm × 50 mm. They should also incorporate a strong kickplate with a height of at least 150 mm.

Self-closing gates should be provided where fixed edge protection is absent due to a ladder or stairway, unless the openings are a safe distance from the working platforms.

Where there is not a guardrail or a midrail provided adjacent to the working face of a building or a structure, the face should:

- Be less than 225 mm from the platform edge.
- Extend at least 900 mm above the top surface of the platform.
- Have a strength and rigidity of not less than that of a guardrail.
- Perform the function of a guardrail and midrail in all other respects.

Where a toeboard or kickplate is not provided adjacent to the working face of a building or structure, the gap between the platform edge and face should be less than 225 mm. A safeguard to prevent anyone being endangered by debris falling from the platform should also be installed.

Where necessary, cantilevered catch platforms should be provided to prevent unintended spillages from falling to the ground. (See Figure 8.8(B)).

8.9 ACCESS AND EGRESS

Every working platform must have safe and suitable access and egress (see Figures 8.9(A) and 8.9(B)).

Common means of access and egress include existing floor levels, permanently installed platforms, ramps, stairways, ladders, personnel hoists, temporary access ways, temporary stair systems and portable ladders.

The form of access and egress for working platforms on a scaffold depends upon the nature of the work, the site conditions and restrictions, the height of the platforms, the number of people required for the work and the time the scaffold will be standing.

Where access is provided by mechanical means, such as a personnel hoist, an alternative form of egress, such as a ladder or stair tower, should also be provided for emergency use.

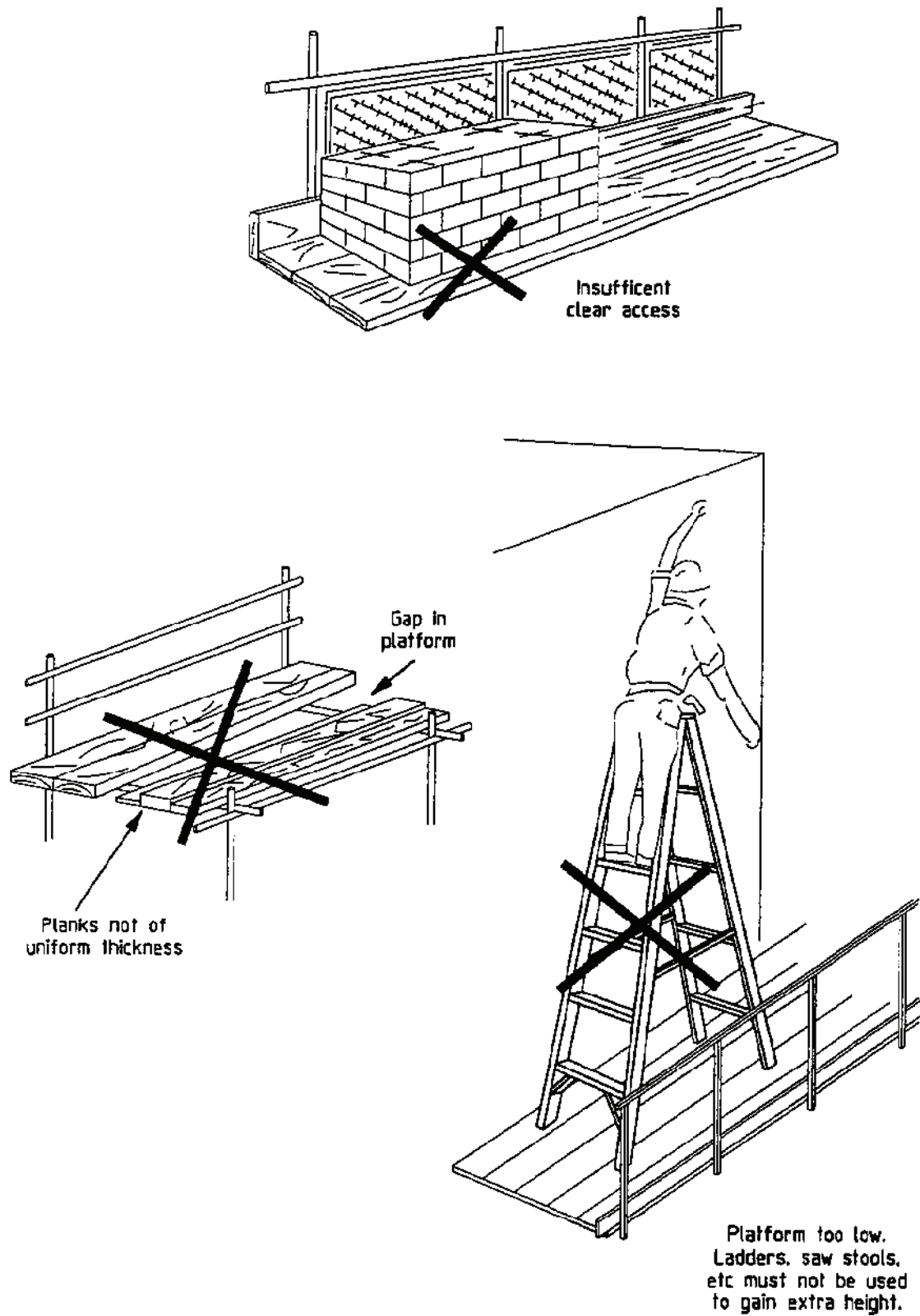


FIGURE 8.7 UNSAFE WORKING PLATFORMS

Access ways can be flat or ramped so as to give access from permanent floors. They must be built to the same requirements as heavy duty working platforms, except that their width can be reduced to:

- 675 mm, for transporting materials.
- 450 mm, when used by persons with hand tools only.

The slope of a ramped accessway, such as a barrow ramp must be not greater than 20° (or 1:3) to the horizontal. Where the slope exceeds 7° (or 1:8), the upper surface must be cleated to stop people from slipping.

Cleats should be 25 mm thick, 50 mm wide and extend across the full width of the platform. A 100 mm gap may be left in the centre of the platform for the wheel of a barrow. Cleats should be fixed every 450 mm along the length of the ramp.

Temporary stair systems can be constructed using tube-and-coupler scaffolding. Most prefabricated modular scaffolding systems include stairway parts. Modular stair systems are generally available in multi-bay forms with landings every 2 m and in single-bay forms with landings every 1.5 m.

Ladder access may be used where access to the working platform is needed by only a few persons and where tools and equipment can be delivered separately to the working platform (such as by a rope and gin wheel, materials hoist or crane). Portable ladders intended for access to or within scaffolds must be industrial grade single ladders.

Ladders should be within a separate ladder access bay of the scaffold wherever space permits. Ladders may be fixed to the external face of a scaffold, provided that the stability of the scaffold is not adversely affected and there is unobstructed access to and from the ladder without having to climb over or through guardrails.

Rules for ladder access systems include:

- Pitch ladders at a horizontal to vertical slope of not less than 1:4 nor more than 1:6.
- Secure ladders against displacement in any direction.
- Unless the base is at ground level or on a fully-covered supporting structure, provide ladder landings at the head and at the base of each ladder.
- Make sure the ladder extends at least 900 mm above the landing.
- Make sure the height between successive ladder landings is never more than 6 m.
- Keep openings for ladders as small as practicable and provide trapdoors over or guarding around any openings that may be in or directly beside a working platform.
- Offset the base of a ladder from the head of any ladder that may be directly below it, so that the ladders cannot take the form of a single continuous ladder.
- Make sure there is adequate access to and egress from ladders at each landing.
- Make sure ladders on mobile scaffolds are clear of the supporting surfaces.

Ladder landings and stairway landings require the same level of edge protection adjacent to their open sides and ends as working platforms.

Where a ladder encroaches on a working platform, an unobstructed access of at least 450 mm width along the full length of the working platform should be provided.

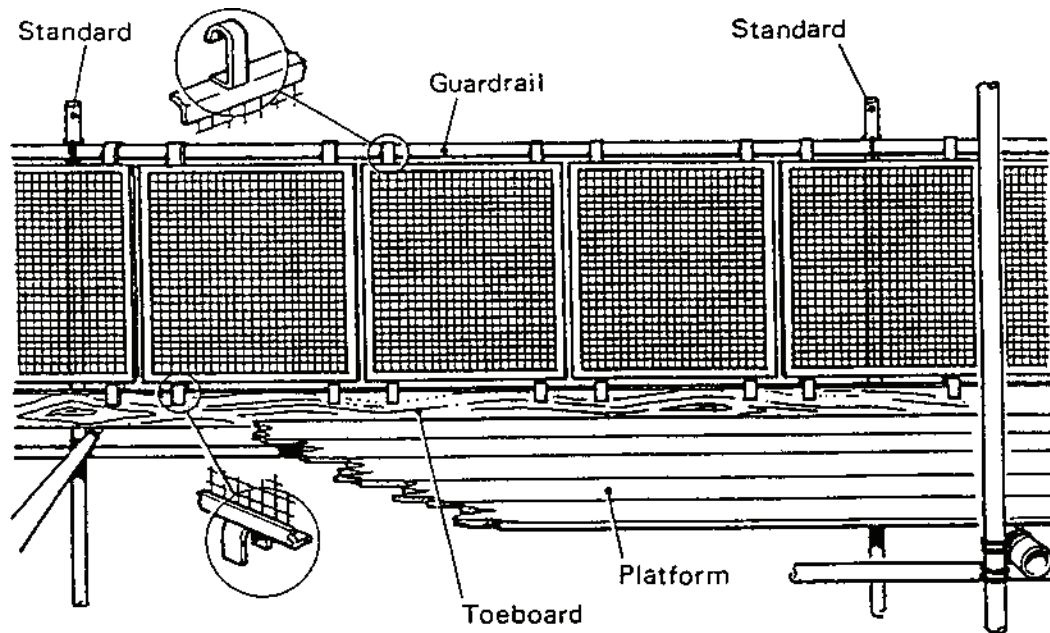
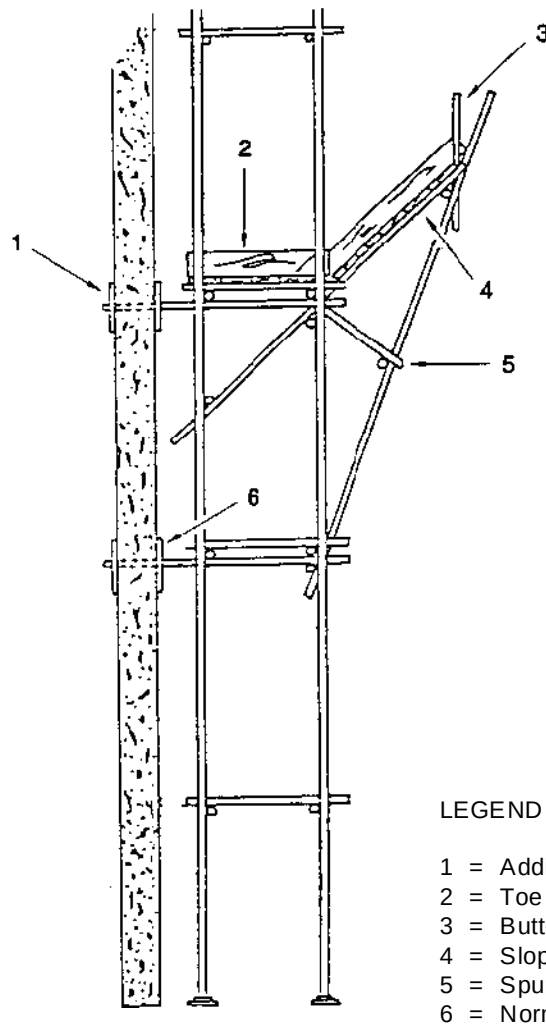


FIGURE 8.8(A) TYPICAL BRICKGUARDS



LEGEND

- 1 = Additional tie
- 2 = Toe board
- 3 = Butt-tube to support toe board
- 4 = Sloping putlog
- 5 = Spur tied at midspan
- 6 = Normal tie

FIGURE 8.8(B) TYPICAL CANTILEVERED CATCH PLATFORM
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8.10 CONTAINMENT SHEETING

Containment sheeting is used for both safety and environmental purposes. Where work is carried out close to pedestrian or vehicular access, scaffolds that are sheeted down to hoarding level can minimize both the risk to the public and the area lost to public access.

When selecting containment sheeting, consider:

- Degree of protection required from rain or washing down operations.
- Ability to contain dust during abrasive blasting operations.
- Need to contain airborne dust and debris from surface treatments.
- Possible reaction to chemicals used in facade cleaning.
- Flammability of the material.
- Need for light transmission.
- Ventilation of the contained space.
- Need for mobility.
- Size of sheet.
- Pattern and frequency of fixing points.

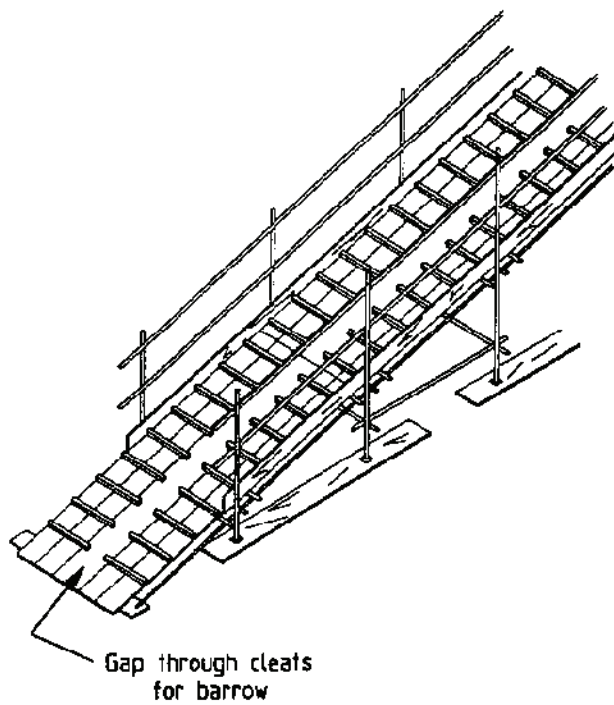
Hessian is not suitable for use as containment sheeting. It has a low strength and is a high fire risk.

Scaffolds fitted with containment sheeting have increased dead loads and are exposed to increased wind and rain loads. The design of such scaffolds and ties must be approved by a competent person, such as an engineer experienced in structural design. The design of a sheeted scaffold should consider:

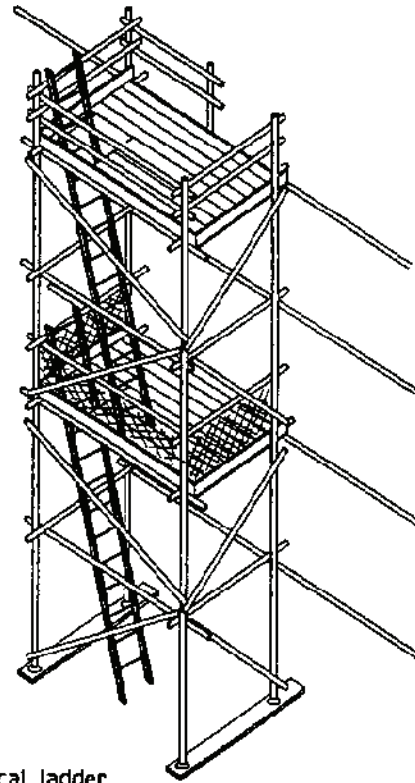
- The weight of the sheeting supported by the scaffold.
- The wind load on the scaffold and the supporting structure.
- Position of the sheeted scaffold in relation to adjacent structures and the possible local increase in wind speeds through narrow gaps between such structures.
- The compatibility of fixings with respect to the strength of the sheeting.
- Specification of additional fixings at returns of the scaffold and at upper levels where wind loads are greater.
- Ability of the fixings or sheeting to fail at wind speeds below those that would cause damage to the scaffold or supporting structure.
- Where a failure of fixings cannot relieve the wind load on the scaffold, whether sheets can be fixed inside the guardrails and toe boards.

The following advice should be followed when fixing containment sheeting to a scaffold:

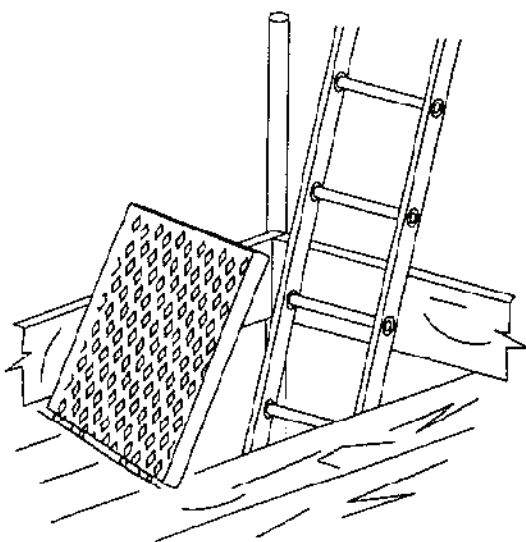
- The advice given in Figures 8.10(A), 8.10(B) and 8.10(C).
- On tube-and-coupler scaffolds, ensure that transoms and putlogs do not project more than 150 mm over the outside ledger. Cap or pack around ends of tubes in contact with sheeting.
- Use fully decked platforms with guardrails while erecting sheeting.
- Sheeting should be fixed to the outside of the scaffold, unless otherwise permitted by the design.
- Sheeting should be continuous, by using sufficient overlap.
- Tightly tie butt joints if overlap is not possible.



(a) Typical barrow ramp

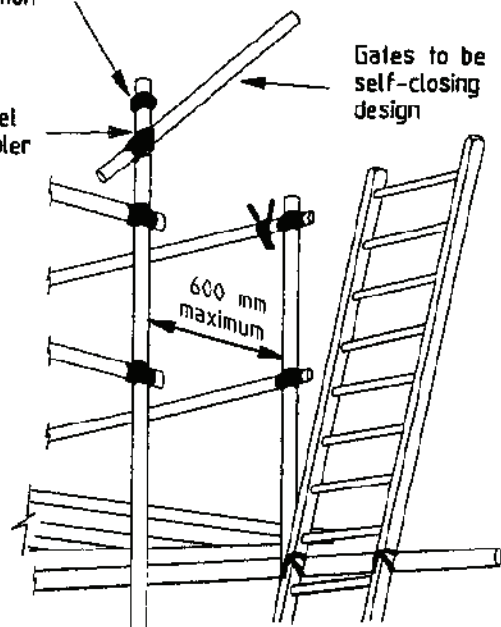


(b) Typical ladder access tower



(c) Typical internal ladder access with hinged trap door

Install coupler above swivel to prevent gate remaining in vertical position



(d) Typical external ladder access

FIGURE 8.9(A) TYPICAL MEANS OF ACCESS

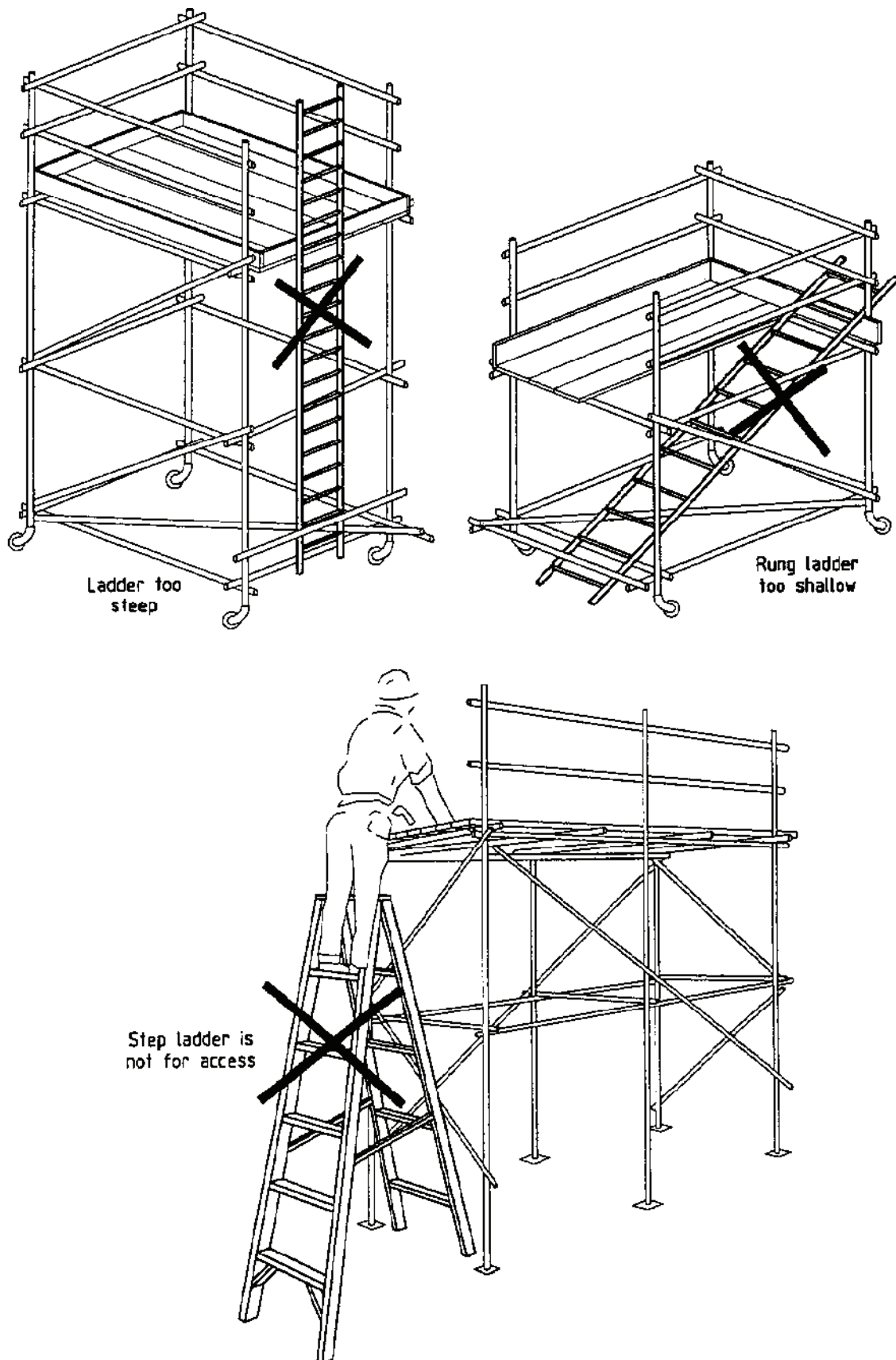


FIGURE 8.9(B) UNSUITABLE FORMS OF LADDER ACCESS

- Begin by securing the top edge then lower the sheet on the outside of the scaffold. In windy conditions, attached tag lines to the bottom edge. Tightly tie the bottom edge to the scaffold.
- Where practicable, the lower edges of sheets should be tucked inside the lower sheets.
- Ensure that sheeting is sufficiently taut and fixed tight against the scaffolding to prevent chafing.
- Where containment sheeting has inbuilt fixing points, ensure that the layout and position of these points is compatible with the scaffolding framework.
- Use every applicable fixing point.
- Ensure that sheeting is in direct contact and tied back to the open edges of the platform and guardrails.
- Ensure that there are as few gaps as possible between sheeting and the platform edge.

8.11 STEEL WIRE ROPE (SWR)

Figure 8.11(A) illustrates a typical construction of SWR. Figure 8.11(B) shows a correct method of measuring the diameter of a SWR.

Where SWR is used as a static load-bearing member in a scaffold, it must not be subjected to a load greater than one sixth of its breaking load. Suspension ropes and secondary ropes for suspended scaffolds require higher safety factors, as detailed in Section 11.

The manufacturer's guaranteed minimum breaking load should be stated on a test certificate, which should come with each new SWR. If this information is not available, assume that the rope is constructed from the lowest quality steel (i.e. normally 1570 MPa).

Most working load tables for SWR are based on one fifth of the breaking load, which is acceptable for general rigging; however, for scaffolding multiply the working loads in these tables by 5/6 (i.e. 0.83), to provide the required factor of safety of one sixth of breaking load.

If a SWR does not have a test certificate, its working load limit (WLL) for a scaffolding application may be attained by calculation, using the formula:

$$WLL = 7.5 \times (D \text{ in mm})^2, \text{ in kilograms}$$

For example, for rope with a diameter of 12 mm

$$\begin{aligned} WLL &= 7.5 \times (D \text{ in mm}) \times (D \text{ in mm}) \\ &= 7.5 \times 12 \times 12 \\ &= 1080 \text{ kgs} \end{aligned}$$

NOTE: This formula should not be confused with the formula used for general rigging applications, which is $8 \times \text{diameter squared}$.

To select the smallest diameter of a rope needed to support a known load, use the formula:

$$\text{Diameter} = \sqrt{[(\text{Load in kg}) \div 7.5]}, \text{ in millimetres}$$

For example for a load of 750 kg

$$\begin{aligned} D &= \sqrt{(750 \div 7.5)} \\ &= \sqrt{100} \\ &= 10 \text{ mm} \end{aligned}$$

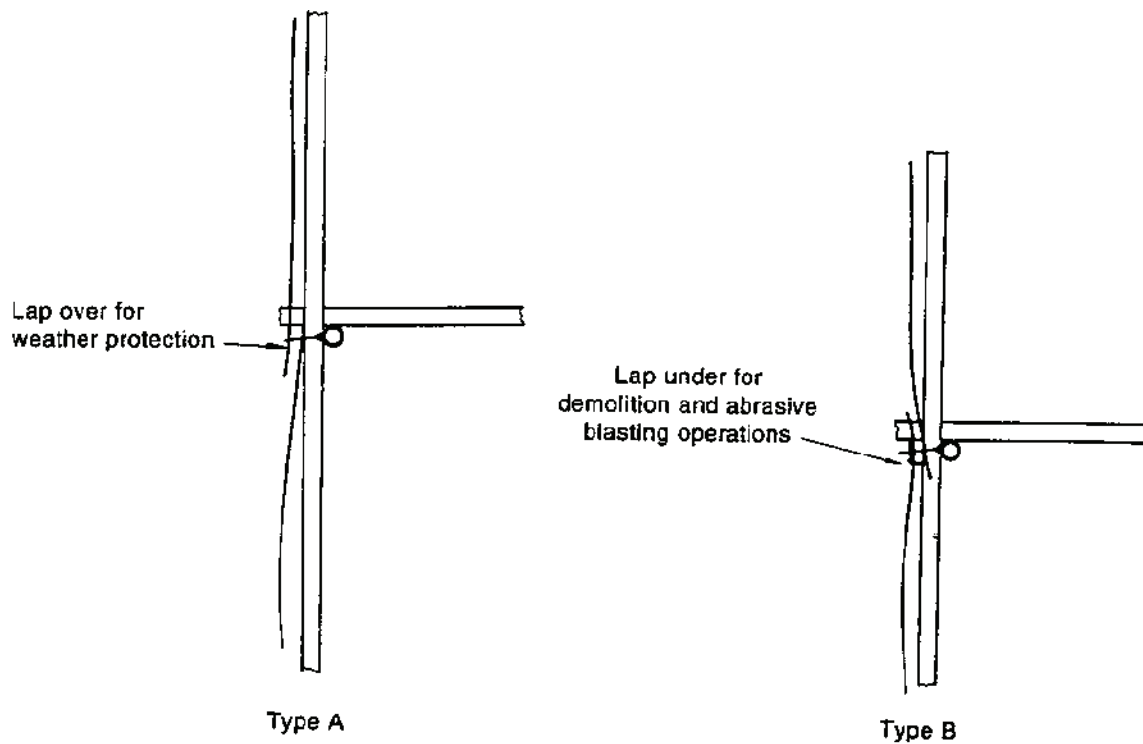


FIGURE 8.10(A) TYPICAL LAP DETAILS OF CONTAINMENT SHEETING

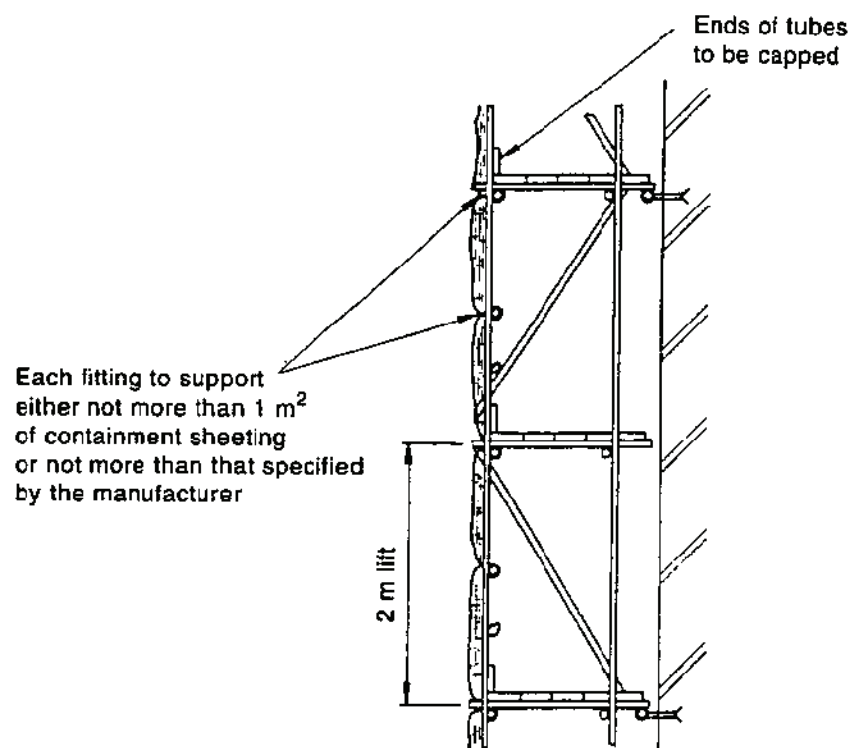


FIGURE 8.10(B) TYPICAL FIXING ARRANGEMENT OF CONTAINMENT SHEETING

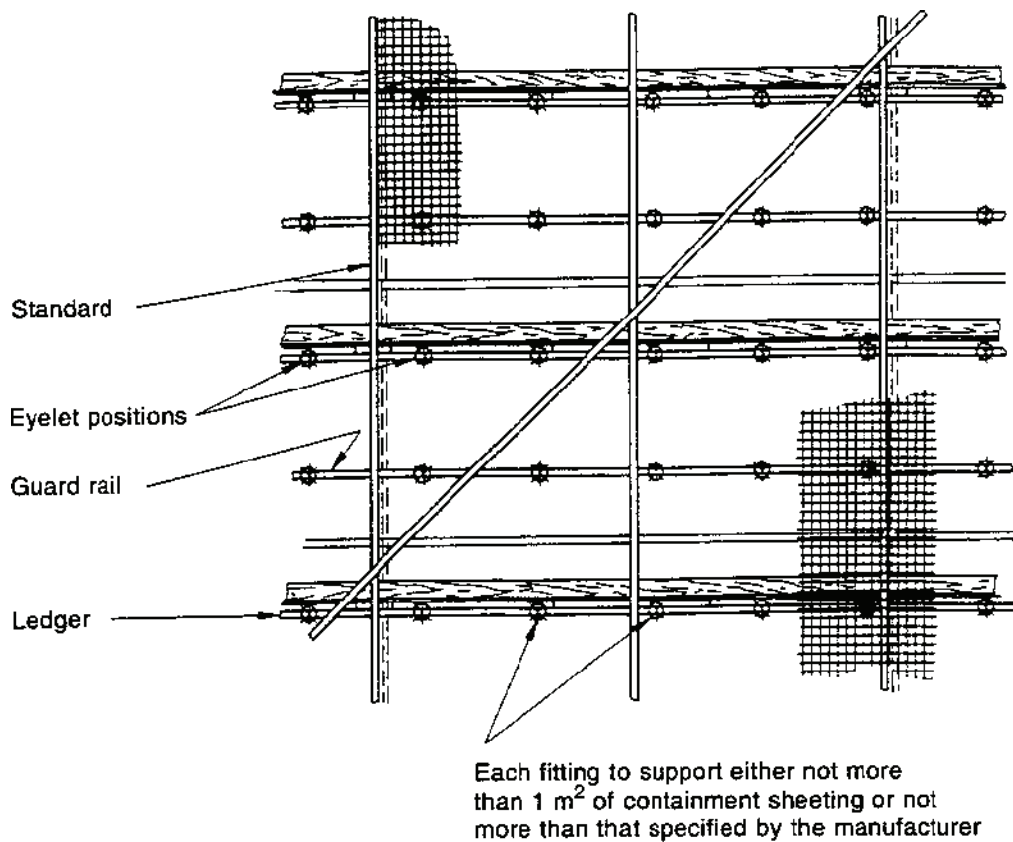


FIGURE 8.10(C) TYPICAL POSITIONING OF EYELETS TO SUPPORT CONTAINMENT SHEETING

Where the SWR is at an angle to the vertical, reeved or choked, the lifting capacity is reduced. In these cases, the WLL is reduced by a load factor. The reduced capacity is called the safe working load (or SWL).

Below are some common load factors:

Vertical unreeved rope	= 1.0
30° to the vertical	= 0.85
45° to the vertical	= 0.70
60° from the vertical	= 0.50
Reeved (around beam)	= 0.50

Therefore, to calculate the SWL of a 12 mm SWR reeved around a beam:

Rope diameter (D) = 12 mm

Load factor = 0.50

$$\begin{aligned}
 \text{WLL} &= 0.50 \times 7.5 \times D(\text{mm}) \times D(\text{mm}) \\
 &= 0.50 \times 7.5 \times 12 \times 12 \\
 &= 540 \text{ kg}
 \end{aligned}$$

Further advice on the care and use of steel wire ropes is given in AS 2759.

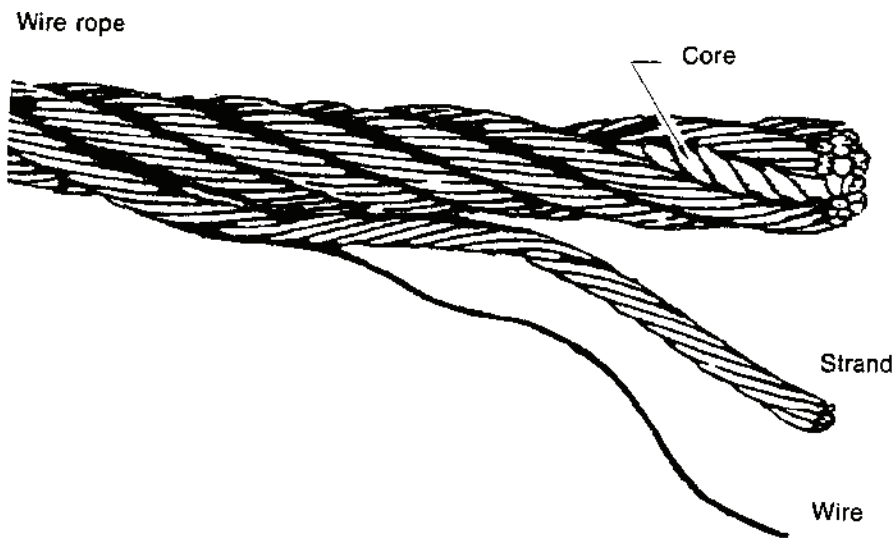


FIGURE 8.11(A) A TYPICAL CONSTRUCTION OF STEEL-WIRE ROPE

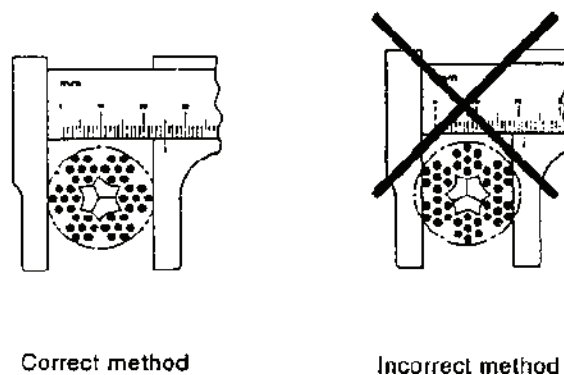


FIGURE 8.11(B) MEASURING THE DIAMETER OF STEEL-WIRE ROPE

8.12 CHAIN

Chain used as a static load bearing member in a scaffold must not be subject to any load greater than one sixth of its breaking load.

Only short-link lifting chains are permitted to be used in scaffolding. These are manufactured in several grades ranging from 315 MPa to 800 MPa.

The grade should be marked on the links. From weakest to strongest, the most commonly used grades are:

Grade 30 is marked L, 30 or 3.

Grade 40 is marked P, 40, 4 or 04.

Grade 60 is marked S, 60 or 06.

Grade 80 is marked T, 800, 80, 8, HA PWB, or CM.

SECTION 9 GENERAL CONSTRUCTION

9.1 SELECTION OF APPROPRIATE METHOD

To select the best method of raising and lowering equipment, consider:

- The size of the scaffold.
- The nature of the scaffold.
- The type of equipment.
- The amount of equipment.
- The proximity of hazards (e.g. electric wires, protrusions, obstructions).
- The ready availability of manual or mechanical lifting appliance (e.g. gin wheels, air or electric winches, on-site cranes).

9.2 ORGANIZING THE WORK SEQUENCE

When two or more scaffolders are working aloft on the scaffold, they should develop a methodical work sequence that:

- Allocates specific tasks to each scaffolder.
- Properly organizes the work.
- Ensures that the work is properly secured.
- Minimizes the risk of accidents.

9.3 ADVERSE WEATHER

Special precautions should be taken before:

- Working aloft in high winds that could cause the scaffolder to overbalance or lose control of the equipment being handled.
- Working from incomplete platforms when the scaffold is wet or coated with ice or snow.

9.4 PLATFORM

Scaffolders should work from a fully-decked platform.

Metal scaffold planks used with modular scaffolding should cover the full width of the framework, where there are no other means to prevent the planks from sliding.

Solid timber scaffold planks less than 44 mm thick should be supported at intervals of not more than 1.8 m.

9.5 GUARDRAILS

Guardrails should be installed as soon as practicable.

Where a scaffolder could fall a distance of more than 2 m, guardrails should be installed during erection of the scaffold. The guardrails should remain in place until that section of the scaffold is dismantled.

9.6 ACCESS

Means of access to the scaffold should be provided from the start of erection, progressed with the scaffold and used by the scaffolder whenever possible.

9.7 WORKING ALOFT

Work practices must minimize the risk of injury from falling. Scaffolders should be trained in all aspects of fall prevention.

The use of safety nets is discussed in Appendix F.

9.8 SAFETY HARNESS

Safety harnesses may be used to provide protection against a fatal or serious fall when a scaffolder is required to work over a void, or lean out from the scaffold or supporting structure without the protection of a guardrail; however, they can increase the risk of injury if used incorrectly or unnecessarily.

Safety harnesses can be used during:

- The erection and dismantling of hung scaffolds prior to or upon removal of the platform.
- The fixing and removal of trolley tracks on suspension rigs.
- The attachment and removal of spurs projecting from the supporting structure.
- The erection and dismantling of cantilever scaffolds prior to or upon removal of the initial platform.

Safety harnesses and lanyards should comply with, be inspected in accordance with and be used in compliance with the relevant requirements of AS 1891, 2626, NZS 5811.1, NZS 5811.2 and the suppliers' information. The inspection check list in AS 2626 is reproduced in Appendix G.

Anchorage points should have a working load limit of not less than 1500 kg and be located above or directly behind the scaffolder. Inertia reels should be used where vertical mobility is needed. The inspection check list in AS 2626 is reproduced in Appendix H.

9.9 EQUIPMENT HANDLING

The scaffolder must have suitable and serviceable lifting gear to raise and lower equipment. Unserviceable gear should either be repaired (i.e. restored to sound condition) before use or be discarded.

9.10 HANDBALLING

Handballing (or chaining) is usually employed for the first few lifts of a scaffold. It involves a gang forming a human chain up the face of the scaffold, to pass equipment from one to another (see Figure 9.10).

Where guardrailing is not yet fixed, it is crucial that scaffolders brace themselves against an unjoined standard while leaving both hands free to pass equipment.

Do not release equipment until the next person in the chain has indicated they have full control. Scaffolders receiving the equipment should call 'mine' when they are ready and in control.

Handballing should only be used where scaffolders do not have to overreach or bend to pass or receive equipment.

9.11 HANDLINES

Handlines are used for raising or lowering equipment where it is not possible to set up a gin wheel.

Scaffolders using a handline should adopt a secure position. On lifts where guardrails have not been installed, use an unjointed standard for support with one leg placed behind the standard to act as an anchor (see Figure 9.11).

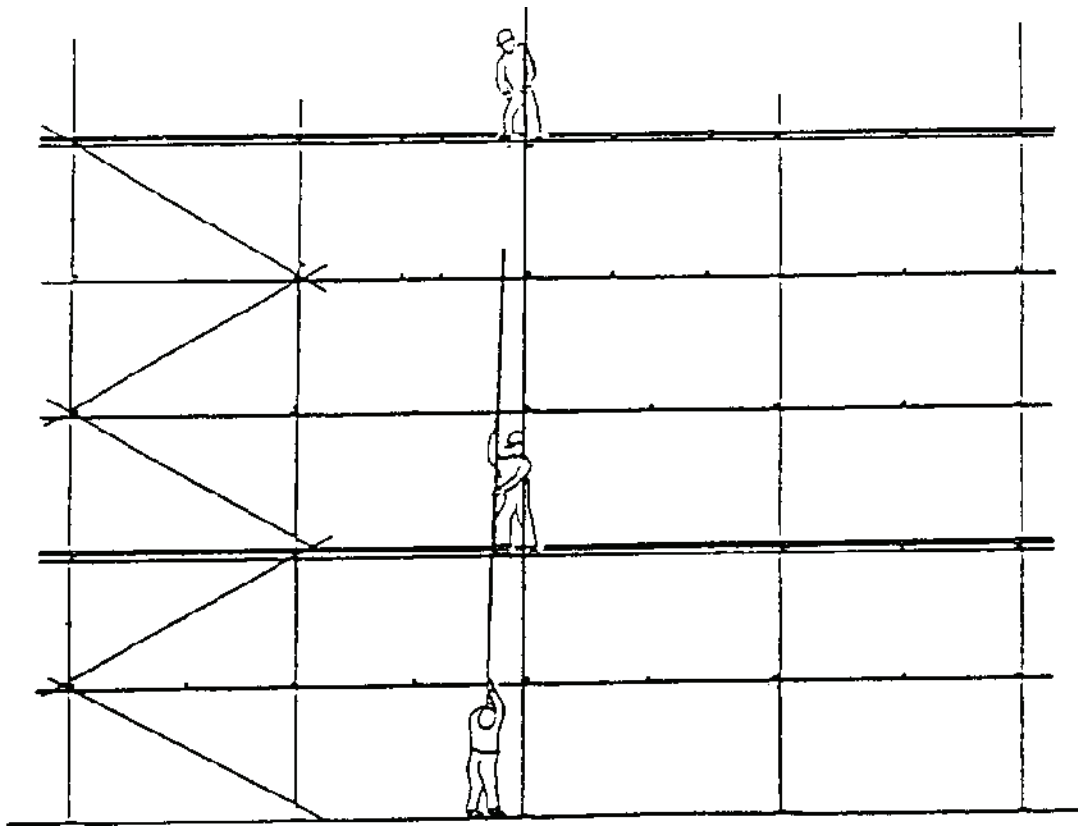


FIGURE 9.10 HANDBALLING

9.12 TOPPING UP STANDARDS

Topping up standards is one of the most hazardous tasks for scaffolders. The dangers of muscle strain and overbalancing can be minimized if the following rules are followed:

- Wherever possible, make sure that the standard joints are positioned immediately above the lift to avoid high top ups.
- Avoid using standards with a length of more than 4 m. Longer tubes are very difficult to top up without strain and should only be used as ledgers, guardrails and braces.
- When topping up a standard, grasp the tube with one hand above your head with a straight arm as a pivot point and while grasping the base of the tube with the other hand, guide the tube to vertical and place it in position.

9.13 USE OF GIN WHEELS

Only blocks that incorporate a rope guide should be used as a gin wheel.

Hook-type gin wheels should not be hooked onto a coupler or onto the end of a tube.

A coupler should be fixed either side of the suspension point, to prevent movement of the gin wheel (see Figure 9.13).

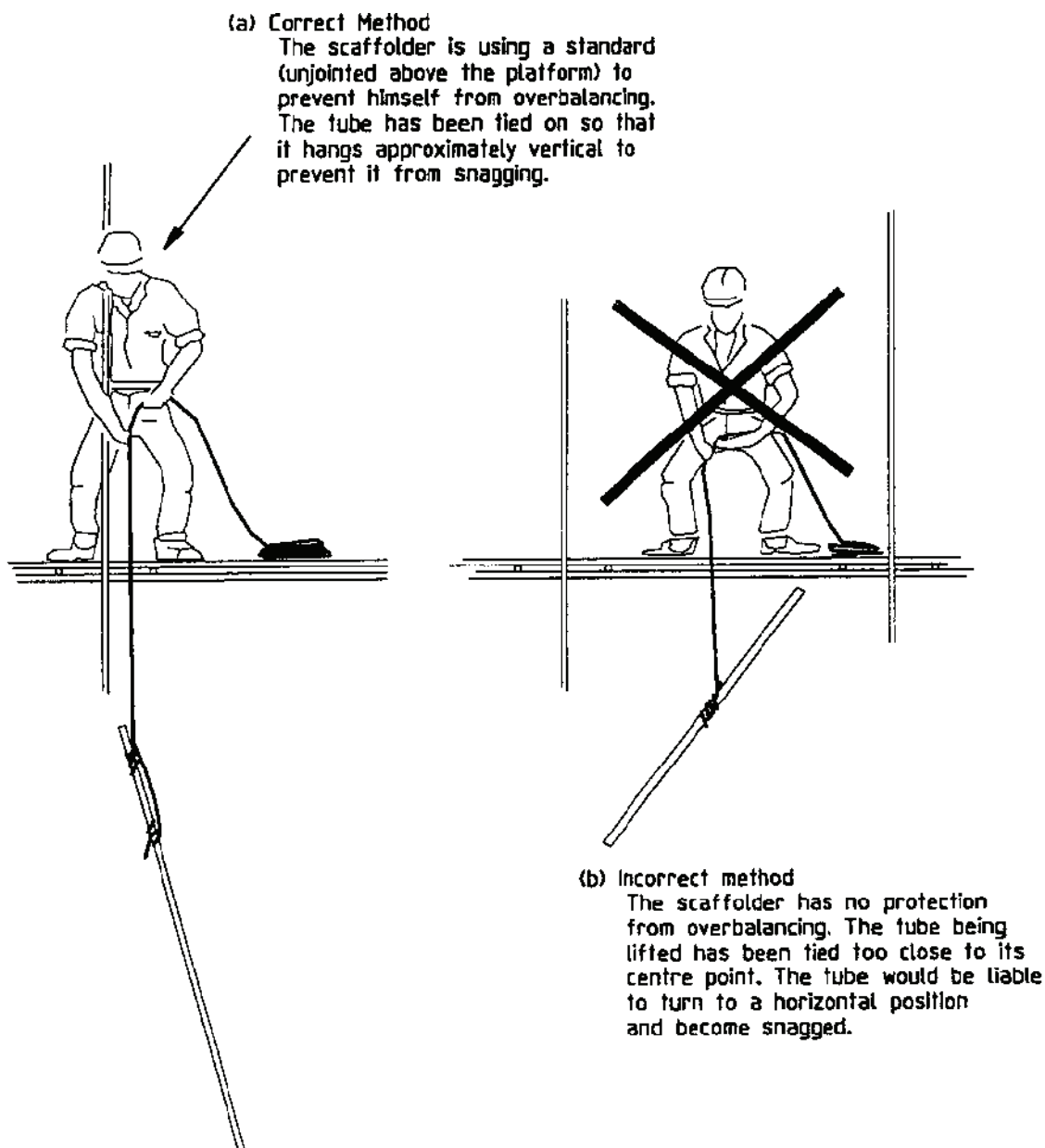


FIGURE 9.11 USE OF A HANDLINE

Where a gin wheel is attached to a cantilevered scaffold tube, the tube should be fixed to a standard with a right-angle coupler. Where the suspension point projects more than 600 mm beyond the standard, the tube should be braced back to the scaffold.

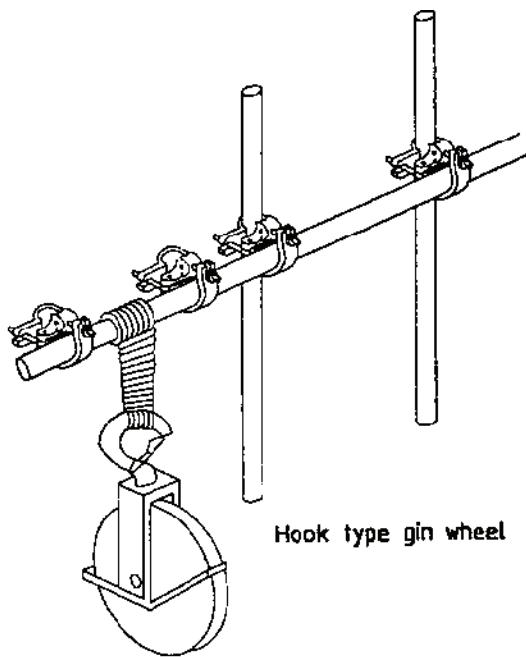
Hook-type gin wheels should be moused or provided with a spring-loaded safety catch.

The rope should normally be joined in a continuous loop using a double sheet bend (or similar), forming a tail for attaching the load. Alternatively, a figure-of-eight knot should be formed in the rope to stop it from running through the wheel.

Loads raised or lowered by the use of gin wheels should not exceed 50 kg.

9.14 CANTILEVER BUILDERS' HOISTS

Information on the erection and use of cantilever builders' hoists complying with AS 1418.7 is given in Appendix I.



Hook type gin wheel

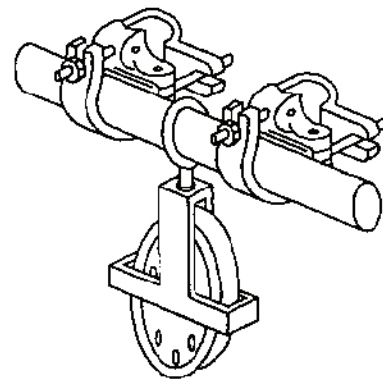
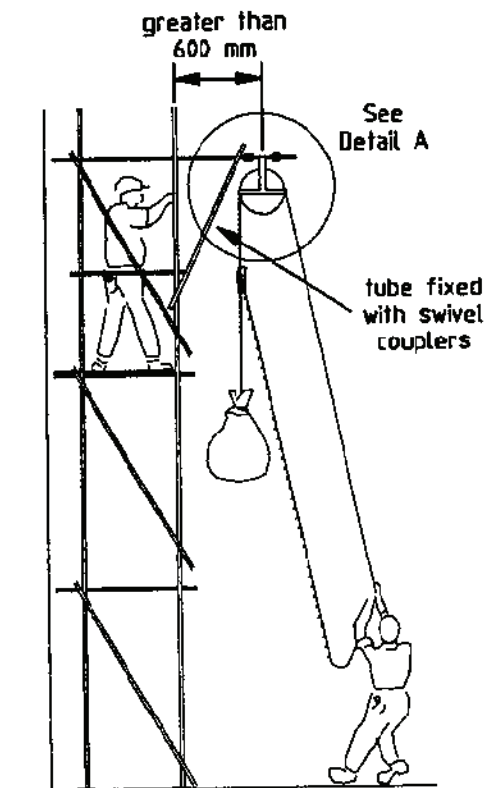
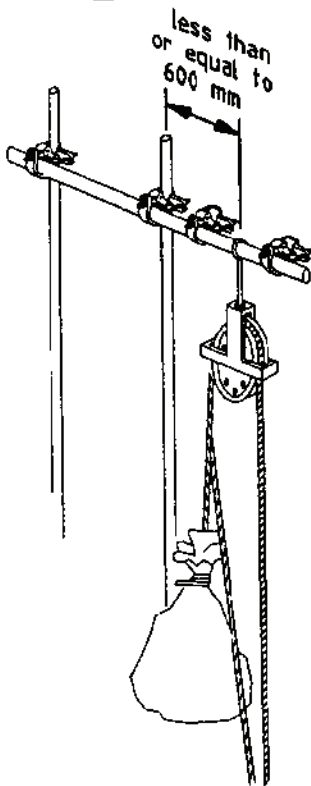
Detail A
Ring type gin wheel

FIGURE 9.13 GIN WHEELS

SECTION 10 ADDITIONAL REQUIREMENTS FOR SPECIFIC TYPES OF SCAFFOLD

10.1 TRESTLE SCAFFOLDS

Trestle scaffolds should be constructed only on a hard level surface, with each trestle in the fully-opened position. Figure 10.1(A) illustrates typical splithead and trestle scaffolds.

Frame trestle scaffolds that do not incorporate height adjustment should be constructed only on a hard level surface.

NOTE: Care should be taken to ensure that height adjustment is achieved only by using the purpose designed pins. Nails or pieces of reinforcing bar must not be used.

The working platform should be not more than 2 m above the supporting surface, regardless of the size of components.

Trestle scaffolds should not be erected where a person or object could fall from the scaffold a distance exceeding 2 m, such as from the edge of an open floor.

Do not piggy-back trestles to construct additional lifts (see Figure 10.1(B)).

Trestle ladder scaffolding should be used for light duty loadings only.

Frame trestle scaffolding may be used for heavy, medium or light duty loadings, depending on the supplier's documented information.

Putlog trestle scaffolding may only be used as heavy duty scaffolding.

The maximum spacings of trestles should not exceed the maximum putlog spacings given in Table 10.1.

For working platforms on trestle scaffolds, light duty platforms should have a width of not less than 450 mm and heavy duty and medium duty platforms should have a width of not less than 900 mm. For all types of work, the full width of the trestles should be planked when using frame trestle and putlog trestle scaffolds. Planks should overhang their end supports by not less than 150 mm nor more than 250 mm and where necessary be secured against uplift. Planks may be lapped to form multiple bays of scaffolding, provided there is no obstruction along the full length of the working platform. The working platform should be horizontal.

Work should only be performed between the trestles. Where a trestle scaffolding is more than one bay in length and being used for bricklaying, blocklaying or similar work, concentrated loads should be placed directly over the putlogs. Incorrect uses of trestle scaffolds are shown in Figure 10.1(B).

TABLE 10.1
MAXIMUM SPAN OF SOLID TIMBER SCAFFOLD PLANKS
COMPLYING WITH AS 1577

Nominal thickness of plank mm	Maximum span between putlogs m
32 (hardwood only)	1.0
38	1.5
50	2
63	2.5

10.2 BRACKET SCAFFOLDS

10.2.1 General The maximum spacing of brackets must not exceed the spacings for putlogs given in Table 10.1.

On straight runs, planks must be butted.

Planks must overhang their end brackets by not less than 150 mm nor more than 250 mm.

Where the brackets do not require guardrails, the planks must be cleated hard against the brackets to prevent displacement, with the cleats on the underside of the plank.

Where planks are lapped or are adjacent to curved or unusually shaped profiles, the upper planks must extend not less than 150 mm past their end brackets.

Where guardrails, midrails and toeboards are required, the brackets must incorporate stanchions, sockets or accessories to which stanchions can be positively fixed.

10.2.2 Tank-bracket scaffolds Make sure that the materials to be welded are compatible.

Each bracket should be constructed of angle steel with a section of not less than 43 mm × 43 mm × 4.5 mm or mild steel tube of equivalent strength and rigidity.

Each bracket should be provided with a hook formed from 50 mm × 6 mm mild steel for attachment to the supporting structure.

Lugs or saddle pieces of mild steel with dimensions of not less than 50 mm × 6 mm × 225 mm and shaped to receive the hooks should be welded to the supporting structure.

The top of the lugs should be fully welded by not less than a 5 mm fillet weld. Welded lugs or saddle pieces must be fully welded.

The working platform should not exceed 675 mm in width and should only be used for light duty loadings.

Temporary access to the work platform should be in the form of independent stairways or ladder-access towers.

10.2.3 Shutter-bracket scaffolds The width of the working platform should be not less than 900 mm. Where this is not practicable, there should be clear access in width of at least 450 mm along the full length of the working platform, unobstructed by ladders or tie rods.

Where it is intended to lift or reposition the shutters with the scaffold in place, planks should be positively fixed to prevent displacement.

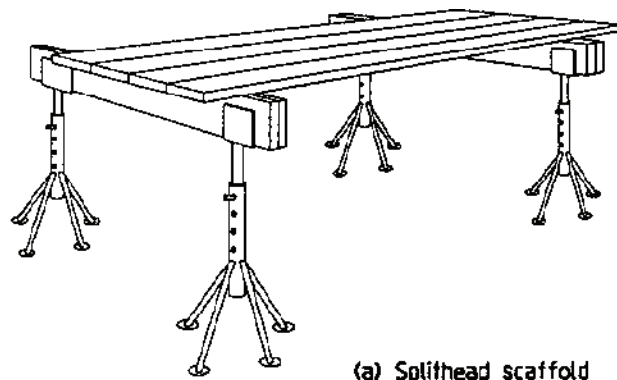
Where external access is provided, it should take the form of independent stairways or ladder access towers.

10.2.4 Stud-bracket scaffolds The working platform should have a width of not less than 450 mm and only be used for light duty loadings.

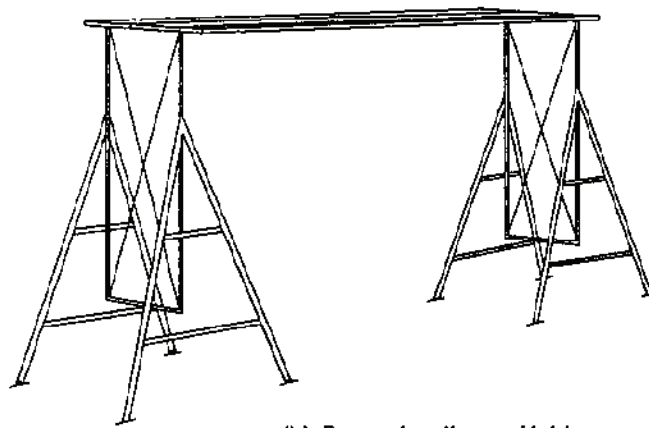
Brackets should not be supported from metal studs, unless otherwise stated in the supplier's information.

10.2.5 Ladder-bracket scaffolds Ladder bracket scaffolds are used for very light work tasks, such as signwriting where an alternative is not practicable. A typical ladder bracket scaffold is illustrated in Figure 10.2.5.

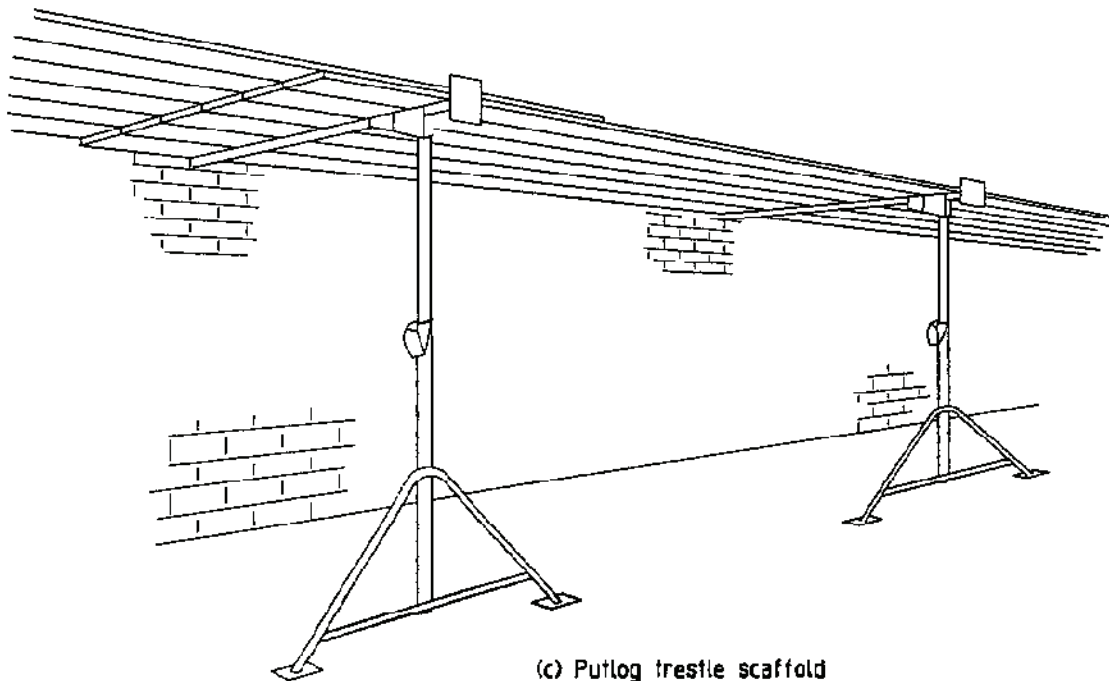
Ladder bracket scaffolds are not suitable for general construction work, because the ladders and brackets are usually not capable of safely sustaining the loads specified in AS/NZS 1576.1 and because the working platforms are usually narrower than the required minimum width of 450 mm.



(a) Splithead scaffold

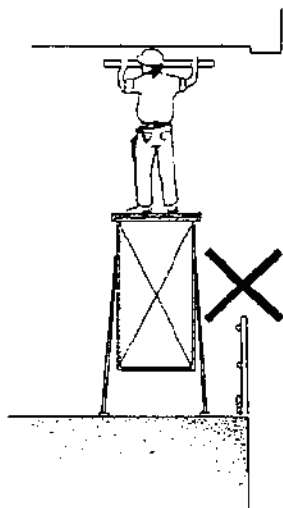
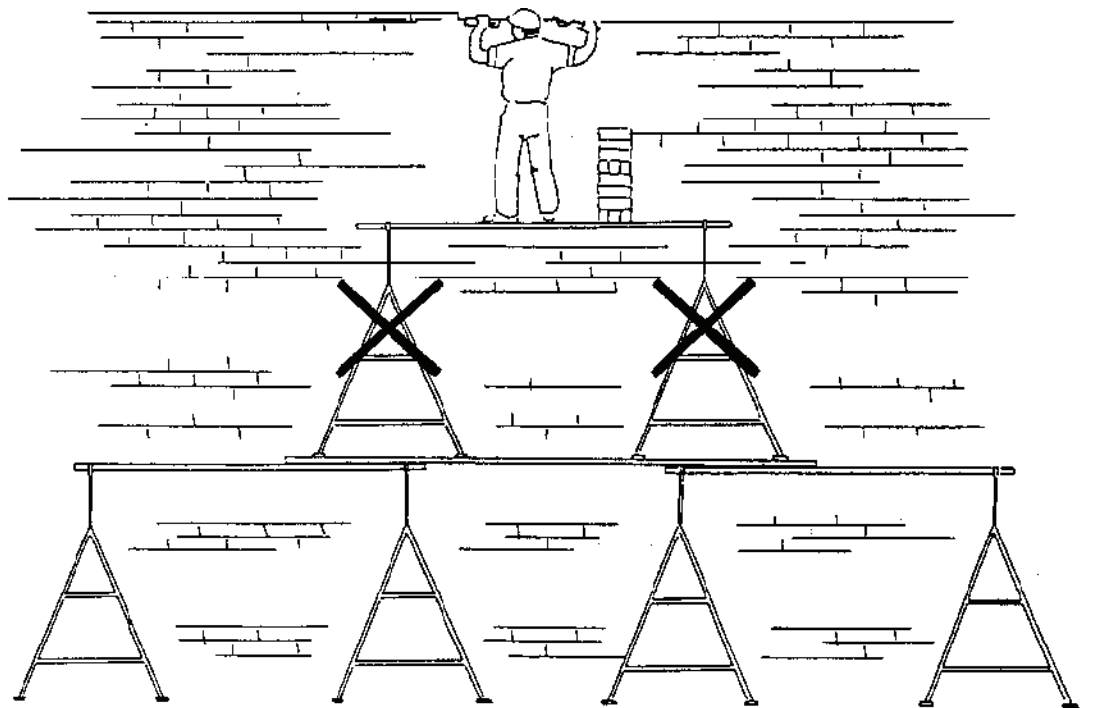


(b) Frame trestle scaffold



(c) Putlog trestle scaffold

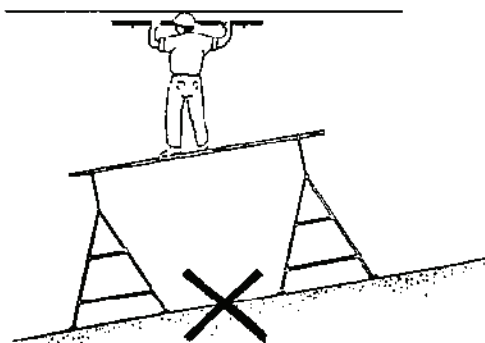
FIGURE 10.1(A) TYPICAL SPLITHEAD AND TRESTLE SCAFFOLDS



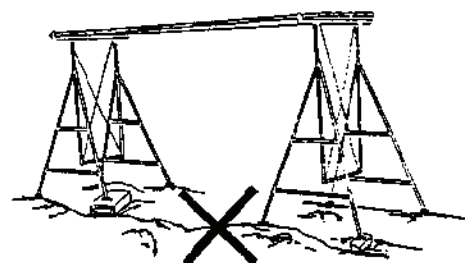
This person could fall over the perimeter guard rail



This person could fall more than 2 m



Trestles have been set up on a sloping surface



Trestles have been set up on soft ground and unsuitable soleplates (such as bricks and concrete blocks)

FIGURE 10.1(B) WRONG USE OF TRESTLE SCAFFOLDS

The documented information provided by the supplier of the ladder-bracket scaffold should clearly state limitations and acceptable uses, and give clear instruction on the construction of the scaffold. The following rules apply to the use of ladder-bracket scaffolds:

- Only use industrial grade single or extension ladders.
- Pitch the ladders at a horizontal to vertical slope ratio of 1:4.
- Make sure the ladders are firmly footed on a hard level surface.
- Secure the ladders against movement in any direction.
- Keep the horizontal distance between brackets to 2.4 m or less.
- Make sure the planks are genuine scaffold planks in good condition.
- Never place a working platform at a height where a person could fall more than 2 m.
- Provide barricades or other suitable controls to prevent traffic damage.
- No more than one person is to be supported in any bay of the scaffold.
- Do not stack materials on the working platform.

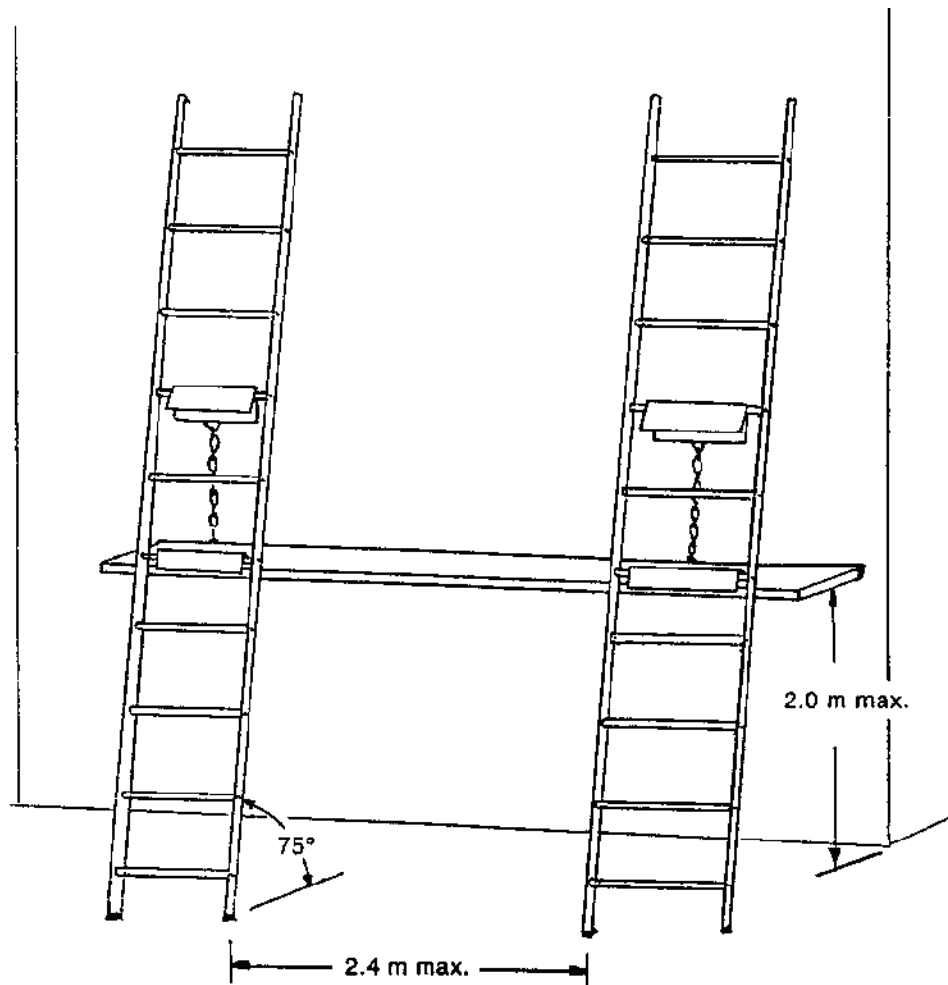


FIGURE 10.2.5 TYPICAL LADDER-BRACKET SCAFFOLD

10.3 PREFABRICATED SCAFFOLDS

10.3.1 Categories Prefabricated scaffolding systems are often divided into the following categories:

- *Modular systems*, consisting of individual standards, ledgers, transoms, braces and other members (see Clause 10.3.6).
- *Frame systems*, consisting of fabricated walk-through frame units (usually incorporating a pair of standards and a transom) that are connected longitudinally with other members, such as scissor braces (see Clause 10.3.7).
- *Tower-frame systems*, incorporating fabricated frame units for use in single-bay tower scaffolds (see Clause 10.3.8).

10.3.2 Registration In Australia, any new or modified prefabricated scaffolding system requires design registration with a State, Territory or Commonwealth regulatory authority. In New Zealand, any new or modified prefabricated scaffolding system requires design registration with the New Zealand Department of Labour.

10.3.3 Supplier's information Suppliers must provide the following information:

- Instructions for erection, dismantling, transportation, storage and maintenance.
- Guidance on the type of scaffolding coupler to use when connecting ties and other accessories.
- The intended duty of the scaffold, including its maximum platform capacity.
- Maximum number of working platforms.
- Maximum height.
- A guide on safe working practices, including the stability of the erected scaffold.

10.3.4 Design Where it is not practicable to follow the supplier's information for the design of a scaffold, the design should be verified for compliance with AS/NZS 1576.1 by a competent person, such as an engineer experienced in structural design.

10.3.5 Construction Do not mix parts from different prefabricated scaffolding systems, unless one of the suppliers both approves of such a mix and guarantees that:

- The components are a compatible size and strength.
- The components have compatible deflection characteristics.
- The fixing devices are compatible.
- The mixing does not lessen the strength, stability, rigidity or suitability of the scaffold.

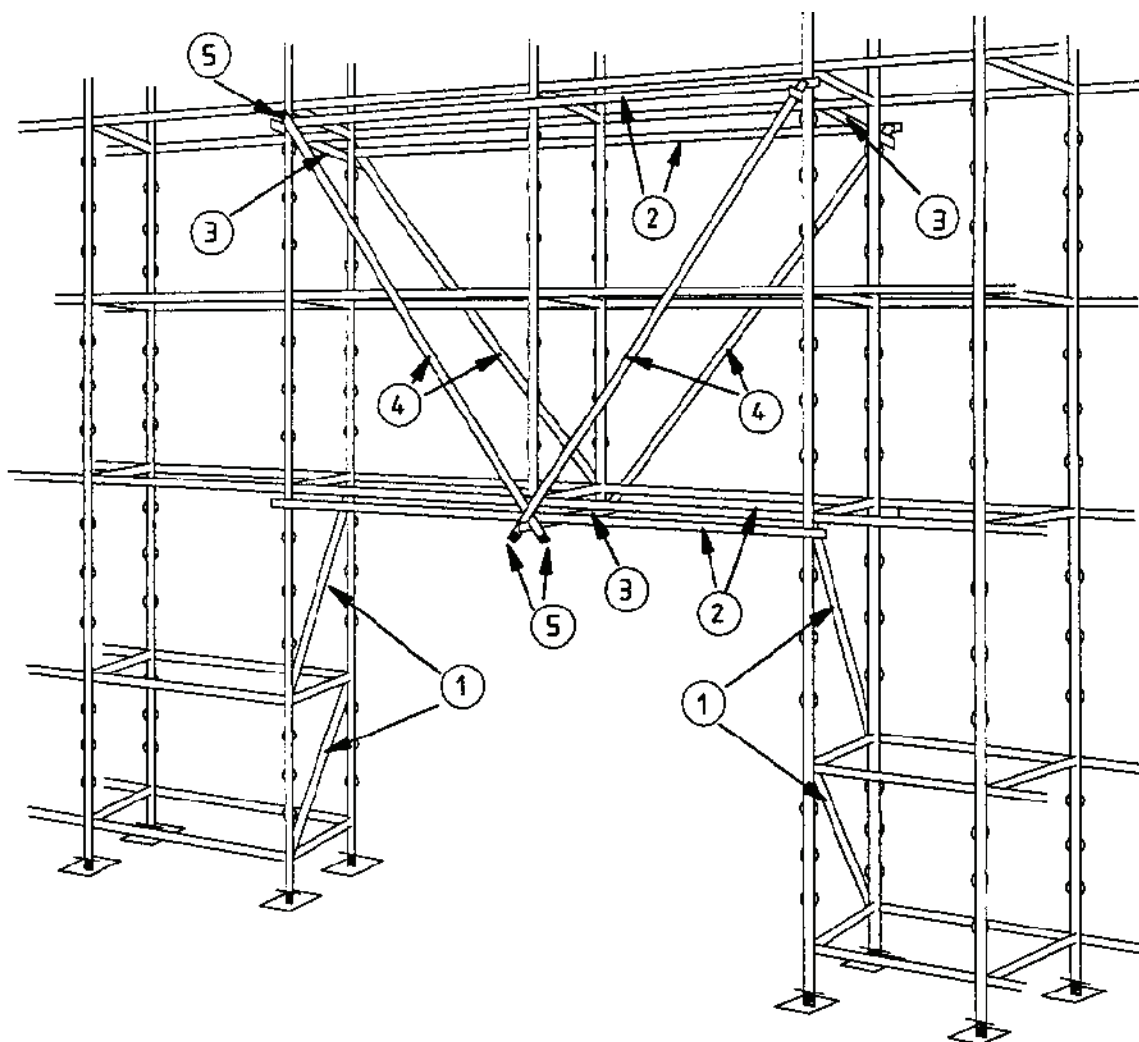
For example:

- Steel prefabricated scaffolding systems should not be used in conjunction with aluminium prefabricated scaffolding systems.
- Aluminium scaffold tubes should not be used as principal structural members in a steel prefabricated scaffold.
- Steel scaffold tubes should not be used as principal structural members in an aluminium prefabricated scaffold. This does not apply to the use of tie tubes or members such as guardrails.

Where a prefabricated scaffolding system does not include purpose-designed components for the construction of access openings, they should comply with AS 1576.3 Supplement 1. Figure 10.3.5 illustrates a typical example.

Horizontal scaffold tubes in single lengths should be fixed with right-angle couplers to the puncheon and the standards at either side of the opening, at the top and at the bottom of the truss.

Where platform brackets are used on modular scaffolds or frame scaffolds, their construction and use should comply with AS 1576.3 Supplement 1. Platform brackets should not be used on freestanding scaffolds, unless advised by the supplier's information or verified by a competent person, such as an engineer experienced in structural design.



LEGEND:

- 1 Transverse bracing on either side of the opening
- 2 Additional tube ledgers
- 3 Additional tube transoms
- 4 Spur or 'thrust' tubes
- 5 Check couplers to prevent slip

NOTES:

- 1 Guardrails, longitudinal bracing and additional bays omitted for clarity.
- 2 The couplers used to construct the truss should be right-angle couplers.

FIGURE 10.3.5 A TYPICAL ACCESS OPENING IN A MODULAR SCAFFOLD

10.3.6 Modular scaffolds Modular scaffolds (see Figure 10.3.6(A)) should be constructed with a lift of ledgers and transoms connected at the lowest fabricated connection points of the standards.

Any joint occurring in a standard below the level of the top-most platform of the scaffold should be staggered so as to be either at least 500 mm above or at least 500 mm below the corresponding joint in the transversely adjacent standards. Alternatively, the joints in the longitudinally adjacent standards can be staggered.

Joints should occur less than 1.5 m above a ledger, to avoid the dangerous practice of high top-ups.

Ledgers should form a continuous line along the full length of the scaffold.

Ledgers should be connected at each lift to each standard at the same level as the transoms (see Figure 10.3.6(C) for the correct method). The hit-and-miss method of replacing the outside ledger with a guardrail is dangerous and must never be used.

For adequate rigidity of modular scaffolding, use longitudinal and transverse braces. Wherever possible, use purpose designed braces. Where appropriate, scaffold tubes and couplers may be used. Spacing of braces should be not less than the supplier's recommendation. Where no recommendation exists, fix a transverse brace to each lift at each end and make sure that there are no more than three bays between longitudinally braced panels in any lift of the scaffold.

Do not fix braces or tie members to:

- Ledgers, where a gravity wedge device fixes the ledgers to the standards, unless the connections are positively fixed by a check-coupler or other suitable means.
- Ledgers incorporated into modular systems that are not circular in cross-section.

Bracing and tie members may be fixed by right-angle couplers to additional horizontal scaffold tubes that are fixed to at least two standards with right-angle couplers.

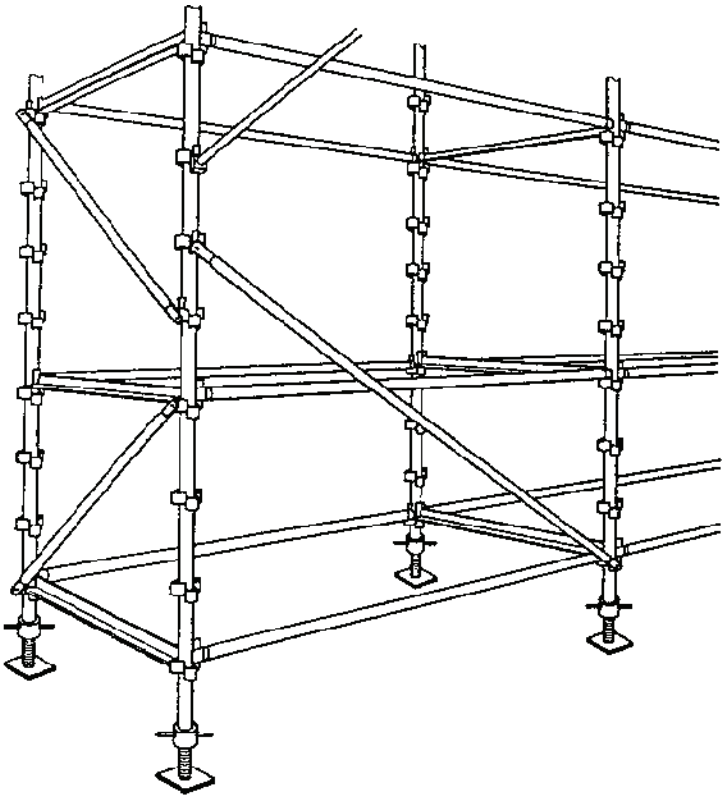
Figure 10.3.6(B) illustrates some common fixing devices for modular scaffolds.

The supplier must provide comprehensive documented erection and dismantling procedures that minimize the risks to scaffolders and persons in the vicinity of the scaffold and ensure that the scaffold can be erected within the design limits of the system.

As a general rule, scaffolds should be erected and dismantled according to the supplier's documented procedures. A scaffold that is not designed in accordance with the supplier's information may need a modified erection and dismantling procedure specified by a competent person. The following steps are recommended for scaffolders who are erecting typical high modular scaffolding systems:

- Step 1** When a lift has been completed, relocate planks to the next lift to provide a platform for further erection. To lift the last plank, brace your back against a standard and keep the feet spread, with one foot securely on the transom and the other foot securely on the ledger.
- Step 2** Pass components up from below or lift up with a handline, gin wheel, winch, hoist or other suitable means.
- Step 3** Top up any standards that break below the height of the guardrail.
- Step 4** Position guardrails at every open side and end where there is a drop of more than 2 m from the platform. Secure the guardrail to the standard at the far end of the bay; then, secure the guardrail to the standard at the near end of the bay. These guardrails are for the protection of scaffolders and should remain in place until the lift is dismantled.
- Step 5** After the guardrails have been placed, top up any standards that break above the height of the guardrail.
- Step 6** Fix the next lift of ledgers and transoms to the standards. Braces and tie assemblies should be fixed where appropriate.

These steps should then be repeated for all further lifts, until the scaffold has reached its required height and the working platforms are installed for the user.



NOTE: Planks and guardrails are omitted for clarity.

FIGURE 10.3.6(A) TYPICAL MODULAR SCAFFOLDING

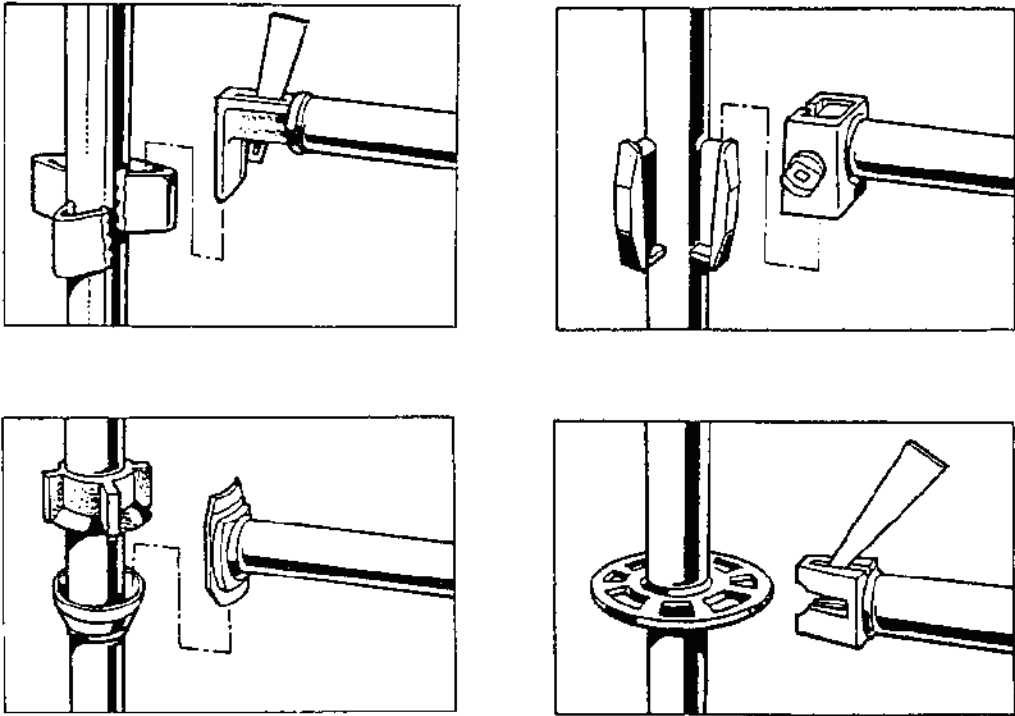
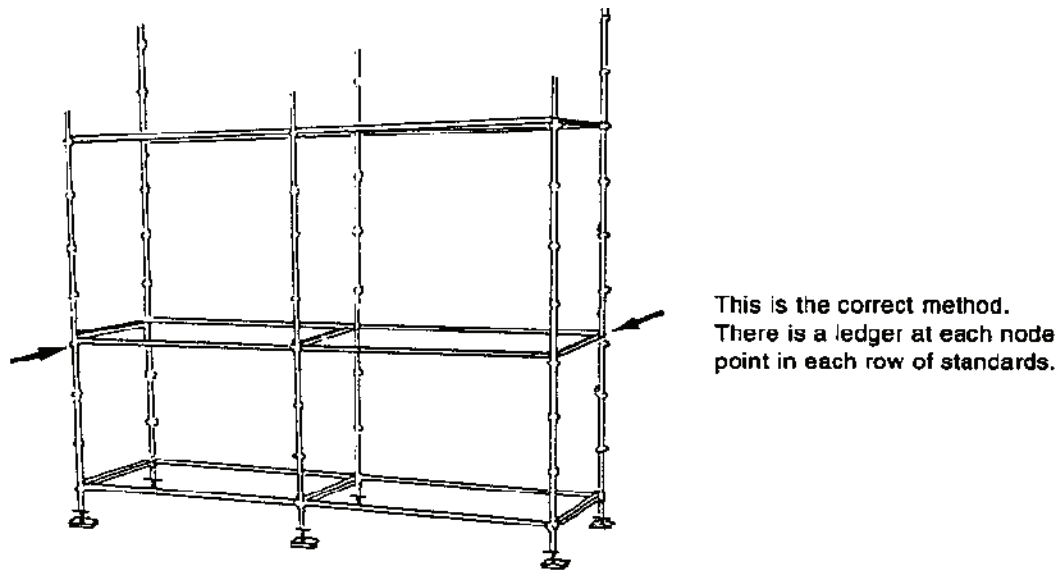
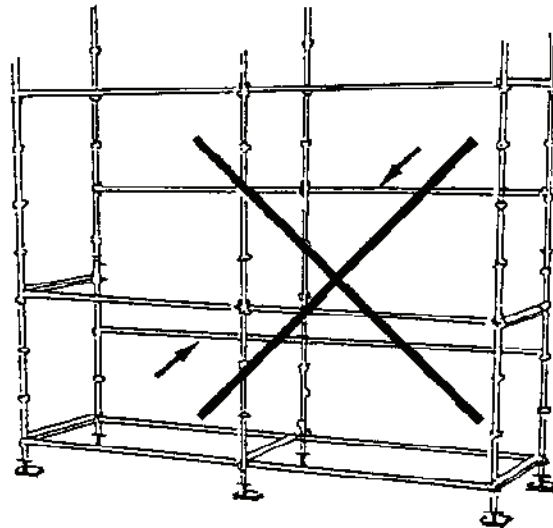


FIGURE 10.3.6(B) SOME COMMON LEDGER FIXING DEVICES FOR MODULAR SCAFFOLDS



This is the incorrect method.
The outside ledger is missing
and has been replaced with
a guardrail.



NOTE: Guardrails and bracing are omitted for clarity.

FIGURE 10.3.6(C) POSITIONING LEDGERS IN A MODULAR SCAFFOLD

10.3.7 Frame scaffolds Frame scaffolding (see Figure 10.3.7) that is not designed to incorporate a base lift should not be set up on adjustable legs where the adjustment exceeds 150 mm, unless either of the following apply:

- The vertical distance from the supporting surface to the first integral transom does not exceed 2.1 m.
- Longitudinal and transverse foot ties are provided, using scaffold tubes and right angle couplers within 700 mm of the supporting surface.

Frame scaffolding without a ledger should be provided with tube ledgers attached to the standards or transoms with right-angle couplers at the level of each platform.

Frame scaffolding without integral captive planks or decking units should be provided with putlogs for the support of platforms.

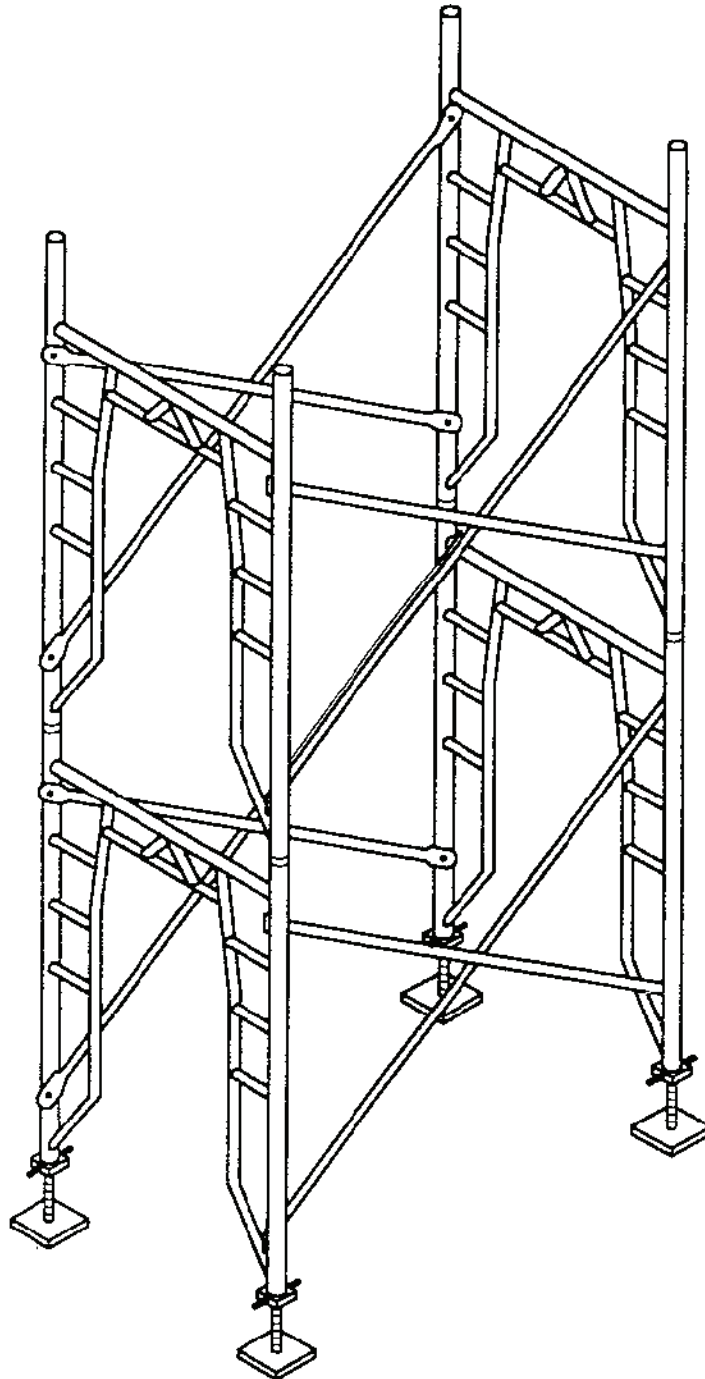


FIGURE 10.3.7 TYPICAL FRAME SCAFFOLDING

10.3.8 Tower-frame scaffolds Tower-frame scaffolds (see Figure 10.3.8) should not be linked together to form a run of scaffold, unless both the following apply:

- It is possible to construct a full length working platform at any lift in the configuration.
- The supplier's supplied documented information specifies such a configuration.

The freestanding height of a light-duty aluminium tower-frame scaffold with a base width of less than 1.2 m should not exceed twice the base width.

The height of light-duty aluminium tower-frame scaffolds must not exceed 9 m (unless specifically stated in the supplier's documented information).

Each light-duty aluminium tower-frame scaffold should not have more than one working platform (unless specifically stated in the supplier's documented information).

Ladder access to a platform supported by a tower-frame scaffold should be constructed within the framework with a hinged trapdoor in the working platform.

Some types of light-duty aluminium tower-frame scaffolding use prefabricated components with outside diameters that are different from scaffold tube.

When tying this type of scaffold to the supporting structure, make sure that couplers:

- Are compatible with the components.
- Will not distort the component.
- Will embrace the component over their full bearing surface, without distortion of the coupler.
- Can also be fixed to normal scaffold tube.

10.4 MOBILE SCAFFOLDS

Typical mobile scaffolds are illustrated in Figure 10.4.

On its acceptance as a working platform, a mobile scaffold must be used only on a hard level surface from which it must not be moved.

A mobile scaffold must not be located closer than 1 m to any slab edge, penetration or other stepdown, unless a positive means to prevent it crossing that point, such as a fixed fence, rail or suitably high upturn, is in place.

A mobile scaffold must not be located any closer to powerlines than the minimum required distance for a static (i.e. non-mobile) scaffold.

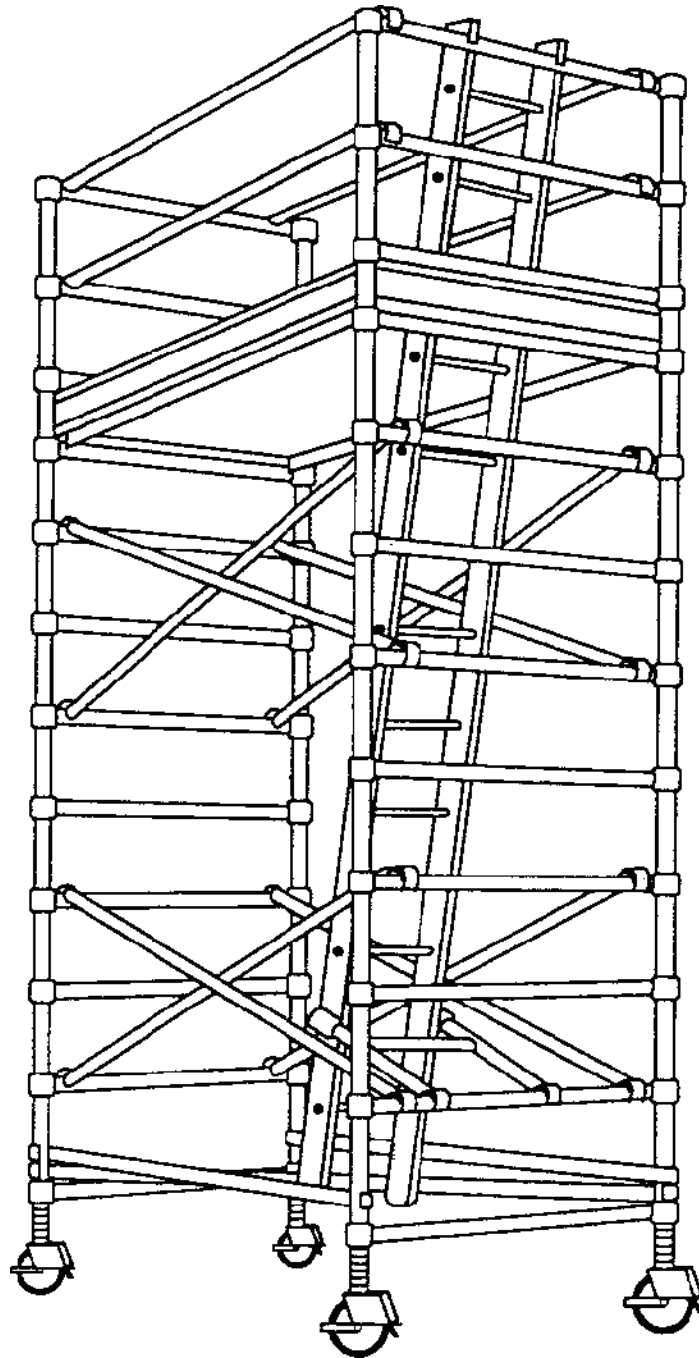
Do not release the wheellocks or relocate the scaffold, unless the scaffold is unoccupied and all items on the scaffold have been secured against falling.

A mobile scaffold should not be relocated by any means other than manpower, unless it has been specifically designed by a competent person.

Where a counterweighted cantilever or spur mobile scaffold is relocated, counterweights should not be removed from it other than by competent persons.

Where a cantilever or spur mobile scaffold is anchored by a reveal propped anchorage, it should only be released and re-set up by competent persons.

Mobile scaffolds should be left with all wheels locked or chocked against any possible movement. Where necessary, portable barriers and lights should be placed around the scaffold.



NOTE: Scaffold may be on baseplates or castors.

FIGURE 10.3.8 TYPICAL TOWER-FRAME SCAFFOLDING

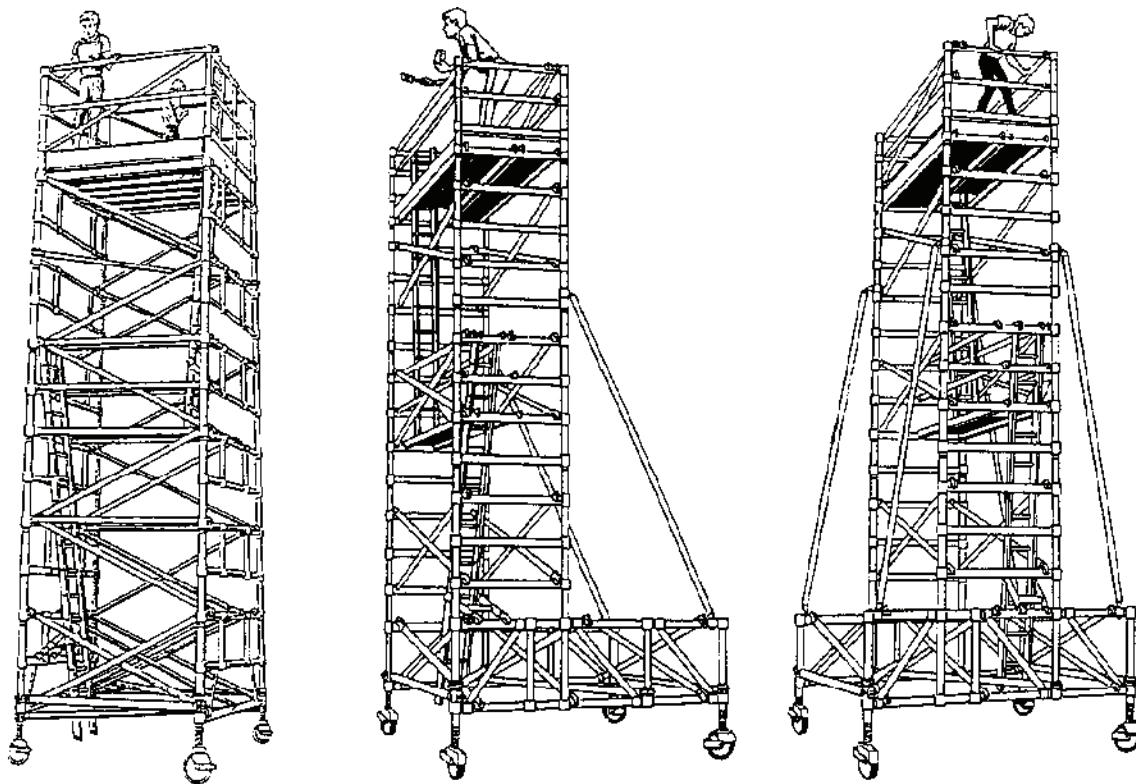


FIGURE 10.4 TYPICAL MOBILE SCAFFOLDS

10.5 TUBE-AND-COUPLER SCAFFOLDS

Tube-and-coupler scaffolds can be adapted to almost any configuration or loading application; however, construction is generally more labour intensive and requires a higher level of skills than does prefabricated scaffolding. Typical tube-and-coupler scaffolds are illustrated in Figure 10.5.

Scaffolds can be constructed according to AS 1576.3 Supplement 1 and do not require design verification where tubes are either:

- steel tubes with a minimum yield stress of 200 MPa, an outside diameter of 48.3 mm and a nominal wall thickness of 4 mm, complying with the dimensional tolerances of AS 1163.
- aluminium tubes with a minimum yield stress of 241 MPa, an outside diameter of 48.4 mm and a nominal wall thickness of 4.5 mm, complying with the dimensional tolerances of AS 1866.

AS 1576.3 Supplement 1 should be regarded as a comprehensive set of design specifications for a wide range of unsheathed light, medium and heavy duty tube-and-coupler scaffolds up to 45 m in height. Configurations outside the scope of AS 1576.3 Supplement 1 should have their designs verified for compliance with AS/NZS 1576.1 by a competent person such as an engineer experienced in structural design.

Tube-and-coupler scaffolding systems using different materials, different outside diameters or thinner wall sections are required to be tested in the same way as new prefabricated scaffolding systems and must be accompanied with the supplier's documented information.

Except where used as standards, tubes of different wall thicknesses must not be interconnected by spigots or internal-type end-to-end couplers, unless additional measures are taken to positively secure the joint, such as by fixing a short tube with swivel couplers over and parallel to the joint (scarfing), or by fixing a bridle with right-angle couplers to the adjacent members.

In New Zealand, in addition to the above, tubes, couplers and accessories that comply with BS 1139.1.1, BS 1139.1.2, BS 1139.2.1, BS 1139.2.2 or an equivalent overseas standard recognized by Occupational Safety and Health service of the Department of Labour may be used to construct tube-and-coupler scaffolds. Hot-dip galvanized steel tubes with a wall thickness of 3.2 mm that comply with BS 6323.1 may be used to construct light duty and medium duty scaffolds that comply with AS 1576.3 Supplement 1, provided the scaffold height does not exceed 33 m.

10.6 SPUR SCAFFOLDS

The design of spur scaffolds and the adequacies of structures supporting a spur scaffold must comply with the relevant requirements of AS/NZS 1576.1. A competent person, such as an engineer experienced in structural design, must verify such a design before the spur scaffold is handed over for use.

Spurs should be pitched at an angle not exceeding 45 degrees from the vertical.

Spurs in compression that exceed 2 m in length should be secured at midspan to prevent deflection in any direction.

Spurs should be fixed to ledgers or transoms with right-angle couplers and provided with check couplers. Swivel couplers should not be used.

A single set of spurs should not support more than five lifts.

10.7 CANTILEVERED SCAFFOLDS

The design of cantilevered scaffolds and the adequacies of their supporting structures must be verified for compliance with the relevant requirements of AS/NZS 1576.1 by a competent person, such as an engineer experienced in structural design.

Needles should be secured by through-bolting, cast-in inserts or propping. Steel beams used as needles should have a width of not less than 75 mm and be fixed with their greater dimension vertical. Where practical, the inboard portion of the needle should be at least three times the length of the outboard portion.

Bolts and threaded rods used to secure the inboard portion of the needle should have a diameter of not less than 15 mm and lock nuts to prevent any loosening from vibration.

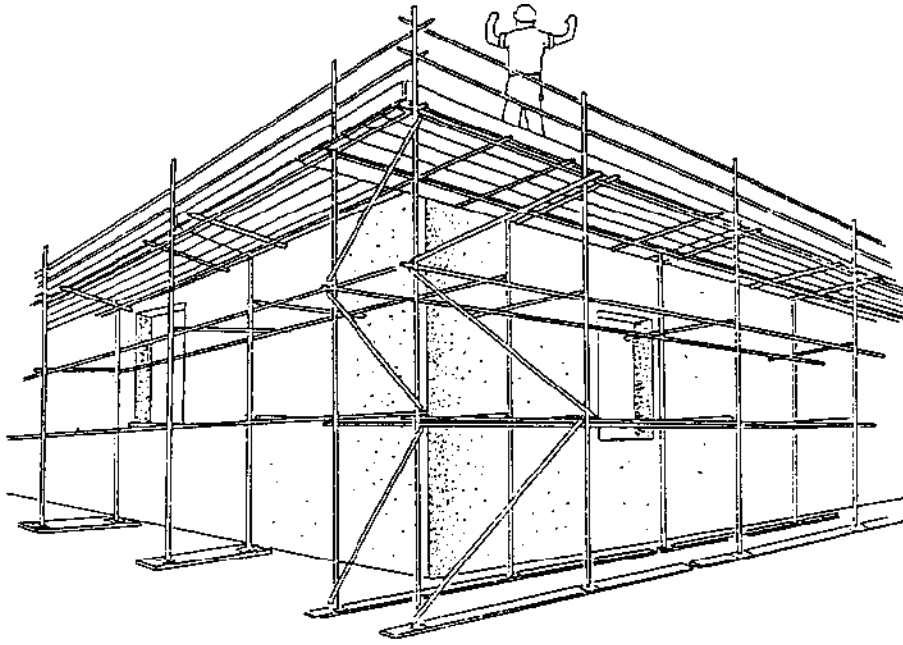
Where reveal props are used to secure the inboard portion of needles, at least two rows of vertical members should be used with the rows tied longitudinally and transversely. They should be braced to form an independent framework, with the props secured over the beams with fork-heads or other suitable means to prevent dislodgment (see Figure 10.7).

Scaffold planks should be laid directly on the outboard portion of the needles, to provide a platform for the scaffolder to commence erecting the scaffold.

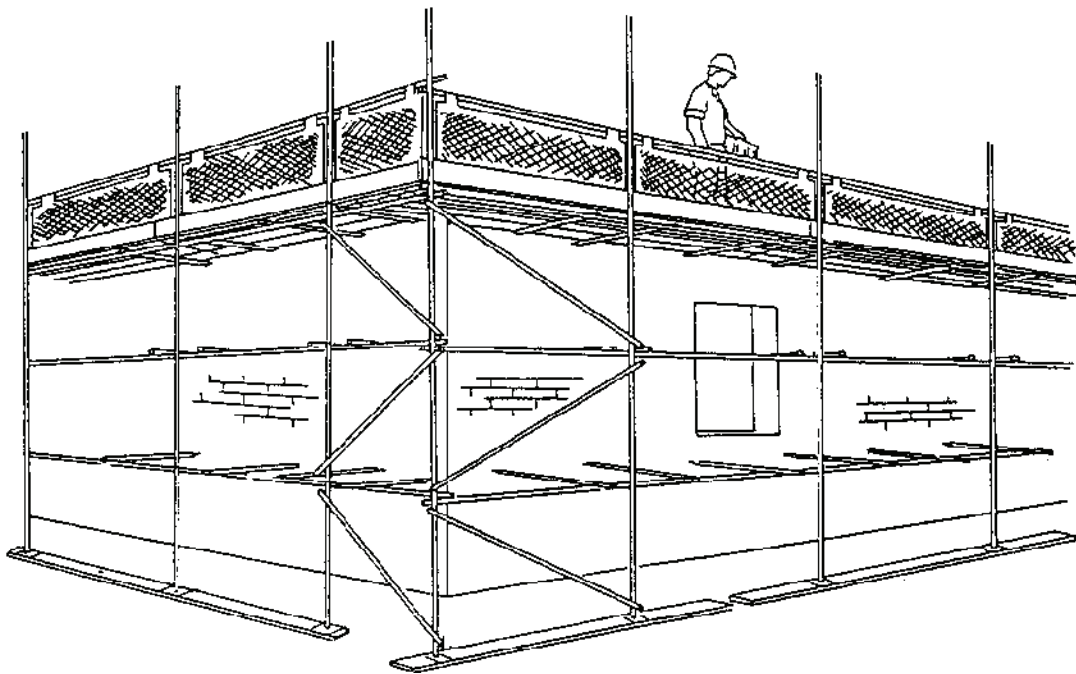
The standards should be secured over the needles with fork-heads or other suitable means to prevent dislodgment.

A lift of ledgers and transoms should be constructed as close to the needles as possible.

A protection deck should be left in place either directly on the needles or on the base lift. When the deck is on the needles, the planks should be held captive by lashing, spiking, cleating or other suitable means.



(a) Independent type

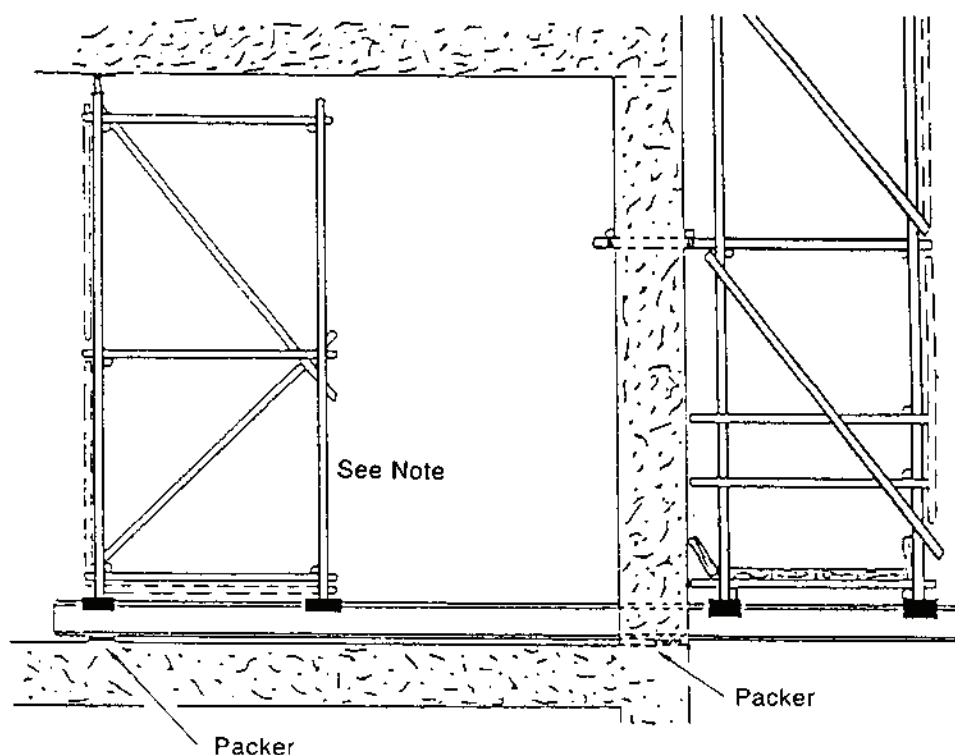


(b) Single-pole type

FIGURE 10.5 TYPICAL TYPES OF TUBE-AND-COUPLER SCAFFOLD

The scaffold should be tied to the building or structure at the first lift above the base lift. Drilled-in anchors should not be used in tension to anchor needles, unless the following conditions apply:

- An engineer has verified the design and the adequacy of the surrounding material.
- The anchors are torque-controlled non-sleeved types or undercut types.
- Any chemical anchors have been proof loaded before use.



NOTE: This inside vertical member provides strength and stability to the reveal props and does not share the vertical load.

FIGURE 10.7 TYPICAL CANTILEVERED SCAFFOLD

10.8 MAST-CLIMBING WORK PLATFORMS

Mast-climbing work platforms are available for use as either freestanding units or in single or multiple tower configurations progressively tied to the supporting structure as they are erected (see Figure 10.8).

Mast climbing systems in Australia must have their design registered with a regulatory authority.

The supplier's information for a particular model of mast-climbing work platforms should:

- Confirm that its design has been registered according to regulatory requirements.
- Provide sufficient information that will enable the scaffolder or rigger to erect, alter and dismantle the unit safely within design limitations.
- Include testing requirements, pre-operational checks and servicing requirements.

Any variation to the supplier's general recommendations should be verified in writing by the manufacturer, importer or supplier, or by a competent person such as an engineer experienced in structural design.

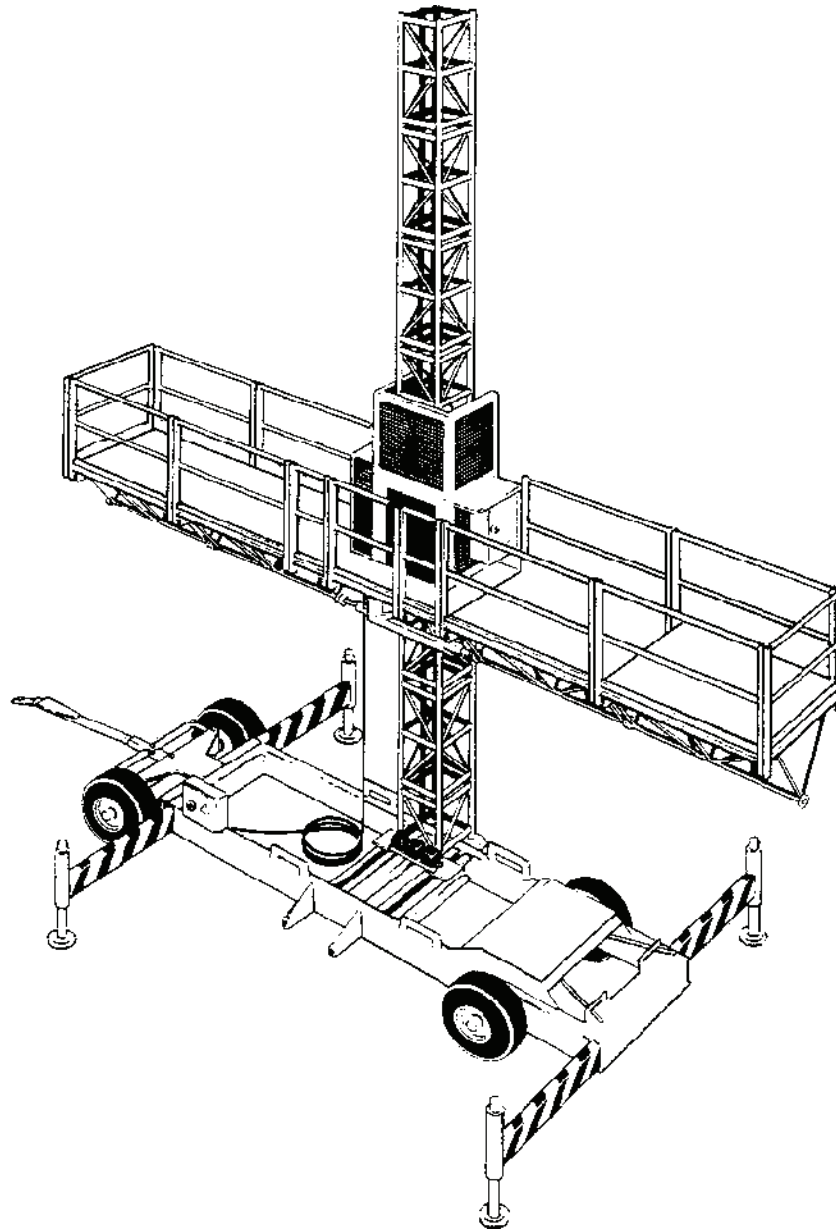


FIGURE 10.8 TYPICAL SINGLE-MAST-CLIMBING WORK PLATFORM

10.9 HUNG SCAFFOLDS

The design of hung scaffolds and the adequacies of their supporting structures must be verified for compliance with the relevant requirements of AS/NZS 1576.1 by a competent person, such as an engineer experienced in structural design.

An independent scaffold may be hung at a fixed height from beams, grid-mat flooring, structural-roofing members or other overhead structure, whether permanent or temporary (see Figures 10.9(A) and 10.9(B)). Open-ended hook-rods must not be used. Couplers with speed threads or multi-start threads should not be used to secure hung standards or to fix ledgers to hung standards.

Beam clamps, trolleys and shackles should have a rated working load limit of not less than 500 kg.

Standards may include:

- Scaffold tubes in single lengths.
- 6 × 24 flexible steel-wire rope slings with a nominal diameter of not less than 11 mm.
- Grade T chains with a nominal diameter of not less than 8 mm.

The method of securing the standards to the supporting structure should prevent the scaffold from becoming dislodged or displaced by the scaffold swaying. Scaffold tubes used as hung standards should have check couplers immediately above the suspension points and immediately below the lowest ledgers. Steel wire rope (without eyes) used as a hung standard should be fixed to a shackle by a wedge-type socket, together with a wire-rope grip fixed to the rope tail. Alternatively, a thimble-eye splice can be made using a double-base clamp or three fist-grip rope clamps.

NOTE: Wire-rope grips should not be used to secure wire ropes that support persons or substantial loads.

Steel wire ropes or chains secured around the sharp edges of beams should be protected from damage by beam chaffers.

Ledgers supported from the eyes of slings or from shackles should be provided with a coupler on each side of each support point, to prevent the ledger from sliding. Pins of shackles should be moused, to prevent inadvertent unwinding.

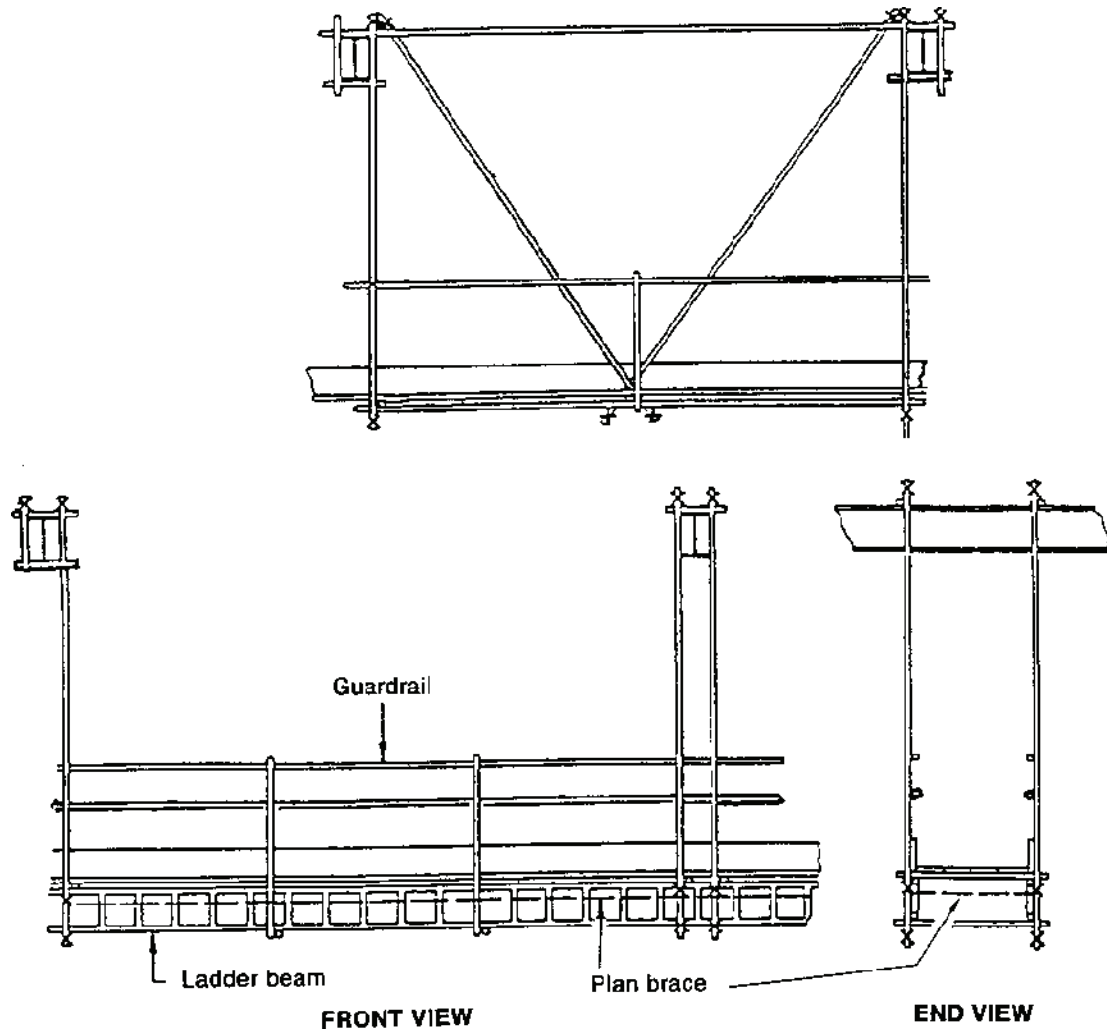
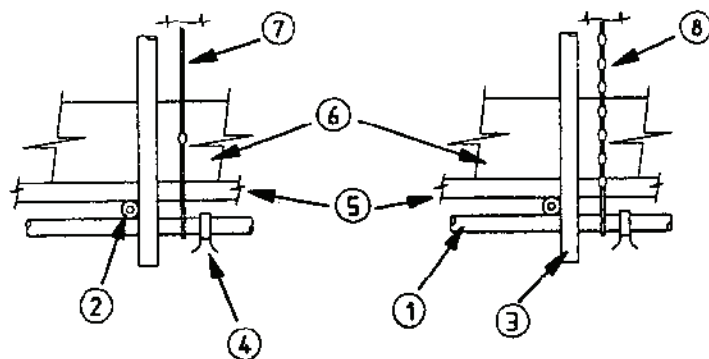
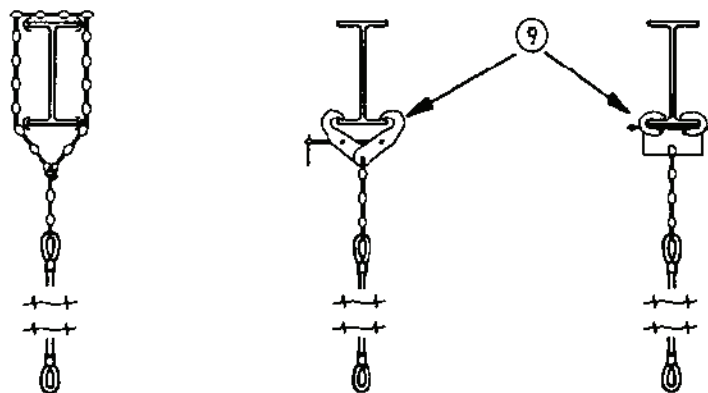
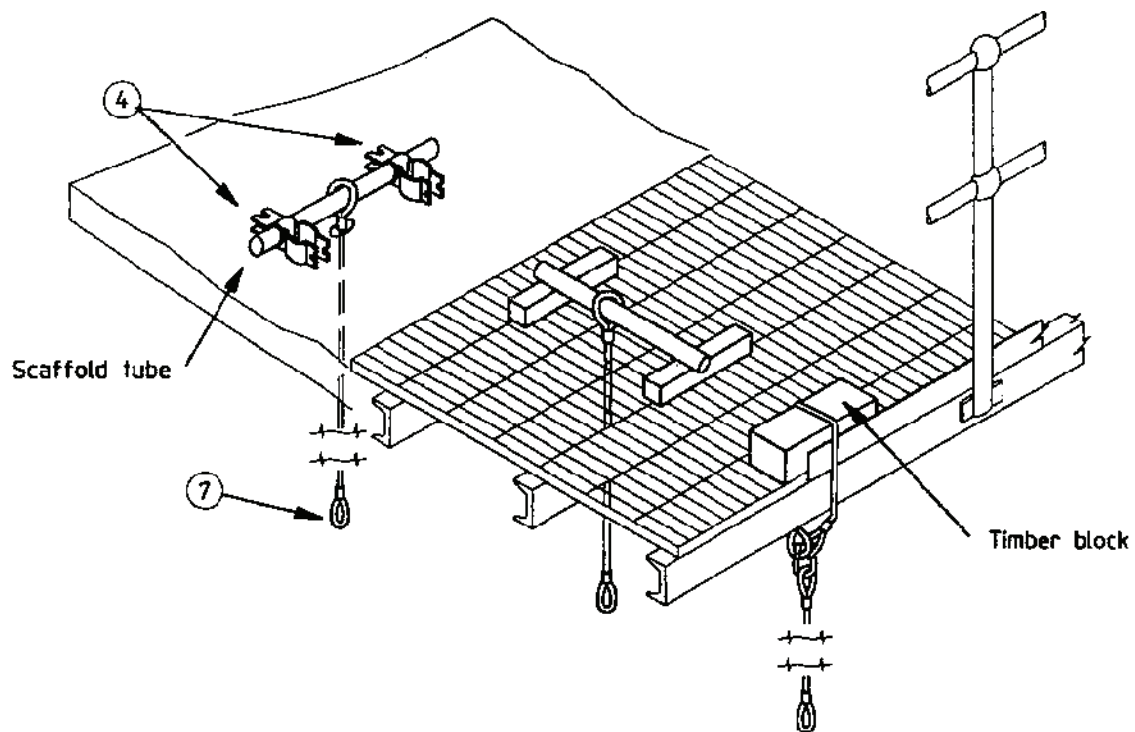


FIGURE 10.9(A) EXAMPLES OF HUNG SCAFFOLDS



LEGEND:

- 1 Ledger
- 2 Putlog
- 3 Guardrail stanchion
- 4 Keeper clip
- 5 Platform plank
- 6 Toeboard
- 7 FSWR sling with soft eye
- 8 Short-link chain with lifting ring
- 9 Beam clamp

FIGURE 10.9(B) TYPICAL DETAILS OF HUNG SCAFFOLDS USING CHAIN AND ROPE

SECTION 11 ADDITIONAL REQUIREMENTS FOR SUSPENDED SCAFFOLDS

11.1 EQUIPMENT

Suspended scaffolds must comply with AS 1576.4 and be supplied with clear instructions about the safe configurations for erection and the intended uses and load combinations.

Associated electrical equipment must comply with relevant legislative requirements.

Each scaffolding hoist must be tested by a competent person.

Each scaffolding hoist and protective device must be fitted with a data plate that is permanently and legibly marked with details of correct use. Data plates for scaffolding hoists and protective devices should include:

- Type.
- Model.
- Serial number.
- Name or identification mark of the manufacturer.
- Details of steel wire rope to be used, including nominal diameter, grade (quality) and construction.
- Rated load of hoist.
- Reeving requirements, where applicable.
- Power supply requirements, where applicable.

Typical suspended scaffolds are illustrated in Figures 11.1(A) and 11.1(B).

Each type of cradle and boatswain's chair must be tested by a competent person. A copy of the test report must be retained for the design life of the equipment.

11.2 ERECTION, ALTERING AND DISMANTLING

The erection, alteration or dismantling of a suspended scaffold should take place only under the control of a person holding the appropriate certificate of competency, who should examine all components for defects.

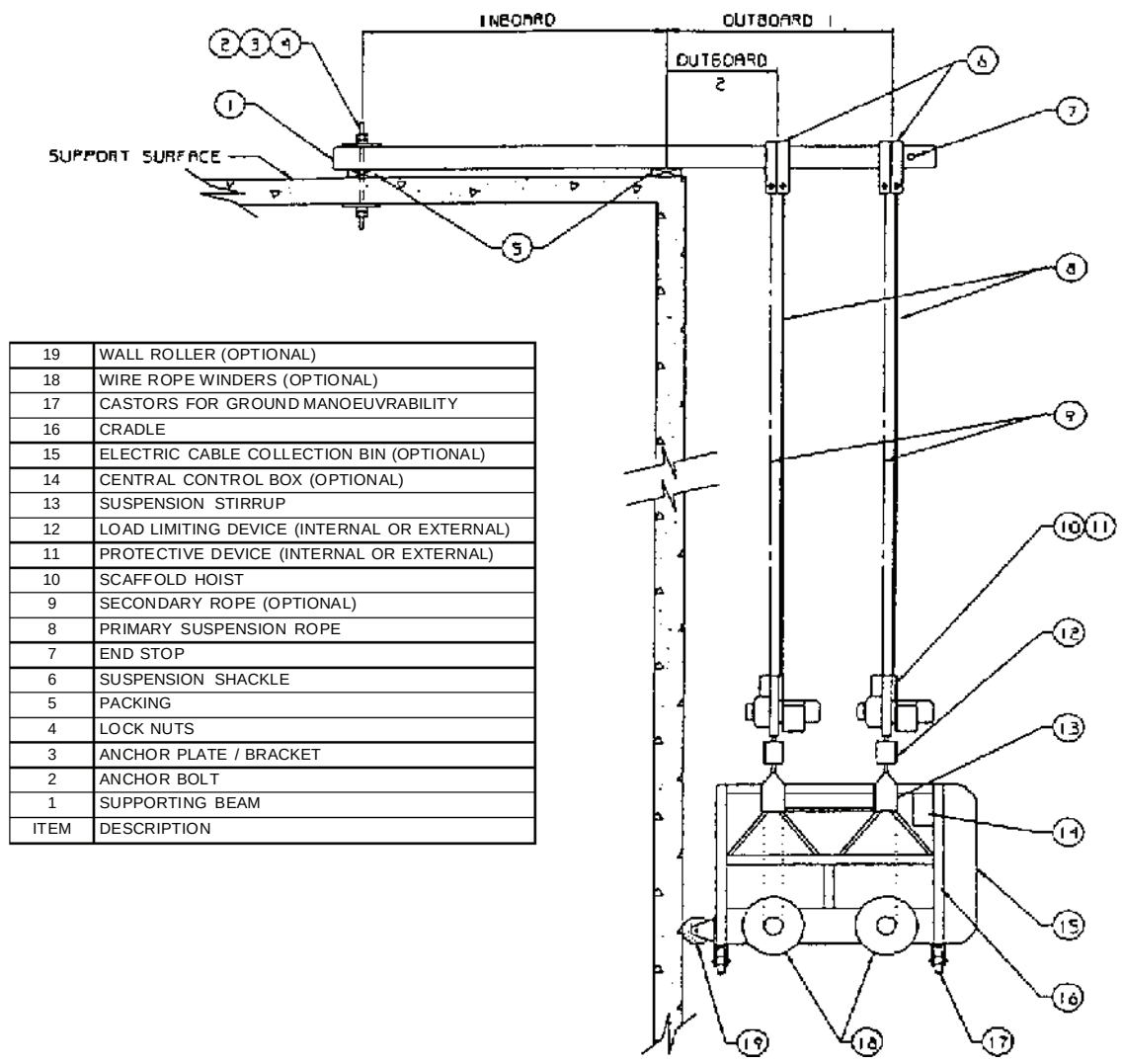
Suspended scaffolds and suspension rigs should only be erected or altered in accordance with their design specifications. At the completion of the installation or alteration, the competent person in control of the work should issue a written statement of completion to the person in charge of the workplace. The statement should be retained until the suspended scaffold is further altered or dismantled.

Thoughtless or unauthorized alteration to the cradle structure or supporting equipment can endanger the lives of those working from the cradle. Persons who are to work in the vicinity of a suspension rig or other supporting structure must not inadvertently alter or damage the supporting equipment.

Purpose-made weatherproof covers should be fitted to all scaffold hoists, to minimize contamination of the hoist mechanism by construction materials.

Those working on suspended scaffolding must be protected from coming into contact with unprotected electric wires (see Clause 5.4) or dangerous plant or substances.

Where there is any likelihood of debris falling from above the cradle, install overhead protection. Meshed or solid overhead debris protection fixed to the cradle should be used only where debris would not cause instability to the suspended scaffold.



19	WALL ROLLER (OPTIONAL)
18	WIRE ROPE WINDERS (OPTIONAL)
17	CASTORS FOR GROUND MANOEUVRABILITY
16	CRADLE
15	ELECTRIC CABLE COLLECTION BIN (OPTIONAL)
14	CENTRAL CONTROL BOX (OPTIONAL)
13	SUSPENSION STIRRUP
12	LOAD LIMITING DEVICE (INTERNAL OR EXTERNAL)
11	PROTECTIVE DEVICE (INTERNAL OR EXTERNAL)
10	SCAFFOLD HOIST
9	SECONDARY ROPE (OPTIONAL)
8	PRIMARY SUSPENSION ROPE
7	END STOP
6	SUSPENSION SHACKLE
5	PACKING
4	LOCK NUTS
3	ANCHOR PLATE / BRACKET
2	ANCHOR BOLT
1	SUPPORTING BEAM
ITEM	DESCRIPTION

FIGURE 11.1(A) A TYPICAL DOUBLE-ROPE SUSPENDED SCAFFOLD

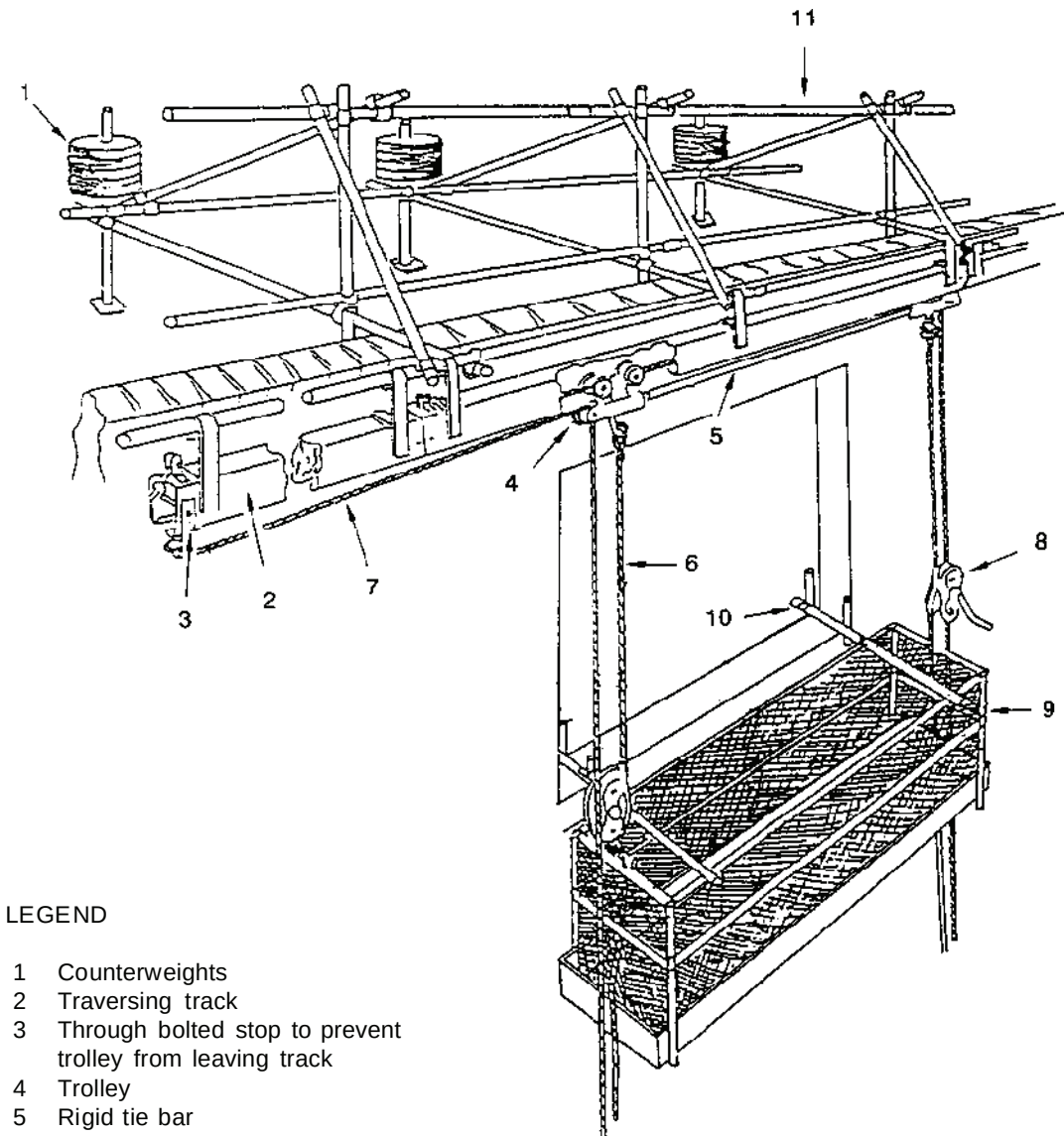
11.3 ACCESS

Install safe access and egress points for persons using the suspended scaffold. The approaches to the suspended scaffold should be illuminated by either natural or artificial light. Where access and egress is not from the ground or from a protected landing, safety harnesses and lanyards should be provided for all those entering and leaving the cradle. Where harnesses are used during access and egress, a suitable tying method should be used to secure the cradle against movement. Harnesses should be properly anchored.

11.4 SUPPORTING STRUCTURES

An assessment of the structure supporting a suspended scaffold and the intended loads should be made by a competent person, prior to the erection of a suspended scaffold.

A copy of the statement of assessment or design should be made available to the person erecting, altering or dismantling the suspended scaffold.



LEGEND

- 1 Counterweights
- 2 Traversing track
- 3 Through bolted stop to prevent trolley from leaving track
- 4 Trolley
- 5 Rigid tie bar
- 6 Suspension rope
- 7 Traversing rope
- 8 Scaffolding hoist
- 9 Cradle
- 10 Tie
- 11 Tubular suspension rig

FIGURE 11.1(B) A TYPICAL TRAVERSING SWING-STAGE SUSPENDED SCAFFOLD

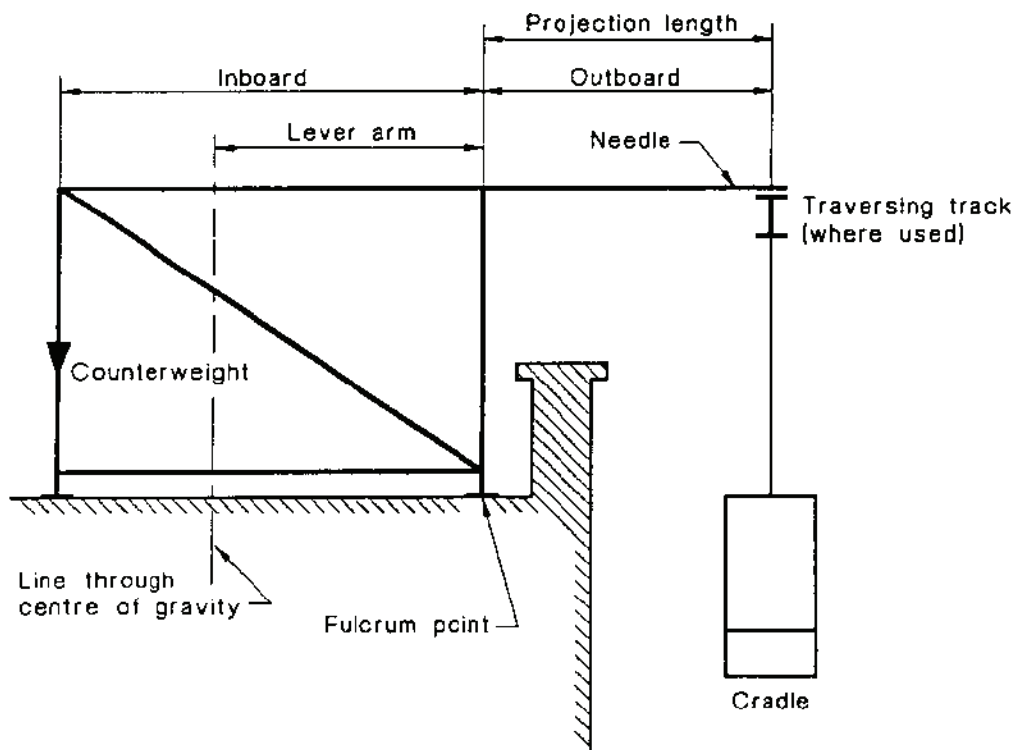
11.5 SUSPENSION RIGS

11.5.1 General Typical suspension rigs for suspended scaffolds are illustrated in Figure 11.5.1. The design of suspension rigs should take into account any likely lateral forces, including wind forces and surging.

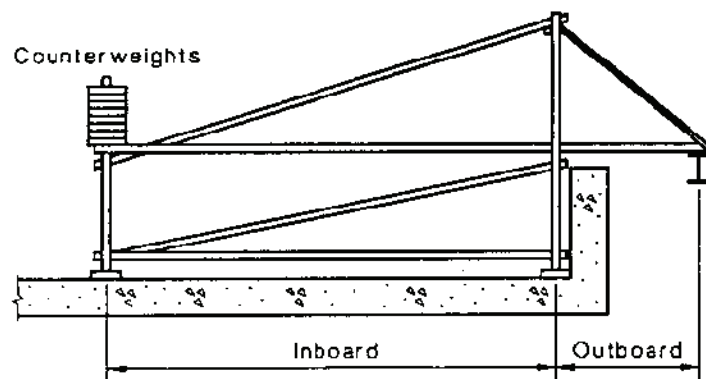
The outboard end of a needle should never be lower than the inboard end.

A beam spanning between only two supports should be horizontal.

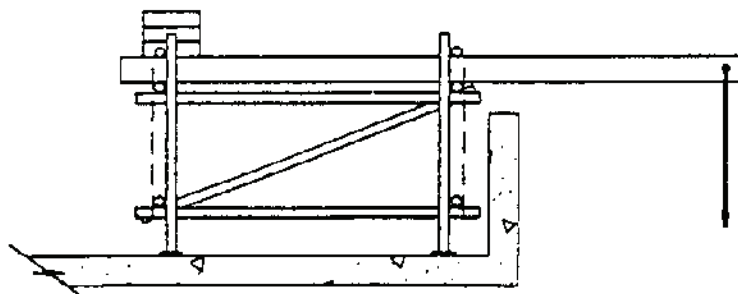
A needle or supported beam should always be mounted with the greater dimension of its cross section in the vertical position.



(a) Traversing track supported by needle

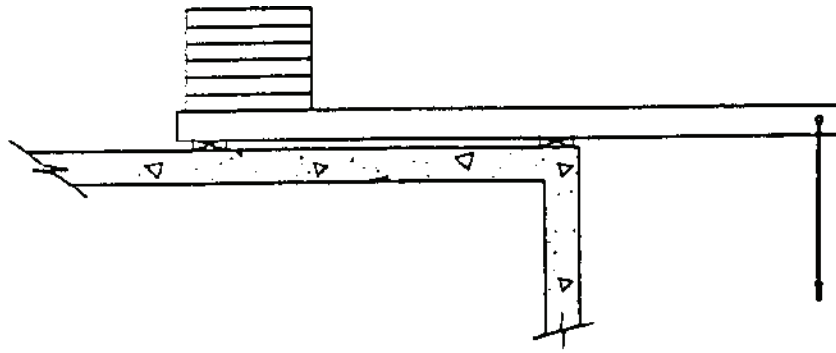


(b) Traversing track supported by truss

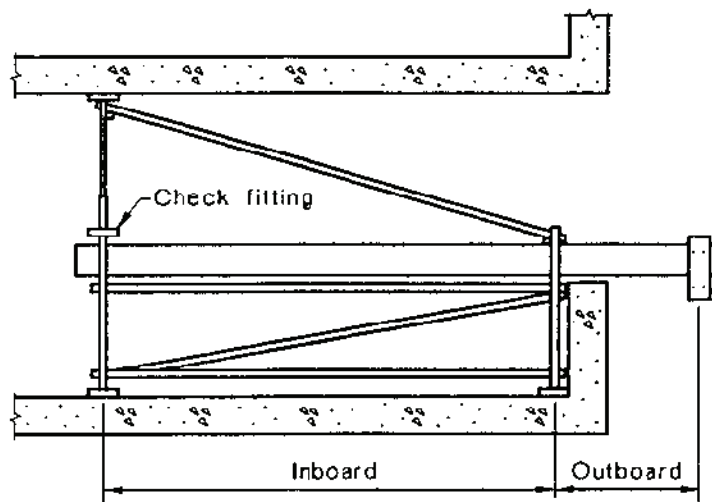


(c) Needle supported by tube and coupler rostrum

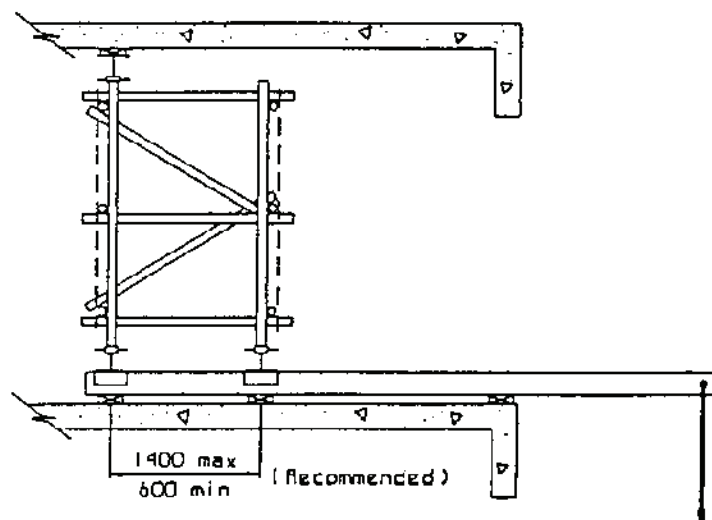
FIGURE 11.5.1(in part) TYPICAL SUSPENSION RIGS



(d) Needle anchored with counterweights



(e) Propping needle between floors



(f) Propping needle between floor and soffit

DIMENSIONS IN MILLIMETRES

FIGURE 11.5.1(in part) TYPICAL SUSPENSION RIGS

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11.5.2 Stability for cantilevered suspension rigs The ratio of stability of a suspended scaffold incorporating a cantilevered suspension rig must be not less than three, where the ratio of stability is the sum of the moments acting on the inboard portion of the suspension rig divided by the sum of the moments acting on the outboard portion of the rig.

The formula for calculating the number of counterweights needed on each needle of a cantilevered suspension rig is—

$$\text{Number of counterweights} = 3 \times (\text{rope tension in kg}) \times (\text{outboard in mm}) \div (\text{inboard in mm}) \div (\text{mass of each counterweight in kg}),$$

where rope tension is calculated in accordance with Clause 11.8.

For example:

Counterweights	= 25 kg each
Maximum rope tension	= 700 kg
Outboard	= 900 mm
Inboard	= 3600 mm

Therefore: $3 \times 700 \times 900 \div 3600 \div 25 = 21$ counterweights per needle

Include the following factors in calculations of the inboard moments:

- The self-weight of the inboard portion of the suspension rig, including any counterweights.
- The design load of anchorages and props.
- The strength of the supporting structure.
- The distance between the fulcrum and the inboard end of the suspension rig.

Consider the following in calculations of the outboard moments:

- The self-weight of the outboard portion of the suspension rig, including trolley tracks and trolleys.
- The mass of secondary ropes, traversing ropes, electrical cables and compressed air cables.
- The distance between the fulcrum and the suspension rope attachment points.
- The maximum rope tension.
- Where suspended scaffolds incorporate trolley tracks, the most adverse horizontal position of the cradle should be considered when calculating the ratio of stability.

11.5.3 Needles and supporting beams Needles and supporting beams should consist of either scaffold tubes and couplers that comply with AS 1576.2 or AS/NZS 1576.3, or steel or aluminium beams designed for the intended loads.

A beam that spans between only two supports often is used to support a suspended scaffold in a shaft, boiler or chimney, or through grid-mat flooring and like situations. The supporting beam should be fixed or located to prevent the possibility of dislodgment or slippage.

11.5.4 Anchorage Anchorage design should take into account the nature of the material to which the anchorage is fastened.

Anchorage bolts should be provided with lock-nuts or other suitable means that will prevent loosening.

Friction anchors and chemically-inserted anchors must not be used in tension in anchorage systems.

11.5.5 Reveal propped needles Needles should be positively fixed under or to the reveal props. In the case of needles that are rolled steel joists or universal beams, close fitting fork-heads may be used.

11.5.6 Counterweighted needles Counterweighted needle suspension rigs should not be used to stabilize a needle attached to two or more suspension ropes.

Counterweights should be placed directly on the needles or on the innermost supporting components to the needles in the designed location. Counterweights should be secured to the suspension rig in such a manner that they cannot be displaced or removed without the use of a tool. Typical counterweights are illustrated in Figure 11.5.6.

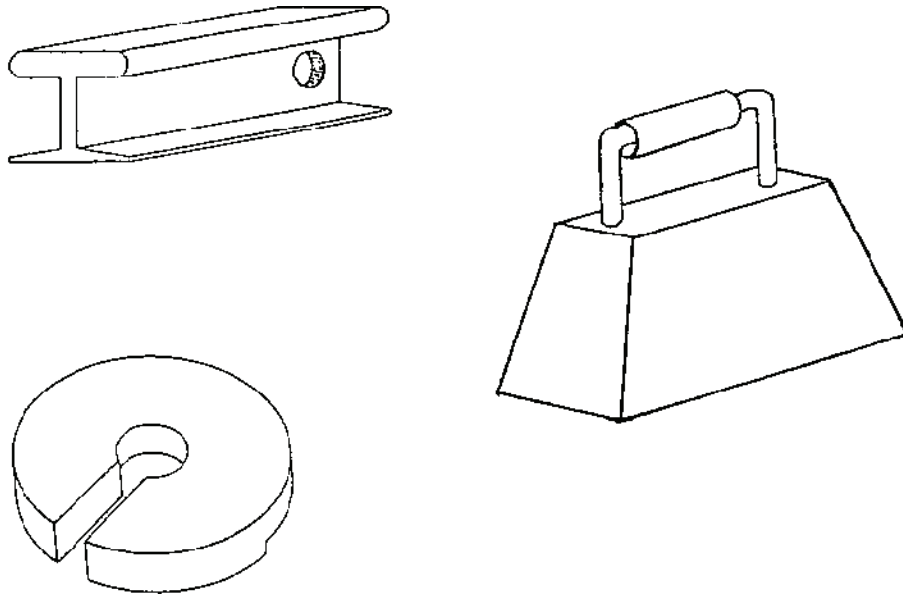


FIGURE 11.5.6 TYPICAL COUNTERWEIGHTS FOR COUNTERWEIGHTED NEEDLES

11.5.7 Built-up framework The built-up framework of a suspension rig should be purpose built to engineering principles or constructed from scaffold tubes and couplers that are tied together with braced ledgers and transoms, to form a structure that is rigid and stable under working conditions.

Each suspension rig should incorporate at least two rows of uprights, fixed with ledgers and transoms and provided with longitudinal, transverse and plan bracing systems. Couplers must comply with AS 1576.2 and tubes and modular components must comply with AS/NZS 1576.3.

11.5.8 Suspension ropes Do not use wire-rope grips on suspension ropes or secondary ropes, because of potential damage to the rope. Suspension ropes and secondary ropes should:

- Have a diameter, grade and construction that conforms to the data plate of the scaffolding hoist and protective device with which they are to be used.
- Be marked with a recorded identification number.
- With climber-type scaffolding hoists, be bullet-headed to help reeving.
- Have an eye that is swaged and thimbled. No part of the rope construction is to be removed to facilitate the swaging operation.

For drum-type scaffolding hoists, there should be at least three full turns of the suspension rope on the drum when the suspended scaffold is at the lowest point, and the flange of the fully-loaded winch drum should extend not less than two rope diameters clear of the built-up rope, to prevent the rope jumping over the drum flange.

Ropes for climber-type scaffolding hoists should comply with the following:

- There should be at least 1 m to spare of suspension rope while the suspended scaffold is at its lowest point.
- Excess rope should be coiled carefully and tied to hang freely below the cradle, or inserted into a rope winder, to avoid kinking.
- Where the lowest point of the suspended scaffold is neither on the ground nor on an impassable base, and excess rope is not coiled below the cradle, the tail end of the rope must be fitted with a suitable rope clamping device to prevent the rope passing through the hoist.
- Do not fix rope ends to the suspended scaffold, as this can cause kinking or birdcaging and rope failure.

11.6 OVERHEAD FIXINGS

Where shackles are used to secure suspension ropes and secondary ropes to suspended scaffolds, scaffold tubes, beam clamps and various other devices, the shackle pin should be moused to the body of the shackle with wire, to prevent accidental unwinding.

Chains or slings supported over a beam with sharp edges should be protected with beam chaffers.

A positively fixed stop should be fixed at the end of each needle, to prevent ropes from sliding off.

A check coupler should be fitted on either side of the suspension point of suspended scaffold tube needles, to prevent movement.

In the case of a steel or aluminium beam, a bolt with a diameter of not less than 12 mm should be fitted through the needle with pipe washers.

The maximum rope tension applied to a shackle, strap, bolt, sling, chain, trolley, beam clamp or other device used to attach a suspension rope or secondary rope to overhead support should not exceed 80 percent of the rated working load.

The maximum rope tension applied to a choked sling should not exceed 40 percent of the rated working load of the sling.

Where a strap is used around a needle or supporting beam, it should be made to an engineered design.

11.7 TRAVERSING EQUIPMENT

Where traversing tracks for suspended scaffolds are suspended beneath needles or beams that span between two supports (to help horizontal movement of a suspended scaffold), the ends of the traversing track should be fitted with through-bolted stops, to prevent any trolley from running off the track.

Trolleys supporting suspended scaffolding must have a rated working load of at least 500 kg.

Trolleys supporting a swing stage should be connected with a spacer tie at the same centre-to-centre distance as the suspended scaffolding hoists, to prevent spreading.

Trolleys supporting a double-rope suspended scaffold should be rigidly connected, both longitudinally and transversely, and braced to prevent twisting.

To prevent cradles from colliding on the traversing track or imposing an excessive load on the rig and supporting structure, fit a buffer zone with intermediate stops to the traversing track.

Ropes used for traversing a suspended scaffold should be of fibre and at least 12 mm diameter.

11.8 ROPE TENSION

For electrically-powered scaffolding hoists, the maximum rope tension should be assessed as the sum of the following loads:

- The mass of the suspension rope.
- Any stabilizing weights attached to the suspension rope.
- The rated working load of the scaffolding hoist, as limited by the load limiting device.

For example, consider the following arrangement:

- Electrically-powered scaffolding hoist.
- Reeved with 30 m of 8 mm suspension rope.
- Rope weighs 0.26 kg/m.
- Rope has a 10 kg stabilizing weight attached to its tail.
- Rated working load is 400 kg.
- Load-limiting device is set to 1.25 times the rated working load.

Thus, the maximum rope tension = $(30 \times 0.26) + 10 + (400 \times 1.25)$
 $= 517.8 \text{ kg}.$

For pneumatically- or manually-powered scaffolding hoists, the maximum rope tension should be the sum of the following six loads:

- The mass of the suspension rope.
- Any stabilizing weights attached to the suspension rope.
- The self-weight of the scaffolding hoist.
- The self-weight of any secondary-rope device.
- That portion of the cradle weight supported by the rope.
- The rated live load of the cradle taking into account any grouping of live loads.

11.9 ELECTRICAL EQUIPMENT AND CONTROLS

Electrical equipment and controls of suspended scaffolds must be safe to install, use and maintain, and should comply with the relevant requirements of AS 1418.2. Any central control box should be fully enclosed, lockable, shatterproof and weatherproof and should include:

- Socket outlets for scaffolding hoists.
- An emergency stop button.
- A power-on light.
- A Type I or Type II residual current device complying with AS 3190. In New Zealand, Type II devices must be of a design that is unaffected by the presence of direct currents. Residual current devices complying with VDE 0664.1, VDE 0664.2 or JIS C 8371 may also be used in New Zealand.

Operating buttons and levers should be of the spring-loaded deadman type.

The electrical control system should fail to a safe condition, in the event of an electrical failure or malfunction. Protection also should be provided against phase reversal in three-phase systems.

Electrical cables should be purpose designed and should be suspended only from built-in thimbles.

Where a central control box is fitted, the main supply cable should be plugged into the box. From this box, separated yokes should lead to the scaffolding hoists. The control box should be removable for safety and security. When in use, it should be attached securely to guardrails of the cradle on the side away from the working face.

Do not use electrical cable with a damaged outer covering that has exposed any insulation of the wires.

Do not use electric cable coated with material from a previous usage that is likely to conceal damage.

11.10 PNEUMATIC EQUIPMENT

Pneumatic equipment, including air-supply hoses, on suspended scaffolds should be of a type and capacity that are recommended by the manufacturer of the scaffolding hoist. Air supply hoses should be suspended from built-in thimbles with an effective bending diameter of not less than eight times the nominal internal diameter of the air supply hose.

Incoming air should pass through a suitable filter and lubricator set, which is fitted as close as possible to the air motor powering the scaffolding hoist, but does not supply air to a second motor.

Oil in the lubricator should be maintained within the specified levels. Oil should be of the type recommended by the lubricator manufacturer.

11.11 PROTECTIVE DEVICES

Single-rope suspended scaffolds must be provided with protective devices that are capable of holding the suspended scaffold, in the event that the suspension hoist rope should fail. Double rope suspended scaffolds do not require protective devices.

Secondary ropes fitted to protective devices should be independently attached to the overhead support.

NOTE: A protective device may be an overspeed device, a slack-rope device or a combination of the two. It may be integrated into the scaffolding hoist or be a separate unit.

11.12 LOAD-LIMITING DEVICES

Each electrically-powered scaffolding hoist must have a device that will prevent the hoist from lifting a load of more than 1.25 times the rated load of the hoist.

The load-limiting device may be set to limit the loads being lifted to below the rated load of the hoist (e.g. the supporting structure may not be capable of carrying such a load).

The load-limiting device should be clearly marked with a load setting of 75 percent of the maximum activating load (i.e. taking into account operational tolerances).

11.13 CRADLES

Where netting is specified for cradles on suspended scaffolds, it should be either galvanized wire mesh netting having wires with a diameter of not less than 1.5 mm and spaced apart at not more than 25 mm, or of other fire-resistant materials of an equivalent strength. It must be securely fixed between the guardrail and the toeboard on all sides and ends of the cradle.

A sign clearly displaying the working load limit should be securely fixed to the inside of the cradle.

For articulated cradles, a sign should display the working load limit in the inside of each bay.

The width of the platform of swing stage cradles should be not less than 450 mm nor more than 900 mm. The cradle should not be cantilevered past a scaffolding hoist by more than the designed distance.

For double suspension rope cradles, the platform width should be not less than 900 mm nor more than 1.7 m.

Each of the platform dimensions of work cages should be not less than 750 mm nor more than 1.5 m. Stabilizing sheaves for suspension ropes and secondary ropes should be mounted at least 2.0 m above the platform.

11.14 MULTI-TIERED CRADLES

Safe access should be provided between platforms in each multi-tiered cradle. Alternatively, provision should be made to ensure that suspended scaffolding hoists can be operated from each level, including the operation of the manual descent facility on power-operated hoists.

Access between levels should be protected on both sides with securely fixed mesh, be fitted with hinged trapdoors and allow clear access for at least 450 mm along each working platform.

A sign displaying the working load limit should be fixed to the inside of each bay at each level.

11.15 TRAINING

Suspended scaffolding can be the most hazardous type of scaffolding.

Persons who are to operate suspended scaffolding equipment should be provided with sufficient information, instruction, training and supervision to enable them to use the equipment correctly and undertake emergency procedures. Operators must have written authorization to operate that type of suspended scaffold.

Because workers in the cradle of a suspended scaffold are isolated from the rest of the workplace, they must have adequate training to avoid the risk of injury, and to enable them to work from the suspended scaffold in a manner for which it was designed. Before working from a suspended scaffold, workers should be able to demonstrate the safe work practices learnt in training. Retraining of users should occur each time there are changes in work practices that are likely to affect health and safety, particularly with suspended scaffolding.

Records of training of operators and workers on suspended scaffolding should be up-to-date and available.

11.16 NOTIFICATION OF USE

The local regulatory authority may require notification before using suspended scaffolds that use fibre suspension ropes (e.g. boatswain's chairs that incorporate abseiling equipment). In New Zealand, erection and use of suspended scaffolds is classified as notifiable work, which must be notified to Occupational Health and Safety service before work commences.

11.17 SAFETY EQUIPMENT

Adequate personal protective equipment should be present in the cradle and correctly used.

Any fire extinguishers that are fitted to the stage must be full and bear a stamped dated tag.

A first aid box with adequate equipment, including eyewash and means to stem heavy bleeding, should be provided and checked daily. The contents will depend on the type of work being carried out and the type of materials and substances being used.

There should be a reliable and efficient communication system between the cradle and outside. Communication systems include persons being in sight of the cradle or chair at all times at a distance from which hand signals, whistles or bells can be used effectively. Radio communication may be used.

In the event of an emergency, there should be a means for rescuing those in the cradle, such as the fire brigade on standby and crane-lifted work boxes.

11.18 OPERATION

Use suspended scaffolds only in the way in which they were designed and intended. Each scaffolding hoist should be directly operated by an authorized person. It should not be operated from a remote point via an improvised means such as lengths of rope or wire or by means of wedging or tying the deadman's handle in the 'on' position. Where remote control is a design feature of the equipment, it should comply with the manufacturer's recommended procedure.

Care should be taken to ensure that, apart from any purpose designed buffer rollers, no part of the cradle or hoist comes into contact with the structure (including projecting scaffold tubes, timber, open swivel or pivot windows) while it is climbing, descending or traversing.

Where automatic levelling devices are not fitted during climbing or descending operations, the cradle should be maintained as close to the horizontal as practicable by synchronized operation of the hoists.

Do not operate an emergency descent while power is connected to the scaffolding hoist.

Take precautions to prevent damage to the suspended scaffold or its supporting structure by traffic, cranes or other plant.

Use lateral restraints to prevent instability of the cradle during use. Instability may come from the work procedures or from wind. Lateral restraints include:

- Lanyards.
- Tensioned wire rope systems.
- Removable ties.
- Fan units.
- Suction units.

11.19 WORK PRACTICES

Suspension ropes and secondary ropes should not be contaminated with construction materials resulting from work activities.

The cradle should be maintained in a tidy condition with unobstructed access along the entire length.

Materials or equipment in the cradle must not be in excess of the working load limit of the suspended scaffold.

Take all reasonable precautions to prevent material or debris falling from the suspended scaffold or from the area where work is being performed from the suspended scaffold. To protect those passing below, install a catch platform that has been designed to support a uniformly distributed load of not less than 5 kPa, and is sufficiently long and wide so that it will trap any falling debris. Independently supported catch platforms may be used. Alternatively, install a guardrail or a fence, to protect persons by preventing them from passing below the scaffold.

11.20 WHILE UNATTENDED

Unattended cradles on suspended scaffolds should be left so that they cannot be accessed by unauthorized persons and so that wind cannot affect the equipment in a way that will cause damage to adjacent structures or equipment.

While a cradle is unattended, appropriate safety procedures, including the following, should be adopted:

In short periods, such as meal breaks:

- The cradle should be secured to the serviced structure, to prevent damage to property or equipment from the effects of wind.
- The power supply should be disconnected from the scaffolding hoists, either at the units or at the supply point.

For overnight or longer periods:

- The cradle should be left in an inaccessible position, preferably 3 m above the ground or adjacent deck level.
- The cradle should be secured to the serviced structure, to prevent damage to property or equipment from the effects of wind.
- Trailing ropes and cables should be stored on the cradle.
- Mobile suspension rigs should be effectively chocked or otherwise prevented from moving under the effect of wind.
- Protective devices should be trip locked onto their respective wire ropes.
- Power supply leads should be disconnected from the power supply outlets on the supporting structure.
- Air supply hoses should be disconnected from any permanent air supply line, or the supplying air compressor should be shut down and the air reservoir pressure released.

SECTION 12 GENERAL USE

12.1 SPECIFIC TYPES OF SCAFFOLD

Additional requirements for use of specific types of scaffolds are included in Sections 10 and 11.

12.2 BEFORE USE

The scaffold must be suitable for the intended use.

Prior to a scaffold being used, make sure there is positive indication that the scaffold is ready for use. The scaffold should not have any tag, notice or sign that advises that the scaffold should not be used or should be limited in its use.

The working platform must contain no litter and must not have been adversely affected in any way by previous users.

The approaches, accesses, walkways and egresses must be suitable for transportation of the materials and equipment intended to be used. Means must be provided for the movement of materials to and from the working platforms with minimum risk.

Effective lighting of all approaches, accessways, walkways, working platforms and exits must be provided. Lighted areas should be free from glare and deep shadow.

The scaffold and its supporting structure must not be subject to instability, as a result of any tunnelling, drilling, trenching or other excavation work.

12.3 HANDOVER CERTIFICATE

If required by the statutory authority or user or if recommended by the supplier or the designer, a handover certificate must be provided before a suspended scaffold, cantilevered scaffold, hung scaffold, or scaffolding over 4 m high in Australia or over 5 m high in New Zealand can be used. Handover certificates should be signed by, or on behalf of, the organization responsible for the erection or alteration of the scaffold and should include:

- The name and address of the person requiring the scaffold.
- The name and address of the organization that erected or altered the scaffold.
- The address and location of the scaffold.
- A description of the type of the scaffold.
- The size of the scaffold.
- The duty loadings applicable to the scaffold.
- The maximum number of platforms that can be loaded or worked from at any one time.
- The intended purposes of the scaffold.
- The date and time of handover.
- Confirmation that the scaffold complies with any design specifications, complies with any supplier's information, and is suitable for its intended tasks.

12.4 DURING USE

12.4.1 Responsibilities of users Once a scaffold has been accepted as a working platform, the users are responsible for using a scaffold in a way that minimizes risk to themselves and others. Their employer must ensure that they possess sufficient knowledge and have undergone adequate training to use the scaffold correctly. Also, they must know all emergency procedures relating to the scaffold, especially those relating to suspended scaffolds and mast-climbing work platforms.

12.4.2 Load limitations Users must understand the loading limitations and restrictions relating to the scaffold and the need to keep within them.

Scaffolds and scaffold members should not be used as anchorage points for any chain block, rope block, purchase, snatch block, deflection sheave, barrow hoist or other lifting device, unless specifically designed by a competent person to withstand the forces involved.

12.4.3 Work practices Accesses, walkways and egresses should be maintained in a clear unobstructed condition at all times.

Working platforms should be maintained in a tidy condition. Any build-up of material or debris should be monitored and progressively removed, avoiding any overload situation. Falling material or debris generated during use should be confined within the working platform or catch platform. Where molten metal cannot be contained entirely within a scaffold, such as during welding or oxyacetylene cutting, any areas affected should be isolated and a spotter stationed during the work. Any containment sheeting on such a scaffold must be resistant to combustion.

Users must understand the importance of not working from improvised platforms to gain extra height or length, such as the bare framework of the scaffold structure, planks placed on guardrails, or stepladders placed on decks.

12.4.4 Electrical hazards Monitor electrical leads and associated equipment for potential hazards.

Make sure that water used during operations, such as water blasting or hosing, does not cause an electrical hazard from contact with any wiring, particularly wiring associated with or connected to the scaffold.

Where the scaffold stands in close proximity to power lines, the handling of reinforcing steel rods should be carried out in a manner that preserves a safe minimum distance, even though such power lines are protected against brush contact with 'tiger tails'.

12.5 AFTER USE

Unused material, scraps and rubble should be removed from the scaffold and the working platform cleaned prior to hand over.

SECTION 13 INSPECTION, TESTING AND MAINTENANCE

13.1 RECORD KEEPING

Records of all tests, maintenance, inspections, commissioning and alterations must be kept by the employer and made available upon request to employees or health and safety representatives. Appendix J gives a convenient check list that may be photocopied and used. Appendix K gives convenient record sheets that may also be photocopied and used.

13.2 FREQUENCY OF INSPECTIONS

Scaffolds must be inspected a number of times during use to determine the need for any modifications or repairs that may be required to keep the scaffolds in a serviceable condition.

Scaffolding hoists, protective devices, load limiting devices and wire ropes must be regularly inspected, serviced and tested. The relevant information must be recorded in an appropriate register. All inspections, servicing and tests must be carried out by a competent person.

The appropriate intervals between inspections depend upon the site conditions, the nature of the work, the degree of risk associated with a failure of the scaffold, and the recommendations or specifications given by the scaffold designer and the equipment supplier. Minimum inspection intervals are prescribed in Australia by occupational health and safety laws and in New Zealand by the Approved Code for Scaffolding.

13.3 AUSTRALIAN REQUIREMENTS

In the absence of more specific requirements of the local regulatory authority, inspections should be carried out on all suspended scaffolds, cantilevered scaffolds, spur scaffolds and hung scaffolds and to any other scaffold from which a person or object could fall more than 4 m. These scaffolds and their supporting structures will require inspection:

- Before the first use.
- At intervals not exceeding 30 days.
- As soon as practicable and prior to further use following an occurrence that could have affected the stability or adequacy of the scaffold, such as severe storm conditions or an earthquake.
- Prior to its use following repairs.

In each case, the employer must ensure that no work is carried out from the scaffold, unless written confirmation has been obtained that the scaffold, or its relevant portion, is complete.

The person who carries out compliance inspections and provides the written confirmation must be competent. Suitable persons are those with an appropriate scaffolding certificate of competency, a recognized scaffolding inspection qualification, or the designer. However, where complex scaffold designs are used, or expert judgement regarding the strength, rigidity or deterioration of supporting structures is needed, the person carrying out inspections may require the advice of an engineer experienced in structural design, a geotechnical engineer or a geologist, as appropriate.

It is recommended that the handover certificates, records of inspection and repairs and maintenance details be held on site in an accessible format, until the scaffold has been completely dismantled.

Inspection records should include:

- The individual identification number or mark of the scaffold.
- Any relevant design or specification reference.
- The location of the scaffold.
- The purpose for which the scaffold is provided.
- The date and time of each inspection.
- Comments on each inspection.
- The name and signature of the person conducting the inspection.

13.4 NEW ZEALAND REQUIREMENTS

In New Zealand, all suspended scaffolds and any other scaffolds that have a height of 5 m or more require inspection:

- Before the first use.
- On each day (in the case of a suspended scaffold), or weekly (in the case of other scaffolds) while the scaffold is in use.
- Monthly while the scaffold is set up but not in use.
- After any storm or occurrence that could have adversely affected the safety of the scaffolding or, in the case of suspended scaffolding, the method and means of suspension and the anchorage.
- After each structural alteration, addition or change to the nature of any anchorage.

The inspections must be carried out by the employer, a competent person or a certified scaffolder of the appropriate class.

Inspection details must be entered into an on-site register, which must be signed by the person carrying out the inspection to confirm the continuing compliance and suitability of the scaffold.

13.5 INSPECTIONS

The appropriate inspection method will depend on the site conditions and the type of scaffold. The inspection checklist given in Appendix E may be a useful tool for the inspection of common scaffold configurations. Common defects in some scaffold components are discussed in Clauses 13.7 to 13.15.

Where a very large scaffold has been standing for many months or even years, it may be appropriate to subject a representative sample of the key structural members and tie assemblies to a detailed examination for corrosion and fatigue. AS 1199 provides a recognized methodology for statistical sampling and AS 1399 provides useful guidance.

It is crucial that such an inspection can determine whether:

- The scaffold structure is adequate.
- The supporting structure is adequate.
- Working platforms are secured and protected.
- Access and egress is sufficient.
- The scaffold will enable the relevant work tasks to be performed adequately and safely.

13.6 REPAIRS TO ERECTED SCAFFOLD

Where damaged by other construction activities or by environmental conditions, an erected scaffold should not be used until repairs have been carried out.

Before carrying out repairs, isolate the area around the scaffold to protect other people.

Repair work must not render the scaffold unstable and should comply with the supplier's and designer's information. Repairs should be completed and, before further use, the scaffold must be re-inspected by a competent person.

13.7 COUPLERS AND ACCESSORIES

Couplers and accessories should be maintained so that they can be used as intended. For example, nuts should be free running and hinges should turn freely.

Avoid excessive oil, grease or paint, which can cause a coupler or accessory to slip.

Do not apply uncontrolled heat to couplers and accessories, because excessive temperature will increase the risk of failure.

A bent plate of an adjustable baseplate should be straightened. If straightening is not possible, the baseplate should be replaced. Cracks in welds should be ground out and repaired by re-welding.

13.8 SCAFFOLD TUBES

Tube should be cut by a bench saw or pipe cutter, clean and square to the axis of the tube. Bevelled trimming of tube ends should have a minimum end-face bearing width of 2 mm and sharp edges should be removed. Do not weld or flame cut tube.

Tubes with a surface deformation exceeding 4 mm, or with a cross-sectional distortion exceeding 1.03 times the diameter of the undistorted cross-section, must be reduced in length or scrapped.

Tubes that are bent (i.e. out of straight), so that any section of the tube is more than 1/300 of the length of the tube from the alignment between the ends of the tube (e.g. as determined by a string line stretched between the ends), must be reduced in length or scrapped.

Bent steel tubes may be straightened by using a crow or reeling machine. Do not use reeling to straighten aluminium tube.

Common defects in scaffold tubes are illustrated in Figure 13.8.

13.9 SCAFFOLD PLANKS

Clean materials such as excessive cement, mortar, concrete, oil, grease and paint from scaffold planks.

Do not use timber planks painted in a way that could conceal defects. Replace missing or damaged end-fixings and remove protruding nails. Figure 13.9(A) illustrates faults in timber scaffold planks.

Where a timber scaffold plank is suspected of being too weak for its intended purpose, it can be tested by impact tests as specified in Appendix J.

Figure 13.9(B) illustrates faults in typical metal scaffold planks. Replace damaged or missing end caps or reinforcing straps. Cracks in welds should be ground out and repaired by re-welding. Surface dents should be knocked out if practical, otherwise the plank should be shortened or scrapped.

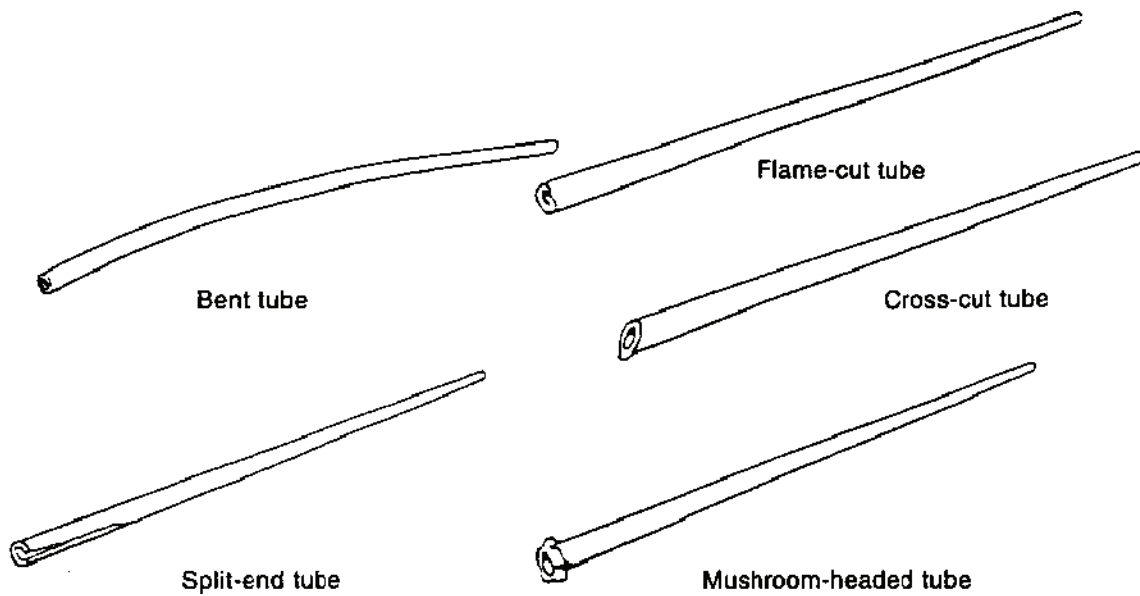


FIGURE 13.8 COMMON DEFECTS IN SCAFFOLD TUBES

13.10 MODULAR SCAFFOLDING

Modular scaffolding components should be maintained to confirm to the original specification in terms of structural capability, material and physical shape.

Any deposit such as concrete, which would restrict a component from performing its function, should be removed.

Cracks in welds should be ground out and repaired by re-welding to the original specification.

Lightly bent tubes (less than 15 mm/m length) should be straightened. If it is not possible to effectively shorten the component by cutting off the damaged part, severely bent tubes, especially where the tube surface is kinked, should be scrapped.

Standards with a circular cross-section must be reduced in length or scrapped in the event that they have a surface deformation exceeding 4 mm or a cross-sectional distortion exceeding 1.03 times the diameter of the undistorted cross-section.

Replace missing or damaged end attachments of ledgers, transoms, braces and similar components. Make sure that brace end-attachments swivel freely.

13.11 ACCESS LADDERS

Any fixed ladder providing access and egress should be inspected as a part of the scaffold by a competent person.

Between uses, test for sound construction by:

- Taking each end of the ladder in turn and trying to push the stiles apart and then together. Movement will indicate insecurely fixed rungs and loose tie rods.
- Laying the ladder flat, raising one end and attempting to push one stile while pulling the other. If the stiles can be moved relative to each other, the rungs are loose.
- Tapping timber rungs with a mallet. A dull sound indicates a defective rung.

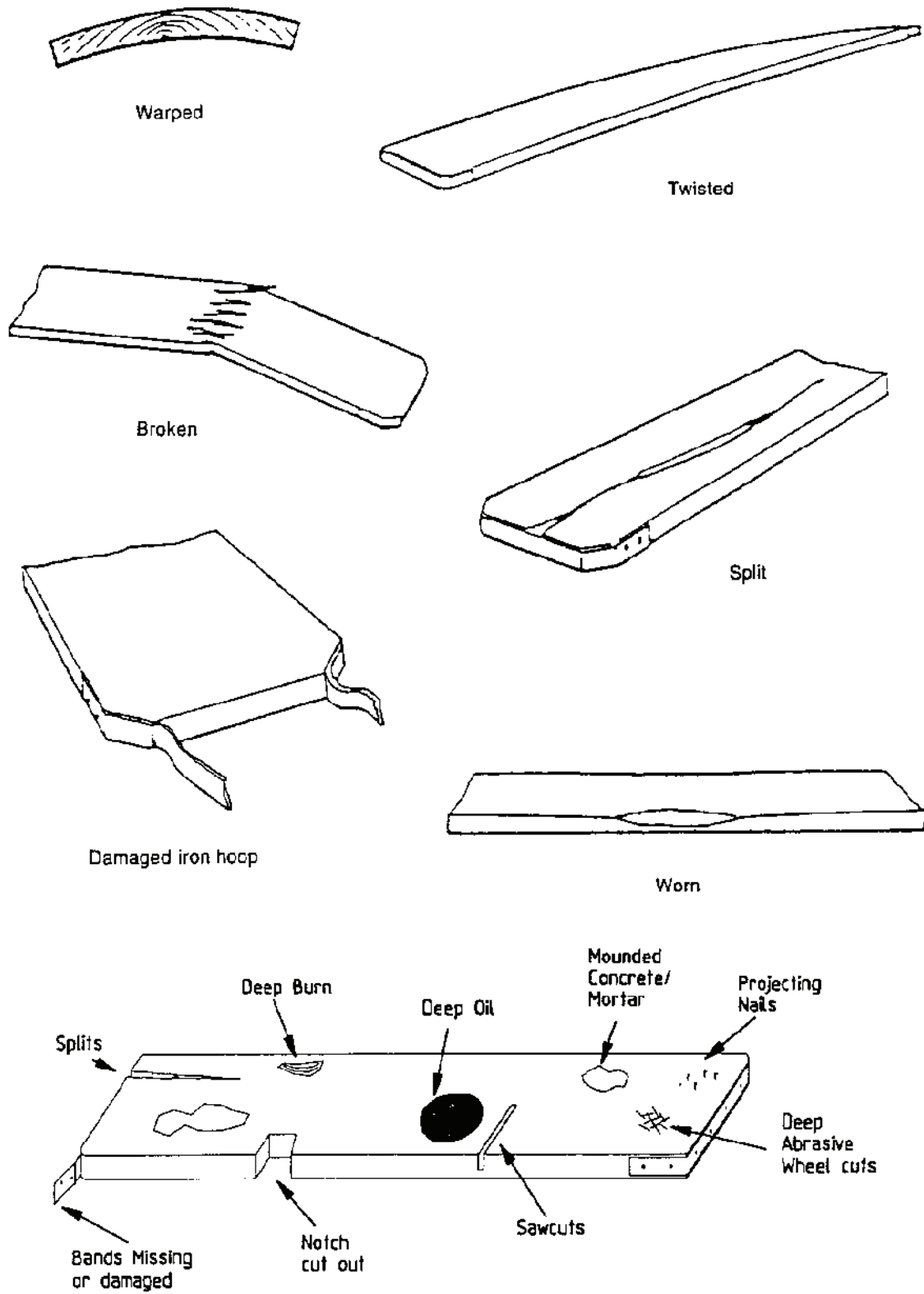


FIGURE 13.9(A) FAULTS IN TIMBER SCAFFOLD PLANKS

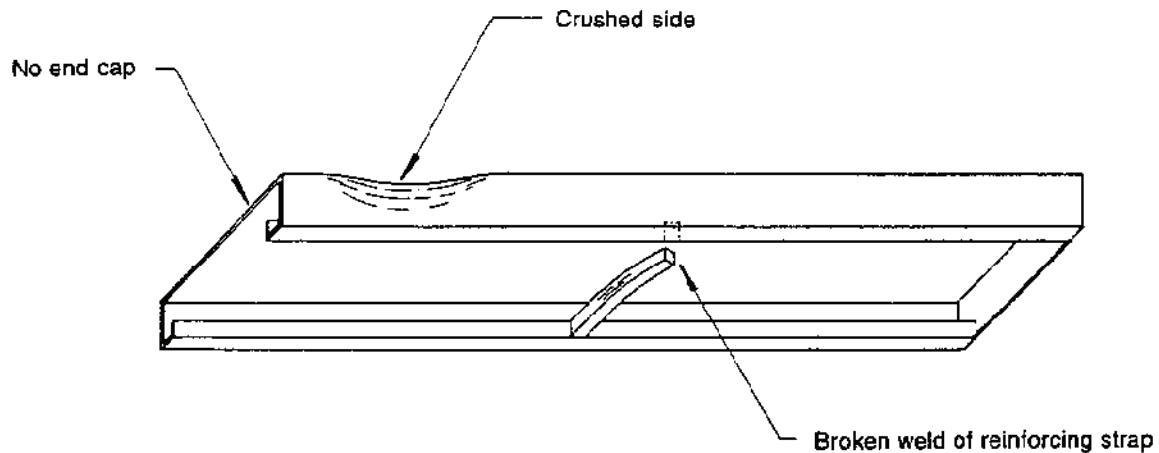


FIGURE 13.9(B) FAULTS IN METAL SCAFFOLD PLANKS

Any ladder that is damaged or unsound should be removed from service, until it has been repaired and has passed re-inspection. Condemn a ladder if any of the following faults exist:

- Timber stiles are warped, splintered, cracked or bruised.
- Metal stiles are twisted, bent, kinked, crushed, have cracked welds or have damaged feet.
- Rungs, steps, treads or top plates are missing, worn, damaged or loose.
- Tie rods are missing, broken or loose.
- Ropes, braces or brackets are missing, broken or worn.
- Timber members, apart from narrow identification bands, are covered with an opaque paint or any other treatment that could disguise faults in the timber.

13.12 SCAFFOLDING HOISTS

Each scaffolding hoist must be inspected and subjected to an operational test by a competent person before being sent to a job. Such a test must include lifting a test load of 1.25 times the rated working load of the hoist (even if the hoist has a load-limiting device) up a distance of not less than 3 m, lowering the load, and arresting and sustaining the load in any position without the hoist showing over-straining of any part.

Any daily on site inspection by an operator must check that:

- The directional switches function correctly.
- Any emergency stop button operates correctly.
- Any top limit switch operates.
- The emergency crank handle is fitted, if there is provision for such an item.
- Where the emergency crank handle has an electrical interlock, loosening of the handle will cut-off power and re-tightening of the handle will return power to the controls. The emergency handle must be left tightened.

Weekly on-site inspections must be made and results must be recorded on an inspection register maintained at the job site, which must include:

- Evidence of having passed an operational test, where the suspended scaffold is run up and down a sufficient distance to ensure that all operational controls on each attached scaffolding hoist function correctly and the mechanism does not produce unexpected noise or vibration.
- Date of test.
- Serial numbers of the scaffolding hoists inspected.
- Name and signature of inspecting persons.

Each scaffolding hoist on a job should be returned to a maintenance shop for a thorough inspection and maintenance program carried out by a competent person, at periods nominated by the manufacturer but not exceeding six months. Such six-monthly inspections should be recorded in a maintenance workshop or base office, which should include:

- Test notations.
- Dates of tests.
- Dates of services.
- List of parts replaced.
- Test load and related working load of the hoist, in kilograms.
- Statement that the hoist performed the test satisfactorily.
- Name and signature of the servicing and testing persons.

13.13 PROTECTIVE DEVICES

Protective devices must be inspected and subjected to an operational test by a competent person before being sent to a job. During such a test, a test load of 1.25 times the rated working load of the associated scaffolding hoist must be applied and the protective device should support the load without any sign of distortion. Protective devices must be inspected daily by the operator and must include:

- An operational check of any overspeed governor.
- Manual tripping of the overspeed device to check correct function followed by resetting.
- An operational check of any slack rope device.

Protective devices must be inspected weekly by a competent person, who must include:

- Checking that the electrical interlock operates correctly, after any overspeed device trips.
- For protective devices not integrated with the scaffolding hoist, ensuring the correcting links do not show any signs of distortion and have correctly sized and fitted connecting bolts and nuts.

Each protective device on a job should be returned to a maintenance shop for a thorough inspection and a maintenance program should be carried out by a competent person, at periods nominated by the manufacturer but not exceeding six months. Such six-monthly inspections and test notations should be recorded in a maintenance workshop or base office and must include:

- Date of service and test.
- List of parts replaced.
- Test load and rated working load of the associated scaffolding hoist, in kilograms.
- Name and signature of the service and testing persons.

13.14 LOAD-LIMITING DEVICES

A load-limiting device must be inspected and subjected to an operational test by a competent person before being sent to a job.

Each load-limiting device on a job, should be returned to a maintenance shop for a thorough inspection and maintenance program carried out by a competent person, at periods nominated by the manufacturer but not exceeding six months, and whenever the load setting is changed.

Operational test for load-limiting devices should ensure that the device stops operation of the scaffolding hoist at a load of not more than 25 percent above the load setting marked on the device. Where the load is limited by an electric current limiting device, the test should be carried out with the supply voltage reduced to typical site conditions.

Daily on-site inspections by the operator must check for visible signs of damage to the equipment and connections to the scaffolding hoist or stirrup.

Weekly on-site inspections by a competent person must check that the device and any visible connections are correctly fitted.

Six-monthly inspections and test notations should be recorded in a maintenance workshop or base office and must include:

- Date of service and test.
- List of parts replaced.
- Load setting of the device in kilograms.
- Supply usage, if applicable.
- Statement that the load-limiting device passed the test.
- Name and signature of the servicing and testing persons.

13.15 WIRE ROPES FOR SCAFFOLDING HOISTS

Wire ropes for scaffolding hoists should be inspected by a competent persons before being sent to a job. Figure 13.15 illustrates common defects in scaffolding hoist wire ropes.

Daily on-site inspection should cover the expected working range of the rope.

Inspection notations should be recorded at the site, in a maintenance workshop or base office and must include:

- Date of inspection.
- Location of inspection (e.g. on site, workshop).
- Comment on rope condition or need to remove damage by shortening.
- Shortened length of rope.
- Name and signature of inspecting persons.

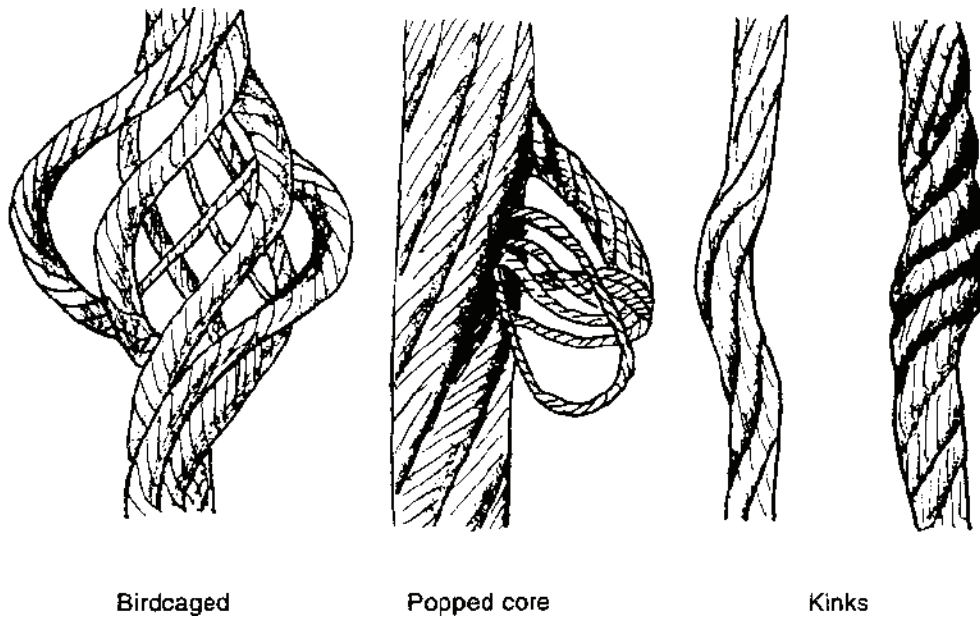


FIGURE 13.15 COMMON DEFECTS IN SCAFFOLDING HOIST WIRE ROPES

APPENDIX A

REFERENCED DOCUMENTS

(Normative)

The following documents are referred to in this Standard.

AS	
1138	Thimbles for wire rope
1163	Structural steel hollow sections
1199	Sampling procedures and tables for inspection by attributes
1353	Flat synthetic-webbing slings
1353.1	Part 1: Product specification
1353.2	Part 2: Care and use
1380	Fibre-rope slings (of natural or synthetic rope)
1399	Guide to AS 1199—Sampling procedures and tables for inspection by attributes
1418	Cranes (including hoists and winches)
1418.2	Part 2: Serial hoists and winches
1418.7	Part 7: Builders' hoists and equipment
1438	Wire-coil flat slings
1576	Scaffolding
1576.2	Part 2: Couplers and accessories
1576.3	Supplement 1: Prefabricated and tube-and-coupler scaffolding — Metal tube-and-coupler scaffolding — Deemed to comply (Supplement to AS 1576.3—1991)
1576.4	Part 4: Suspended scaffolding
1577	Scaffold planks
1666	Wire-rope slings
1801	Industrial safety helmets (incorporating Amdt 1)
1859	Flat pressed particleboard
1866	Aluminium and aluminium alloys—Extruded rod, bar, solid and hollow shapes
1891	Industrial safety belts and harnesses
1892	Portable ladders
1892.1	Part 1: Metal
1892.2	Part 2: Timber
2076	Wire rope grips
2089	Sheave blocks for lifting purposes
2317	Collared eyebolts
2319	Rigging screws and turnbuckles
2321	Short-link chain for lifting purposes (non-calibrated)
2626	Industrial safety belts and harnesses—Selection, use and maintenance
2741	Shackles

AS

- 2759 Steel wire rope—Application guide
- 3190 Approval and test specification—Residual current devices (current-operated earth-leakage devices)
- 3569 Steel wire ropes
- 3610 Formwork for concrete
- 3775 Chain slings—Grade T
- 4142 Fibre ropes
- 4142.1 Part 1: Care and safe usage
- 4142.2 Part 2: Three-strand hawser-laid and eight-strand plaited

AS/NZS

- 1576 Scaffolding
- 1576.1 Part 1: General requirements
- 1576.3 Part 3: Prefabricated and tube-and-coupler scaffolding
- 1576.5 Part 5: Prefabricated splitheads and scaffolds
- 2210 Occupational protective footwear
- 2210.1 Part 1: Guide to selection care and use
- 2210.2 Part 2: Specification
- 2269 Structural plywood

BS

- 1139 Metal scaffolding
- 1139.1 Part 1: Tubes
- 1139.1.1 Section 1.1: Specification for steel tube
- 1139.1.2 Section 1.2: Specification for aluminium tube
- 1139.2 Part 2: Couplers
- 1139.2.1 Section 2.1: Specification for steel couplers, loose spigots and base-plates for use in working scaffolds and falsework made of steel tubes
- 1139.2.2 Section 2.2: Specification for steel and aluminium couplers, fittings and accessories for use in tubular scaffolding
- 3913 Specification for industrial safety nets
- 6323 Specification for seamless and welded steel tubes for automobile, mechanical and general engineering purposes
- 6323.1 Part 1: General requirements
- 8093 Code of practice for the use of safety nets, containment nets and sheets on constructional works

JIS

- C 8371 Residual current operated circuit breakers

NZS

- 1583 Specification for thimbles for natural fibre ropes
- 3609 Specification for timber ladders
- 3620 Scaffold planks
- 5233 Specification for portable ladders (other than timber ladders)
- 5806 Specification for industrial safety helmets (medium protection)
- 5811 Industrial safety belts and harnesses
- 5811.1 Part 1: Specification for industrial safety belts and harnesses

NZS

- 5811.2 Part 2: Code of practice for the selection, use, and maintenance of safety belts and harnesses

NZS/BS

- 302 Stranded steel wire ropes

- 302.2 Part 2: Specification for ropes for general purposes

- 2052 Specification for ropes made from manila, sisal, hemp, cotton and coir

NZS/BS

- 3551 Specification for alloy steel shackles

- 4278 Specification for eyebolts for lifting purposes

- 4429 Rigging screws and turnbuckles for general engineering, lifting purposes and pipe hanger applications

NZS/ISO

- 1835 Short link chain for lifting purposes—Grade M(4), non-calibrated, for chain slings etc

- 3075 Short link chain for lifting purposes—Grade S(6), non-calibrated, for chain slings etc

- 3076 Short link chain for lifting purposes—Grade T(8), non-calibrated, for chain slings etc

- 7531 Wire rope slings for general purposes—Characteristics and specifications

VDE

- 0664 Residual current-operated protective devices

- 0664.1 Part 1: Residual current-operated circuit breakers rated up to 500 V a.c. and up to 63 A

- 0664.2 Part 2: Current-operated earth-leakage circuit breakers with overcurrent protection up to 415 V a.c and up to 63 A

APPENDIX B

GLOSSARY OF TERMS

(Normative)

Term	Definition
Access platform	A platform that gives access to and from places of work to persons, materials and equipment.
Accessory	A fitting that is able to be attached to a structural member of a scaffold, or to join a structural member to something else (see Figure B1).
Adjustable baseplate	A baseplate with an adjustable leg (see Figure B1).
Adjustable castor	A castor incorporating a height-adjusting device that fits inside the standard or vertical member of a scaffold.
Adjustable leg	A threaded bar or tube with nut, designed to fit inside a standard to support the load from the standard, and which is for levelling purposes, in conjunction with a baseplate, forkhead or castor (see (a) and (c) of Figure 10.1(A)).
Articulated cradle	A cradle of several sections that are hinged together and enables persons to walk from one section to another (see Figure B2).
Baseplate	A plate that is able to distribute the load from a load-bearing member to a supporting structure (see Figure B1).
Bay	The space enclosed by four adjacent standards, or the equivalent space in a single-pole scaffold (see Figure B4).
Bay length	The horizontal distance between any two longitudinally-adjacent standards in an independent scaffold or a single-pole scaffold, or the horizontal distance between any two longitudinally-adjacent support points on the cradle of a suspended scaffold (see Figures B3 and B4).
Bay width	The horizontal distance between any two transversely-adjacent standards in an independent scaffold, or the horizontal distance between a standard and a transversely-adjacent wall in a single pole scaffold, or the width of the cradle of a suspended scaffold (see Figure B4).
Beam chaffer	A short length of material with a half-a-tube cross section that is able to be used on the edges of a universal beam, rolled steel joist, or similar, to prevent damage to a reeved sling or chain.
Beam clamp	A fitting that is able to secure a sling or chain to the underside of a universal beam, rolled steel joist, tapered flange beam or similar.
Birdcage scaffold	An independent scaffold consisting of more than two rows of standards connected by ledgers and transoms.

Term	Definition
Boatswain's chair	A suspended scaffold of which the platform is a chair or similar device, suitable for use by one person.
Box tie	A tie assembly that is positively fixed to every side of a column or beam (see Figure 8.5).
Brace	A member fixed diagonally to two or more members of a scaffold, to provide rigidity to the scaffold (see Figure 8.4).
Bracket scaffold	A scaffold that has a platform carried on frames attached to or supported by a permanent or temporary construction.
Brickguard	A meshed panel secured between a guardrail and toeboard, and sometimes incorporating a kickplate (see Figure 8.8(A)).
Bricklayer's scaffold	A single-pole scaffold.
Bridle	A horizontal member spanning between putlogs or transoms, for the purpose of supporting intermediate putlogs.
Butt	A tube fixed to a scaffold and butting to an adjacent structure, to prevent horizontal movement of the scaffold in the direction of the structure.
Cantilever scaffold	A scaffold that is supported by cantilevered load-bearing members (see Figure 10.7).
Cantilever builders' hoist	A builders' hoist (see AS 2549) where the car, bucket or platform is cantilevered from, and travels up and down externally to a face of, the support structure.
Castor	A swivelling wheel attached to the lower end of a standard, for the purpose of supporting and moving a scaffold (see Figure B1).
Catch platform	A platform, attached to a scaffold, to contain falling debris (see Figure 8.8(B)).
Chaffer	A beam chaffer.
Check coupler	A right-angle coupler or swivel coupler that is fixed hard against a load-bearing coupler, to restrict or prevent slippage of that coupler along the tube.
Chord	A principal longitudinal member of a scaffold beam or truss.
Competent person	A person suitably qualified, adequately trained and appropriately experienced for the particular class or kind of work described.
Counterweight	A weight or series of weights that counterbalance a scaffold against overturning (see Figure 11.10.6).
Coupler	A fitting that joins two tubes (see Figure B1).

Term	Definition
Cradle	The portion of a suspended scaffold that incorporates a suspended platform (see Figures 11.10.1, B2 and B3).
Double coupler	A right-angle coupler.
Double-rope scaffold	A suspended scaffold with two rows of suspension ropes (see Figures 11.1(A)).
Double-rope suspension system	A method of suspending a cradle using two hoists and two suspension ropes at each support point (see Figures 11.1(A), B2 and B3).
Drop scaffold	A hung scaffold.
Dropper	A hung standard.
End-to-end coupler	A coupler for joining two tubes end-to-end (see Figure B1).
Extension ladder	A non-self-supporting portable ladder that is adjustable in length and consists of two or more sections with guides or brackets to permit length adjustment.
False standard	A puncheon.
Fan	A cantilevered catch platform.
Fender board	A toeboard.
Finial coupler	A coupler that joins two scaffold tubes in the same plane (see Figure B1).
Fixed cradle	A cradle that does not traverse horizontally, but is capable of being raised and lowered.
Fixed finial coupler	A finial coupler that is able to join two scaffold tubes at right angles.
Flange clamp	A load-bearing clamp (rigid or swivelling) for connecting a tube to the flange of a structural member.
Fork head	A u-shaped housing (fixed or adjustable) for supporting and locating a standard over a bearer or a bearer over a standard.
Frame scaffold	A scaffold assembled from prefabricated frames, braces and accessories (see Figures 10.1(A)(b) and 10.3.7).
Frame-trestle scaffold	A scaffold, other than a trestle ladder, consisting of self-supporting stands that support scaffold planks.
Freestanding scaffold	A scaffold that is not attached to any other structure and is stable against overturning, on its own account or if necessary assisted by stabilizers, outriggers, counterweights or backup bays.

Term	Definition
Fulcrum point	The point of pivot nearest to the outside edge of a suspension rig and about which the balancing moments of a suspension rig are calculated (see Figure 11.5.1).
Gantry	A structure, constructed from structural steel, scaffolding or structural timber, that is primarily intended to support a protection deck or portable buildings such as amenity sheds.
Girder clip	A flange clamp.
Girder trolley	A movable assembly performing the function of a beam clamp.
Guardrail	A structural member to prevent persons from falling off any platform, walkway, stairway or landing.
Half round	A beam chaffer.
Height of a scaffold	The vertical distance from the supporting structure to the highest working platform of the scaffold.
Hoarding clip	A fitting used to fix hoarding panels to a scaffold tube.
Hung scaffold	An independent scaffold that is hung from another structure but is not capable of being raised or lowered when in use (see Figure 10.9(A)).
Hung standard	A suspended member, usually a tube, in a hung scaffold that transmits a load to an overhead supporting structure.
Inboard portion of a suspension rig	The portion of a cantilevered-suspension rig that is on the inside of the fulcrum point (see Figure 11.5.1).
Independent scaffold	A scaffold consisting of two or more rows of standards connected together longitudinally and transversely (see Figures 10.5(a) and B5).
Individual cradle	A cradle of one bay length (see Figure B3).
Internal-perimeter scaffold	A scaffold erected alongside the inside perimeter of a penetration or shaft in a building or other structure.
Joint pin	An internal end-to-end coupler for joining two tubes (see Figure B1).
Keeper clip	A check coupler.
Kickboard	A toeboard.
Kickplate	A plate, usually of metal, that performs the function of a toeboard.
Ladder	An appliance on which a person may ascend or descend, consisting of two stiles joined at regular intervals by cross-pieces (e.g. cleats, rungs, steps, treads).

Term	Definition
Ladder beam	A scaffold beam with chord stiffeners at right angles to the chords (see Figure 10.9(A)).
Ladder-bracket scaffold	A bracket scaffold consisting of two or more single ladders or extension ladders that support planks by means of brackets, to provide a working platform (see Figure 10.2.5).
Ladder clamp	A fitting, incorporating a bolt and a nut, that is used to secure a ladder to a scaffold tube.
Ladder tie	A fitting, incorporating a length of wire rope, that is used to secure a ladder.
Landing	A level area providing access to a stairway or ladder, or located at an intermediate level in a system of stairways or ladders.
Ledger	A horizontal structural member that longitudinally spans a scaffold (see Figures 8.3(A) and 10.3.6(C)).
Lever arm	The distance between the fulcrum point and the centre of gravity of a cantilevered-suspension rig, including any counterweights (see Figure 11.5.1).
Lift	The vertical distance from the supporting surface to the lowest ledger of a scaffold or level at which a platform can be constructed. Also, the vertical distance between adjacent ledgers or levels of a scaffold at which a platform can be constructed. (see Figure B4).
Load-limiting device	A device that limits the lifting capacity of a scaffolding hoist, usually to a load lighter than the stalling load of the motor.
Loading bay	A platform on a scaffold for the storage of materials and equipment.
Longitudinal brace	A brace in a vertical plane on the face of a scaffold.
Mast climber	Mast-climbing work platform.
Mast-climbing work platform	A work platform used for temporary purposes to raise personnel and materials to a working position by means such as a rack-and-pinion drive mounted on an extendible mast, which may be tied to a building.
Maximum total suspended load	The maximum force that can be imposed on a suspension rig, including a load equal to the working load limit on the platform, the self-weight of the platform, suspension ropes and safety ropes, any force from the means of suspension and the rope tensions, any force caused by the operation of the equipment, and the environmental loads.
Midrail	A member fixed parallel to and above a platform, between the guardrail and the platform.

Term	Definition
Mobile scaffold	An independent free-standing scaffold that is mounted on castors (see Figure 10.4).
Modular scaffold	A scaffold assembled from prefabricated individual components, braces and accessories (see Figure 10.3.6(A)).
Needle	A cantilevered structural member that supports a scaffold (see Figure 11.5.1).
Ninety-degree coupler	A right-angle coupler.
Node point	A fixed junction between a ledger or transom and a standard.
Outboard portion of suspension rig	The portion of a cantilevered-suspension rig that is on the outside of the fulcrum point (see Figure 11.5.1).
Outrigger	A framed component that increases the effective base dimensions of a tower and is attached to the vertical load-bearing members (see Figure 10.4).
Overcoat	An end-to-end coupler with a non-adjustable internal spigot and a bearing surface capable of being tightened to the external surface of attached scaffold tubes (see Figure B1).
Panel	The area enclosed by two longitudinally adjacent standards and two vertically adjacent ledgers or levels of a scaffold at which a platform could be constructed.
Parallel coupler	A load-bearing coupler for making a lap or spliced joint between two tubes.
Perimeter scaffold	A scaffold erected alongside the external face of a building or other structure.
Pintle	A projection at the top of a castor that is used to locate the vertical members of a mobile scaffold.
Plan brace	A brace in the horizontal plane that is attached to vertical load-bearing members (see Figure 10.9(A)).
Platform	An elevated surface.
Platform bracket	A bracket attached to a scaffold that enables a platform to be placed between the scaffold and a building or structure.
Podger hammer	A steel tool used for the locking and releasing of typical modular scaffolding fixing devices (see Figure 3.1).

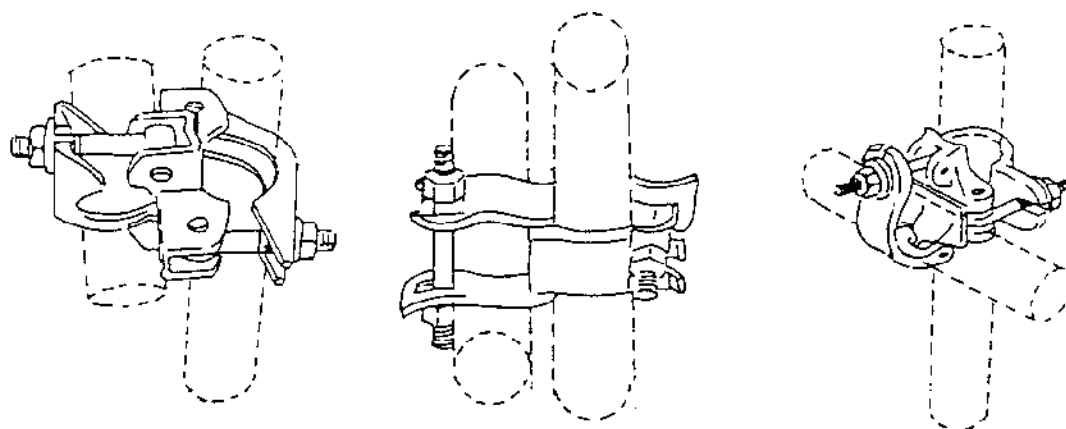
Term	Definition
Prefabricated platform	A framed assembly of one bay length, incorporating a walking surface, that is capable of connecting to its support structure in such a way as not to be inadvertently dislodged. (One or more prefabricated platform units can be used side by side and end to end to form a working platform).
Prefabricated scaffold	A scaffold assembled from prefabricated components and manufactured so that the geometry of the scaffolding is predetermined.
Preventer	A check coupler.
Projection length	The horizontal distance between the fulcrum point and the suspension point of a cantilevered suspension rig.
Protective device	A device that will arrest the descent of and support a cradle or boatswain's chair, in the event of a failure of a suspension rope or a scaffolding hoist.
Puncheon	A vertical supporting member supported from another structural member of a scaffold.
Putlog	A horizontal structural member, spanning between ledgers or between a ledger and an adjacent wall, that is intended to support a platform.
Putlog blade	A fitting fixed to the end of a putlog, so that the putlog may be located and supported in a joint of a wall (see Figure B1).
Putlog coupler	A coupler for fixing a putlog to a ledger (see Figure B1).
Putlog end	A putlog blade.
Putlog head	A putlog blade.
Putlog scaffold	A single-pole scaffold.
Putlog-trestle scaffold	A scaffold consisting of scaffold planks that are supported at one end by non-self-supporting stands and at the other end by a wall (see Figure 10.1(A)(c)).
Rafter clip	A fitting that is able to fix timber bearers to a scaffold tube.
Raker	An inclined tube fixed to a scaffold to keep the scaffold stable.
Reduction coupler	A right-angle coupler or swivel coupler that is able to join two tubes of different outside diameters.
Return	A part of a scaffold set up around the corner of a building or structure.
Return transom	A transom used in modular scaffolding that is able to fix a scaffold return at right angles to a run, by being secured to a ledger.
Reveal	Internal side surfaces of an opening or recess.

Term	Definition
Reveal pin	A fitting used for tightening a tube between a reveal.
Reveal prop	A vertical member that is able to support a needle or other cantilever member. A common example would be an adjustable building prop that is usually tightened between the adjacent floors of a building or other structure.
Reveal tie	An assembly, consisting of a reveal tube, pin, wedge or secured fitting and pads, that is fixed in a reveal, together with any tie tubes.
Right-angle coupler	A non-swivel load-bearing coupler, other than a putlog coupler, for connecting two tubes at right angles (see Figure B1).
Saddle piece	A mild steel plate attached to a steel tank or similar and shaped to receive the supporting hook of a scaffold bracket.
Safety rope	A secondary rope.
Scaffold	A temporary structure for supporting access platforms or working platforms. (See also the definitions for birdcage scaffold, boatswain's chair, cantilever scaffold, freestanding scaffold, hung scaffold, independent scaffold, mobile scaffold, putlog scaffold, single-pole scaffold, spur scaffold, suspended scaffold and tower scaffold and also Figures B4 and B5).
Scaffold beam	A fabricated member consisting of two or more chords and a number of stiffeners between the chords (e.g. a ladder beam, a unit beam).
Scaffold key	A scaffold spanner.
Scaffold plank	A decking component, other than a prefabricated platform, that is able to be used in the construction of a platform supported by a scaffold.
Scaffold spanner	A box or tube-type wrench with a swing-over handle that has been purpose designed for the tightening and releasing of couplers (see Figure 3.1).
Scaffolder	A person engaged in erecting, altering or dismantling scaffolding.
Scaffolding equipment	Any component, assembly or machine used or intended to be used for the construction of scaffolding.
Scaffolding hoist	A lifting appliance (manually operated or power operated) through which the suspension rope passes.
Scaffolding machine	A scaffolding hoist.
Secondary rope	A rope not normally carrying the weight of a cradle or an imposed load, but rigged for use with a protective device.

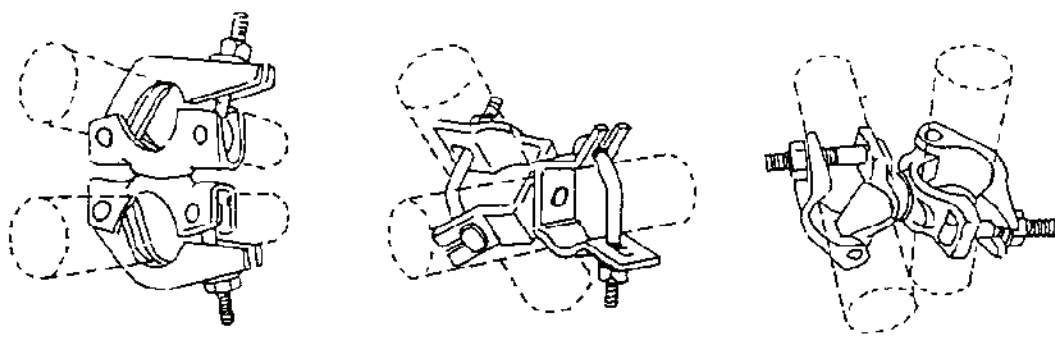
Term	Definition
Shutter-bracket scaffold	A bracket scaffold where the brackets are attached to formwork shutters.
Side rail	A stile.
Single coupler	A putlog coupler.
Single ladder	A non-self-supporting portable ladder of fixed length and consisting of one section.
Single-pole scaffold	A scaffold consisting of a single row of standards that are connected together by ledgers and putlogs fixed to ledgers and built into the wall of a building or structure (see Figure 10.5(b)).
Single-rope suspension system	A method of suspending a cradle using one hoist and a single suspension rope at each support point (see Figures B2 and B3).
Sleeve coupler or external joiner	An external end-to-end coupler (see Figure B1).
Slung scaffold	A hung scaffold.
Soleplate	A member used to distribute a load through a baseplate to the ground or other supporting structure (see Figure 8.3(C)).
Spur	An inclined load-bearing member that transmits a load to a supporting structure.
Spur scaffold	A scaffold that is partially supported by inclined load-bearing members.
Stabilizer	A component with a swivelling friction foot that increases the effective base dimensions of a tower and is attached to vertical load-bearing members.
Stair-tread fitting	A fitting used to fix a stringer, in order to support a stair tread.
Stanchion	A vertical member used to support a guardrail, a mesh panel or similar.
Standard	A vertical structural member that transmits a load to a supporting structure.
Step ladder	A self-supporting portable ladder of fixed length having flat steps or treads and hinged back legs.
Stile	A member in a ladder that supports rungs, steps or treads.
Stop	An attachment that will limit traversing of a cradle.
Strap	A mild steel plate that has been shaped to fit over a square-sectioned or rectangular-sectioned needle, for the support of a suspension rope.

Term	Definition
Stringer	An inclined member used to support stair-treads.
Strut	A scaffold member that supports a compressive force.
Stud-bracket scaffold	A bracket scaffold of which the brackets are attached to studs of a timber-frame building or other structure.
Supporting structure	A structure, structural member or foundation that supports a scaffold.
Suspension gear	An assembly that joins a cradle to a suspension rig.
Suspension-mounted hoist	A scaffolding hoist that is mounted on a suspension rig or trolley track, with one end of the rope attached to the cradle.
Suspension point	A point where a suspension rope is connected to a suspension rig.
Suspension rig	A portion of a structure (including a trolley track) that is mounted at a higher level than that of a cradle and supports and positions the cradle (see Figure 11.5.1).
Suspension rope	A rope carrying the weight of a cradle and supporting an imposed load.
Swing stage	A suspended scaffold with a single row of suspension ropes (see Figure 11.1(B)).
Swivel coupler	A coupler for connecting two tubes at any angle (see Figure B1).
Swivel-finial coupler	A finial coupler that is able to join two scaffold tubes at variable angles.
Tank-bracket scaffold	A bracket scaffold of which the brackets are able to be attached to the face of a metal tank, duct, boiler, structural beam or similar.
Three-to-two	A reduction coupler.
Through tie	A tie assembly that is positively fixed to both sides of an opening through a building or structure (see Figure 8.5).
Thrust	A spur.
Tie	A member or assembly of members used to tie a scaffold to a supporting structure (see Figure 8.5).
Tie bar	A member able to be fixed to the cantilevered end of a pair of platform brackets, to prevent the brackets from spreading and the planks from becoming dislodged.
Toeboard	A scaffold plank or purpose-designed component fixed on edge at the edge of a platform, to prevent material falling from the platform.
Toeboard clip	A fitting used to secure a toeboard to a standard or stanchion.

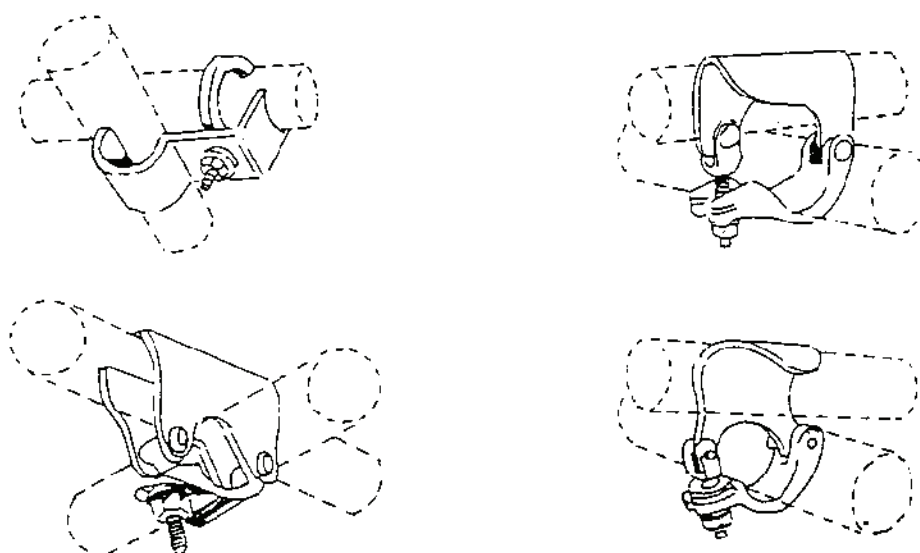
Term	Definition
Tower-frame scaffold	A prefabricated scaffold consisting of fabricated units and only able to be erected in the form of a tower (see Figure 10.3.8).
Tower scaffold	A scaffold of one bay (see Figure 10.3.8).
Transom	A horizontal structural member transversely spanning an independent scaffold between standards.
Transverse brace	A brace in a plane that is vertical and at right angles to the building or structure.
Travelling cradle	A cradle capable of being moved horizontally (see Figure 11.1(B)).
Traversing cradle	A travelling cradle.
Traversing rope	A rigged rope that traverses a cradle, but does not carry the weight of the cradle.
Traversing suspension rig	A suspension rig mounted on wheels or castors and supporting a cradle.
Trestle-ladder scaffold	A scaffold consisting of trestle ladders supporting scaffold planks.
Trestle scaffold	A scaffold consisting of trestles and planks (see Figure 10.1(A)(c)).
Trolley	A wheeled mechanism that is able to support a serial hoist and is capable of travelling along a suspended track.
Trolley track	A suspended rail that supports and guides trolleys in traversing installations.
Tube-and-coupler covered way	An independent tube-and-coupler scaffold that is primarily intended to provide overhead protection.
Tube-and-coupler scaffolding	A scaffold of which the standards, ledgers, braces and ties are circular tubes that are joined together by means of purpose-designed couplers (see Figure 10.5).
U-head	A fork head.
Unit beam	A scaffold beam that incorporates diagonal chord stiffeners.
Work cage	A cradle that is supported by a single suspension rope.
Working load limit	The maximum working load that may be applied to any component or system, under general conditions of use.
Working platform	A platform that is intended to support persons, materials and equipment.
Yoke tie	A box tie.



(a) Right angle couplers

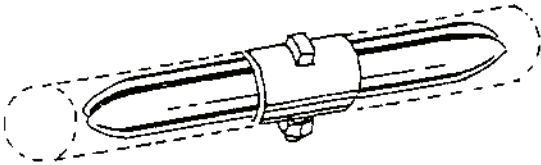


(b) Swivel couplers

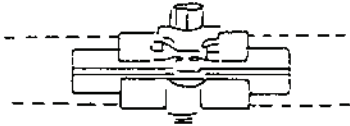


(c) Putlog couplers

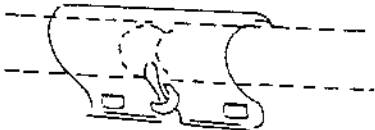
FIGURE B1 (in part) TYPICAL COUPLERS AND ACCESSORIES



(i) Expanding joint pin type end-to-end coupler

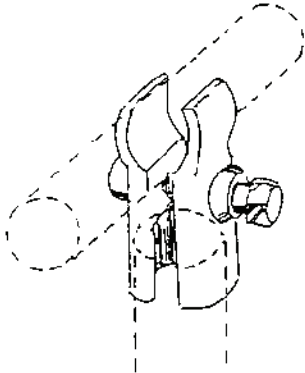


(ii) Overcoat type end-to-end coupler

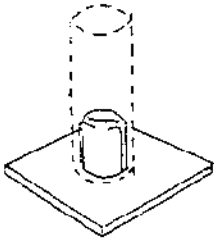


(iii) Sleeve type end-to-end coupler

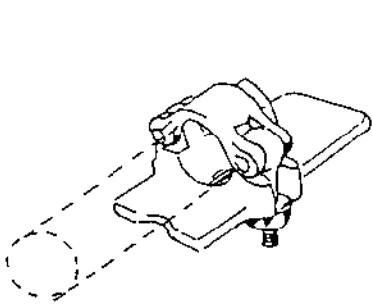
(d) End-to-end couplers



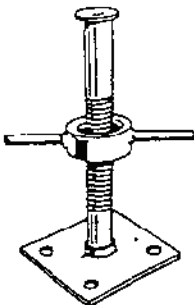
(e) Finial coupler



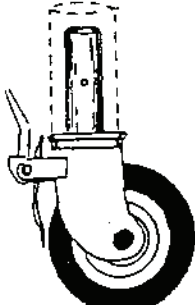
(f) Baseplate



(g) Putlog blade



(h) Adjustable baseplate



(i) Castor

FIGURE B1 (in part) TYPICAL COUPLERS AND ACCESSORIES

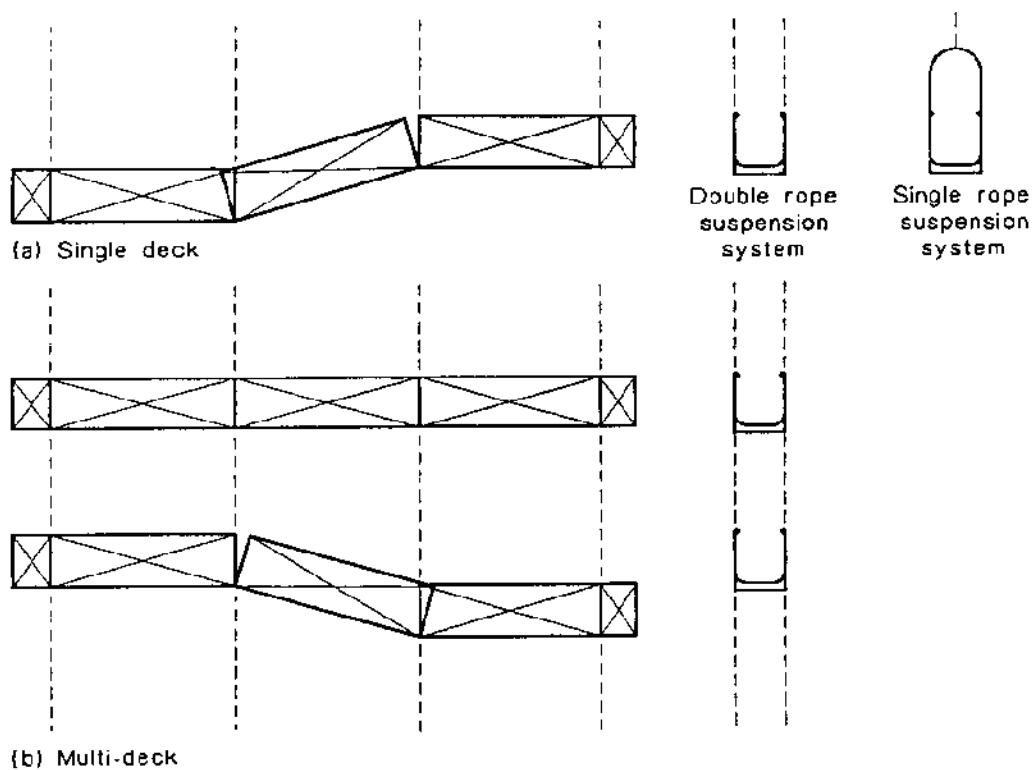


FIGURE B2 TYPICAL ARTICULATED CRADLE

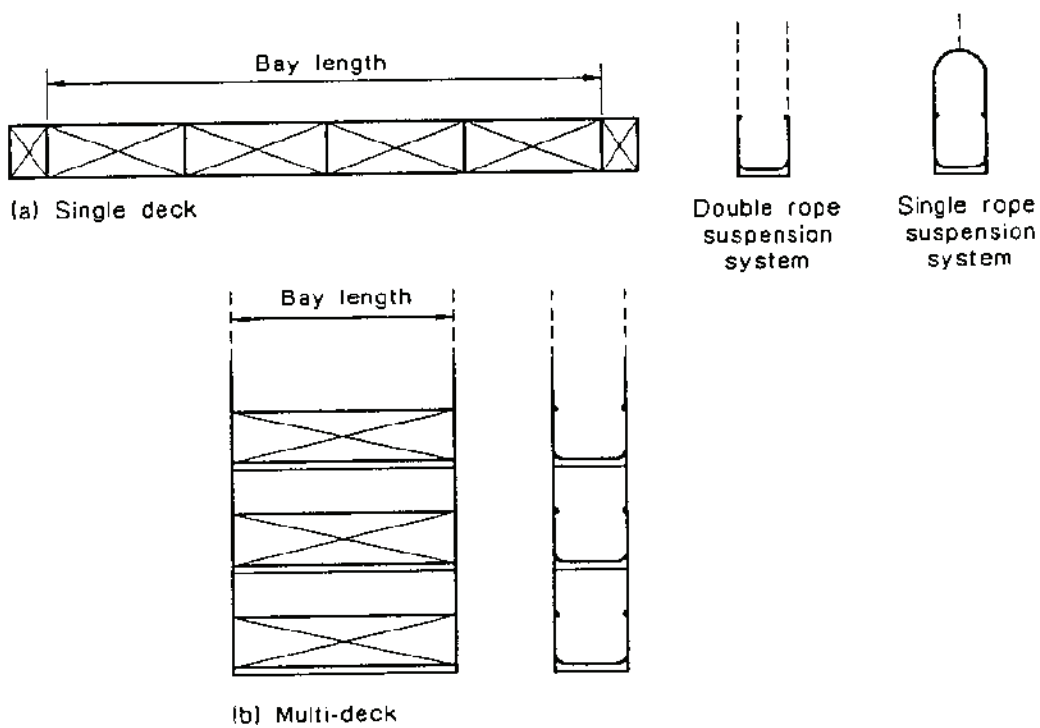


FIGURE B3 TYPICAL INDIVIDUAL CRADLE

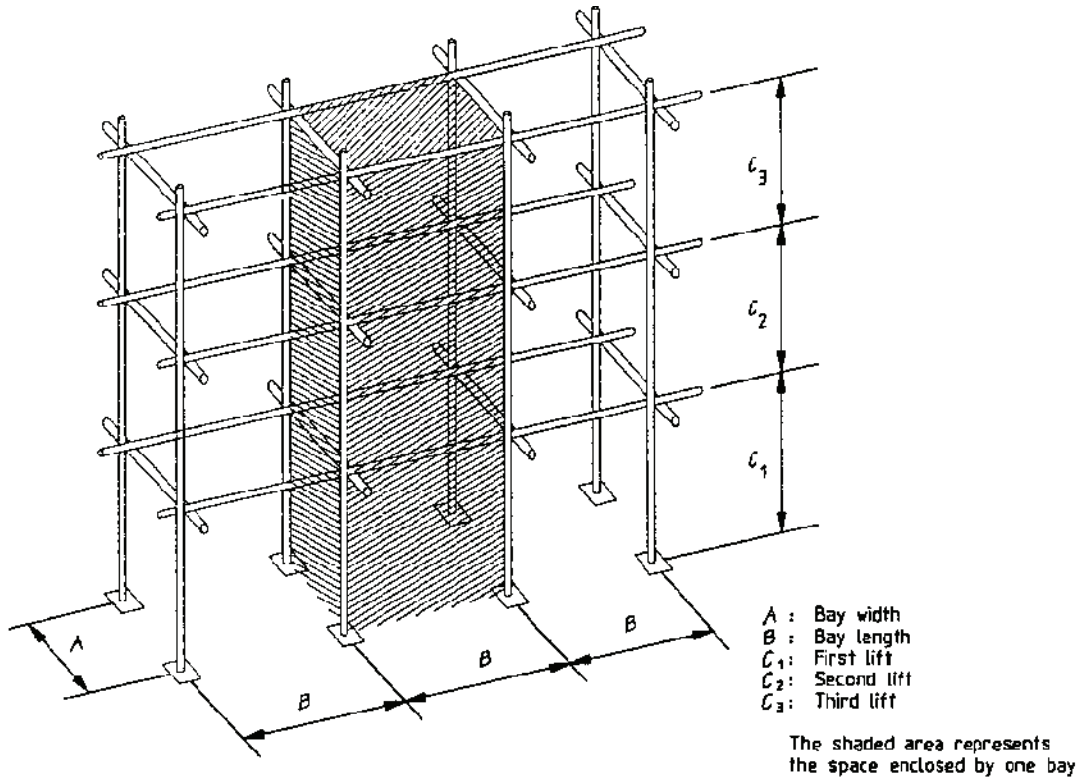
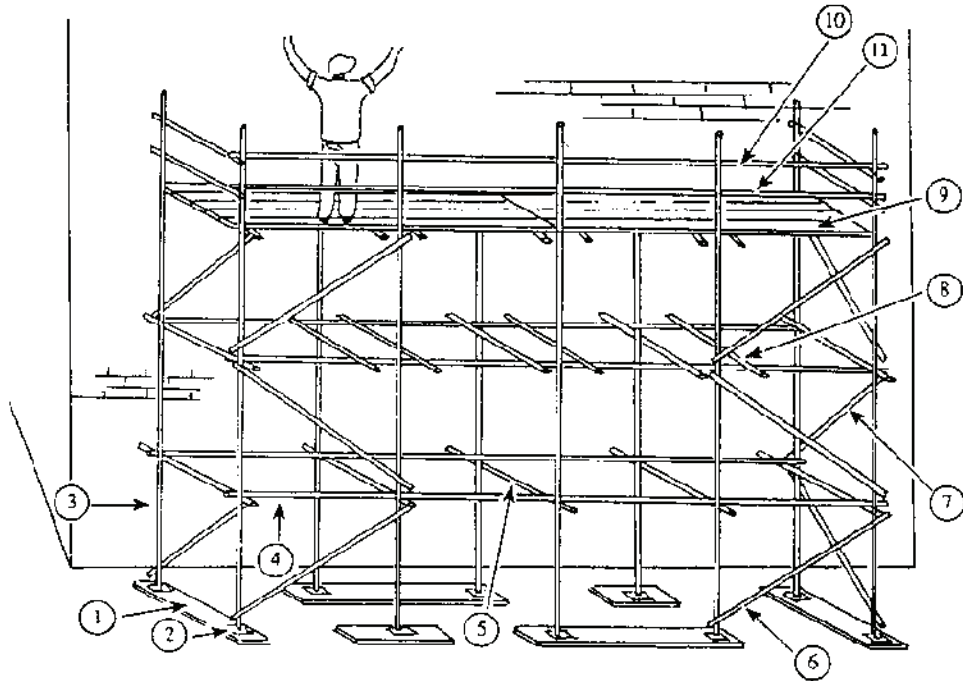


FIGURE B4 MAIN DIMENSIONS OF A SCAFFOLD



- LEGEND
- | | | |
|-------------|----------------------|-------------------|
| 1 Soleplate | 5 Transom | 9 Working station |
| 2 Baseplate | 6 Longitudinal brace | 10 Guard rail |
| 3 Standard | 7 Traverse brace | 11 Mid rail |
| 4 Ledger | 8 Putlog | |

NOTE: Ties, toeboards and access omitted for clarity.

FIGURE B5 EXAMPLE OF AN INDEPENDENT SCAFFOLD

APPENDIX C

SAMPLE QUESTIONS FOR AUSTRALIAN EXAMINATIONS

(Informative)

C1 INTRODUCTION The questions in this Appendix are typical of those set in Australia for scaffolders' certificate exams and are grouped into basic scaffolding, intermediate scaffolding and advanced scaffolding. At the end of each section there is a question requiring calculations. The questions dealing with tube-and-coupler scaffolding are based on the information given in AS 1576.3 Supplement 1. The other questions are based on the information given herein.

C2 BASIC SCAFFOLDING

Certification

- 1 At what height is a scaffolding certificate of competency needed?
- 2 Is a person with a Basic Scaffolding Certificate allowed to construct a tower frame scaffold with outriggers?
- 3 Is a person with a Basic Scaffolding Certificate allowed to construct a swing stage?
- 4 Is a person with a Basic Scaffolding Certificate allowed to install a personnel and materials hoist?

Site hazards

- 5 How close to live unprotected powerlines are you permitted to construct a metal scaffold?
- 6 What could happen if the tie tubes on a scaffold are sticking out too far when a crane is operating?
- 7 What is the danger where a metal scaffold is constructed in a flour mill?

Tools and equipment for scaffolding

- 8 How far above the maximum nut extension must the spindle of an adjustable baseplate extend?
- 9 What is the minimum outside diameter of a common scaffold tube (to the nearest mm)?
- 10 What is the minimum thickness of a hardwood solid-timber scaffold plank?
- 11 What is the maximum load you would lift with a gin wheel?
- 12 Would you suspend a gin wheel from a right-angle coupler?

General scaffold requirements

- 13 What is the maximum load in each bay of a light-duty working platform?
- 14 What is the maximum working capacity of an adjustable baseplate?
- 15 What minimum width of timber would you use as a soleplate?
- 16 What is the minimum width of a medium duty working platform?
- 17 Can planks with different thicknesses be used to deck out a working platform?
- 18 What must be provided between the guardrail and the toe board to complete a platform's edge protection?
- 19 What type of ladder cannot be used for access to a scaffold?

Particular scaffold requirements

- 20 Do castors for mobile scaffolds need wheel locks?
- 21 Would you fix platform brackets on the inside of the scaffold or on the outside of the scaffold?
- 22 What would you do to stop the movement of planks on a crane-lifted shutter bracket scaffold?
- 23 Would you mix components of two prefabricated systems in the one scaffold without a supplier's or engineer's consent?
- 24 If a tie tube was fixed to a wedge-type modular ledger, how would you fix the ledger against uplift?
- 25 How high would you build an unsheeted freestanding steel frame scaffold?
- 26 How is the ladder opening in a working platform of a tower-frame scaffold usually protected?

Associated equipment

- 27 What is the maximum mesh size of a safety net?
- 28 What minimum clearance would you ensure below a safety net?
- 29 What is the minimum and maximum horizontal clearance between the moving platform of a cantilevered hoist and any landing or floor?
- 30 How high would you freestand the tower of a cantilevered platform hoist above its last tie?

Calculation

A tower scaffold with a dead load of 400 kg and one medium duty working platform is set up on 225 mm wide soleplates on a floor with a bearing capacity of 1.5 t per square metre.

What is the minimum length of soleplate needed under each standard?

C3 INTERMEDIATE SCAFFOLDING**Certification**

- 1 Is a person with an Intermediate Scaffolding Certificate allowed to construct a personnel and materials hoist?
- 2 Is a person with an Intermediate Scaffolding Certificate allowed to construct a barrow ramp?
- 3 Is a person with an Intermediate Scaffolding Certificate allowed to construct a mast climber?
- 4 Is a person with an Intermediate Scaffolding Certificate allowed to construct a cantilevered or spurred scaffold?

Tube-and-coupler scaffolds (to the specifications given in AS 1576.3 Supplement 1)

- 5 What is the maximum specified height for a tube-and-coupler scaffold?
- 6 What is the maximum bay length for a light duty independent scaffold?
- 7 What is the maximum lift height for a normal independent scaffold?
- 8 At what lift is the first level of ties fixed on a single-pole scaffold?
- 9 Where double standards are used, what is the maximum first lift height?
- 10 How close to the castors is the first lift of ledgers and transoms on a mobile scaffold?

Particular scaffolds

- 11 What is the maximum allowable slope of a working platform?
- 12 How many lifts would you support with a single set of spurs?
- 13 What type of baseplate would you use to fix the standards of a cantilever scaffold to the needles?
- 14 When a spur is in compression, what would be its maximum length between node points?
- 15 What is the maximum angle at which you would fix a spur?

Calculations

If you are constructing an independent heavy duty scaffold of 6 bays and 4 lifts according to the specifications given in AS 1576.3 Supplement 1, what lengths of tube would you select to start off—

- 16 the first standard?
- 17 the second standard?
- 18 the first ledger?
- 19 the second ledger?

C4 ADVANCED SCAFFOLDING**Hung scaffolds**

- 1 Would you use open-ended hook rods to construct a hung scaffold?
- 2 What is the minimum working load limit for beam clamps, trolleys and shackles used to support a hung scaffold?
- 3 How would you stop shackle pins from unwinding?

Suspended scaffolds

- 4 What is the minimum rated working load for a trolley used to support a suspension rope?
- 5 What is the minimum width of a swing stage cradle?
- 6 What is the minimum height of a stabilizing sheave above the platform of a work cage?
- 7 What is the maximum rope tension allowed on a shackle used to support a suspension rope?
- 8 With a fully-loaded drum-type scaffolding hoist, how far should the drum flange extend beyond the built-up rope?
- 9 When a climber-type scaffolding hoist is at its lowest point, what is the minimum length of spare rope?
- 10 What type of scaffolding hoist must be fitted with a load-limiting device?

Calculations

A swing stage cradle of one bay is set up with two electric scaffolding hoists.

Each hoist has a rated working load of 650 kg using 25 m of suspension rope weighing 32 kg per 100 m. The cantilever needles each have an inboard of 4 m and an outboard of 900 mm. Each counterweight weighs 20 kg.

How many counterweights are required on the inboard end of each needle?

APPENDIX D

COMMON BENDS AND HITCHES FOR FIBRE ROPE

(Informative)

Figure D1 shows some common bends and hitches that are used on fibre ropes.

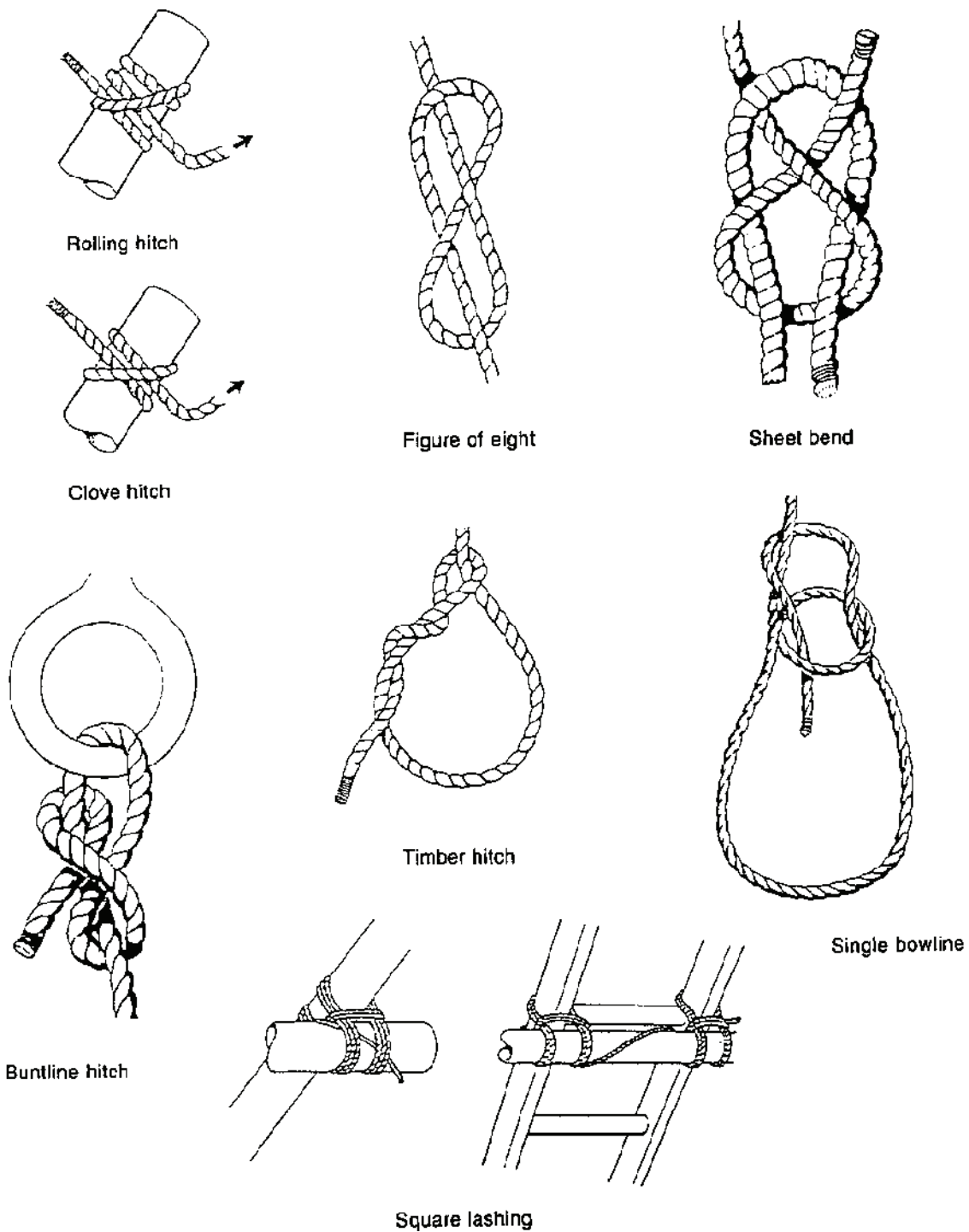


FIGURE D1 COMMON BENDS AND HITCHES FOR FIBRE ROPE

APPENDIX E
GENERAL LOADING INFORMATION
(Informative)

E1 LOADS ON STEEL TUBES

Table E1(A) gives the loads that are permitted to be applied to steel scaffold tube, with a 48.3 mm OD $\times$ 4.0 mm wall thickness complying with AS 1576.3 Supplement 1, and takes into account a 15 percent reduction for reuse and the self-weight of the tube.

TABLE E1(A)
LOADS PERMITTED ON STEEL SCAFFOLD TUBE
WITH A WALL THICKNESS OF 4 mm

Clear span	Tube as a beam				Tube as a strut	
	Simply-supported beam		Cantilever beam*		Strut length	Concentric load
	Point load	UDL†	Point load	UDL†		
mm	kg	kg	kg	kg	mm	kg
225	1 090	2 181	272	545	300	4 780
300	817	1 635	204	408		
450‡	544	1 088	135	271		
600	408	814	101	202	600	4 620
675‡	362	726			900	4 349
900‡	271	543				
1 125‡	216	432				
1 200	202	406			1 200	3 943
1 350‡	189	359			1 500	3 420
1 500	160	322				
1 575‡	153	303				
1 800‡	132	264			1 800	2 809
2 025‡	117	233			2 100	2 234
2 100	112	223				
2 250‡	105	206				
2 400	97	193			2 400	1 780
2 700‡	86	170			2 700	1 437
3 000	76	152				

* Cantilever spans exceeding 600 mm not recommended.

† UDL = uniformly distributed load.

‡ A multiple of 225 mm.

Table E1(B) gives the loads that are permitted in New Zealand to be applied to galvanized steel scaffold tubes 48.3 mm OD $\times$ 3.2 mm wall thickness complying with BS 6323.1 and takes into account a 15 percent reduction for reuse and the self-weight of the tube.

TABLE E1(B)
LOADS PERMITTED IN NEW ZEALAND ON GALVANIZED
STEEL SCAFFOLD TUBES WITH A WALL
THICKNESS OF 3.2 mm

Clear span	Tube as beam				Tube as strut	
	Simply-supported beam		Cantilever beam*		Strut length	Concentric load
	Point load	UDL†	Point load	UDL†		
mm	kg	kg	kg	kg	mm	kg
225	917	1 835	229	459		
300	687	1 373	172	343	300	3 870
450‡	456	914	114	228		
600	343	684	86	171	600	3 740
675‡	304	610				
900‡	230	456			900	3 522
1 125‡	181	363				
1 200	170	341			1 200	3 190
1 350‡	159	301				
1 500	134	270			1 500	2 770
1 575‡	128	254				
1 800‡	110	221			1 800	2 275
2 025‡	98	196				
2 100	94	187			2 100	1 809
2 250‡	88	173				
2 400	82	162			240	1 442
2 700‡	72	143			2 700	1 164
3 000	64	127				

* Cantilever spans exceeding 600 mm are not recommended.

† UDL = uniformly distributed load.

‡ A multiple of 225 mm.

E2 LOADS ON ALUMINIUM TUBES

Table E2 gives the loads that are permitted to be applied to aluminium scaffold tubes 48.4 mm O.D. × 4.45 mm wall thickness complying with AS 1576.3 Supplement 1, and takes into account a 15 percent reduction for reuse, the self-weight of the tube and limits the deflection of beams to 1/150 of the span.

TABLE E2
LOADS PERMITTED ON ALUMINIUM
SCAFFOLD TUBES

Clear span	Tube as a beam				Tube as a strut	
	Simply-supported beam		Cantilever beam*		Strut length	Concentric load
	Point load	UDL†	Point load	UDL†		
mm	kg	kg	kg	kg	mm	kg
225	1 633	3 266	354	816	225	6 750
300	1 224	2 449	199	531	300	6 527
450‡	816	1 632	88	236	450	6 082
600	612	1 224			600	5 637
675‡	544	1 007			675	5 414
900‡	354	566			900	4 746
1 125‡	226	361			1 125	3 595
1 200	198	317			1 200	3 159
1 350‡	156	250			1 350	2 496
1 500	126	202			1 500	2 022
1 575‡	114	183			1 575	1 834
1 800‡	87	139			1 800	1 404
2 025‡	68	109			2 025	1 109
2 100	63	101			2 100	1 032
2 250‡	55	88			2 250	899
2 400	48	76			2 400	790
2 700‡	37	59			2 700	624
3 000	29	47			3 000	505

* Cantilever spans exceeding 450 mm are not recommended.

† UDL = uniformly distributed load.

‡ A multiple of 225 mm.

E3 PERMISSIBLE LOADS ON COMMON COUPLERS AND ACCESSORIES

Table E3 gives permissible loads on commonly used couplers and accessories complying with the minimum requirements of AS 1576.2, unless otherwise specified by the supplier.

TABLE E3
PERMISSIBLE LOADS ON COMMON COUPLERS
AND ACCESSORIES

Type of coupler or accessory	Type of load	Permissible load kg
Right angle coupler	Slip along a tube	630
Swivel coupler	Slip along a tube	630
Putlog coupler	Force to pull tube axially out of the coupler	60
Putlog blades	Shear	230
Adjustable baseplate	Axial load	3 000

E4 MASS OF COMMON SCAFFOLDING MATERIALS

Table E4 gives the weights of commonly used scaffolding materials.

TABLE E4
MASS OF COMMON SCAFFOLDING MATERIALS

Material	Mass
Scaffold tubes:	
black steel 48.3 O.D. × 4.00 mm	4.37 kg/m
galvanized steel 48.3 O.D. × 4.00 mm	4.52 kg/m
black steel 48.3 O.D. × 4.88 mm	5.24 kg/m
galvanized steel 48.3 O.D. × 4.88 mm	5.38 kg/m
galvanized steel 48.3 O.D. × 3.2 mm	3.56 kg/m
aluminium 48.4 O.D. × 4.47 mm	1.67 kg/m
Solid timber scaffold planks:	
225 mm × 32 mm hardwood	8 kg/m
225 mm × 38 mm hardwood	10 kg/m
300 mm × 38 mm hardwood	12 kg/m
225 mm × 38 mm oregon	6 kg/m
225 mm × 50 mm oregon	8 kg/m
225 mm × 63 mm oregon	10 kg/m
300 mm × 50 mm oregon	10 kg/m
Flexible steel wire ropes:	
8 mm diameter	26 kg per 100 m
9 mm diameter	31 kg per 100 m
13 mm diameter	64 kg per 100 m
16 mm diameter	96 kg per 100 m
19 mm diameter	135 kg per 100 m
23 mm diameter	185 kg per 100 m
Couplers	1.5 kg each (average)

E5 MASS FOR PERSONS AND GENERAL MATERIAL

Table E5 gives approximate masses of persons and general materials likely to be supported from scaffolds.

TABLE E5
MASS OF PERSONS AND GENERAL MATERIALS

Item		Approximate mass
Bricks		4.0 kg per brick
Cement		40 kg per bag
Concrete block:	400 × 200 × 200 mm hollow	19 kg per block
	400 × 200 × 150 mm hollow	16 kg per block
	400 × 200 × 100 mm hollow	13 kg per block
	400 × 200 × 100 mm solid	16 kg per block
Concrete:	ready mixed wet	2550 kg/m ³
	in wheelbarrow	140 kg
Drums	empty 200 litre	13 kg
Marble		2700 kg/m ³
Paint	(except red and white lead) 5 litre	10.5 kg
Person:	singly	80 kg
	plus wheelbarrow with concrete	220 kg
Plaster:	fibrous	1.6 kg/m ²
	bag	38 kg
Plywood	17 mm	10 kg/m ²
Sand		2000 kg/m ³
Shale		2600 kg/m ³
Steel rods:	6.5 mm diameter	25 kg per 100 m
	10 mm diameter	67 kg per 100 m
	12 mm diameter	89 kg per 100 m
	16 mm diameter	158 kg per 100 m
	20 mm diameter	247 kg per 100 m
	25 mm diameter	358 kg per 100 m
Tiles:	terra cotta	3.5 kg per tile
	concrete	3.75 kg per tile
Timber:	ironbark	1400 kg/m ³
	other hardwoods	1100 kg/m ³
	softwoods	640 kg/m ³
Water (excluding container)		1 kg/L

APPENDIX F

INDUSTRIAL SAFETY NETS

(Informative)

F1 INTRODUCTION

Australian basic scaffolding certificates and the basic rigging certificates entitle a holder to install industrial safety nets even though they are not scaffolds. For this reason, this Appendix has been included.

Industrial safety nets are sometimes used as an effective means of fall protection for persons working at heights, particularly where it is not practicable to provide scaffolds or temporary guardrailings.

Industrial safety nets do not apply considerable loads to the supporting structure, unless they are arresting a fall. Also, they have a distinct advantage over individual fall arrest systems, such as safety harnesses, because they allow unrestricted movement for workers and because they do not rely on workers 'doing the right thing' by always keeping their harnesses correctly anchored.

Where combined with overlay nets of finer mesh, industrial safety nets can also be used as an inexpensive means of containing falling debris.

F2 DESIGN

As a general rule, safety nets are required by regulatory authorities to comply with the design, manufacturing and test requirements of BS 3913.

They are generally manufactured from synthetic fibre knotted lines with a 100 mm mesh size. These lines are attached to perimeter cords. Safety nets are usually available in sizes of 4 m × 3 m or larger, and the nets can be joined to cover larger areas.

The label attached to each net will state a maximum fall distance for which the net has been designed. This is normally 1 m or 6 m.

F3 INSTALLATION

Although safety nets can be attached to or supported from scaffolds, they are often secured directly to the structural framework of buildings, bridges, towers and similar structures.

Prior to the installation of a safety net, its intended configuration, its method of attachment and the strength of the supporting structure should be verified as adequate by a competent person, such as an engineer experienced in structural design. It is recommended that this verification be given in writing and retained on site until the net has been dismantled.

The installation design should include detailed information regarding the exact positioning of the net, the fixing and tensioning methods and the erection and dismantling procedure. In particular, where nets are to be cantilevered from scaffolds, the scaffold must be designed for the additional loads and additional ties to the scaffold's supporting structure may be required (see Figure F3(A)).

Where possible, fabrication of the net assembly should be carried out on the ground or on an adjacent floor and lifted into place with a crane, hoist or purchase.

Nets should be installed as close as possible to the working levels and in any case not below the maximum fall distance stated on the label of the net.

The gap between a net and the building or structure should be as small as practicable, but never greater than 200 mm.

Nets should project beyond the outermost working point by a horizontal distance of at least the sum of 2 m plus two-fifths of the maximum fall height.

It is important that nets are sited so that a person who has fallen can be quickly rescued. For example, nets erected adjacent to a working platform, floor or other access point are easily accessible. Where this is not possible, it may be necessary to cut the net, to quickly and safely retrieve a fallen person. Where there is any possibility of debris falling into a net, the debris should be able to be removed without the need to walk on the net.

Because they will stretch when a person falls into them, it is essential that sufficient clearance is maintained at all times below the net. This clearance should be not less than the greater of 2 m and two-thirds of the length of the shortest side (see Figure F3(B)).

Nets should not be stretched taut when erected. They should have an unloaded sag of between a quarter and a fifth of the length of the shortest side (see Figure F3(B)).

Nets should be securely attached to the supporting framework using tie cords, hooks, rings or thimbles equally spaced at intervals of not more than 750 mm along each side and at the corners. They should be fixed to the border cords and adjacent mesh cords of the net (see Figure F3(C)).

Where cords are wrapped around sharp edges, they should be packed to prevent damage.

F4 USE AND MAINTENANCE

A net that is used for the safety of persons should never be subjected to a loading test. The net should incorporate test cords that can be removed and tested in accordance with BS 3913. After a net is put into use, these tests should be undertaken at regular intervals as dictated by site conditions, subject to a maximum interval of three months. The test cords should not be used for any other purpose and should remain fixed to the net until they are required for testing. A record of the test results should be kept.

Nets should be thoroughly examined on both sides by a competent person immediately before they are erected. Where a net is spread out for examination, particular care should be taken to avoid it coming into contact with cement mortar or other corrosive substances.

The net, the supporting framework and the anchorages should be inspected by a competent person immediately following erection, at weekly intervals thereafter, and immediately following any incident that may affect the strength of the net.

It is important that nets are kept free of any debris that may cause injury to persons falling into them.

The following situations should be avoided wherever possible:

- Dragging the net over rough surfaces.
- Allowing cords to come into contact with sharp edges.
- Stacking materials on the net.
- Allowing debris to accumulate in the net.
- Jumping on the net.
- Throwing objects into the net.
- Allowing sparks or flames from welding or oxy-cutting, hot gases from blow torches or hot ashes from chimneys or furnaces to damage the net.
- Allowing chemicals to degrade the net.
- Allowing moving loads to strike the supporting framework.
- Interfering with any part of a net and its assembly.

Expert advice from the manufacturer, the supplier or a competent person should be sought where there is any doubt regarding the suitability of the net following contamination or severe shock loading. Expert advice should also be sought on the serviceability of any net that has been in use for two years or more.

Damaged nets should be repaired only by specialists.

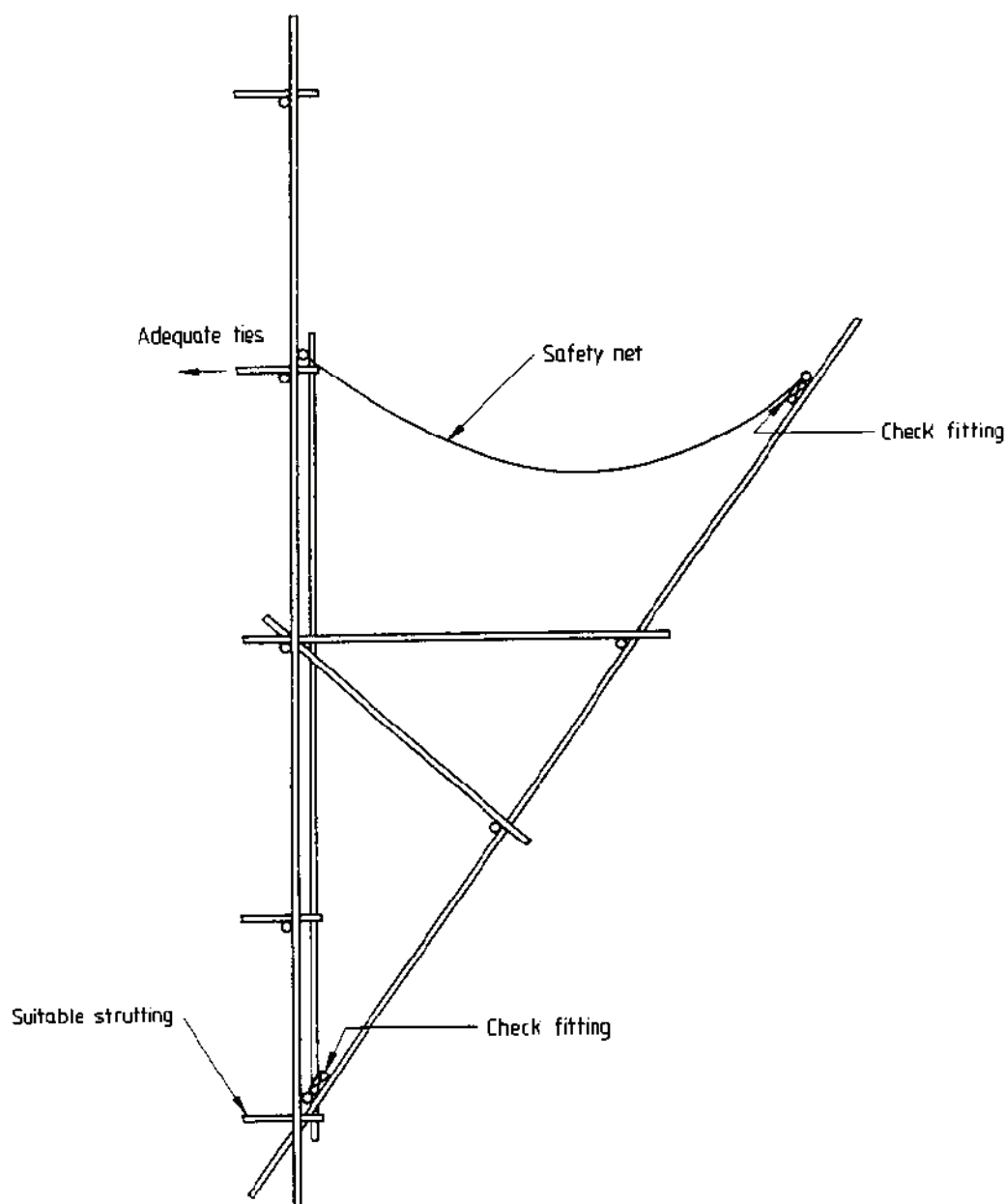
While they are not in use, safety nets should be stored under cover where they are protected from the weather and strong sunlight.

F5 FURTHER INFORMATION

Persons engaged in the design, manufacture, testing, importation or supply of industrial safety nets should consult BS 3913.

Persons engaged in the design of safety net installation and persons responsible for the inspection, maintenance and storage of safety nets should consult BS 8093.

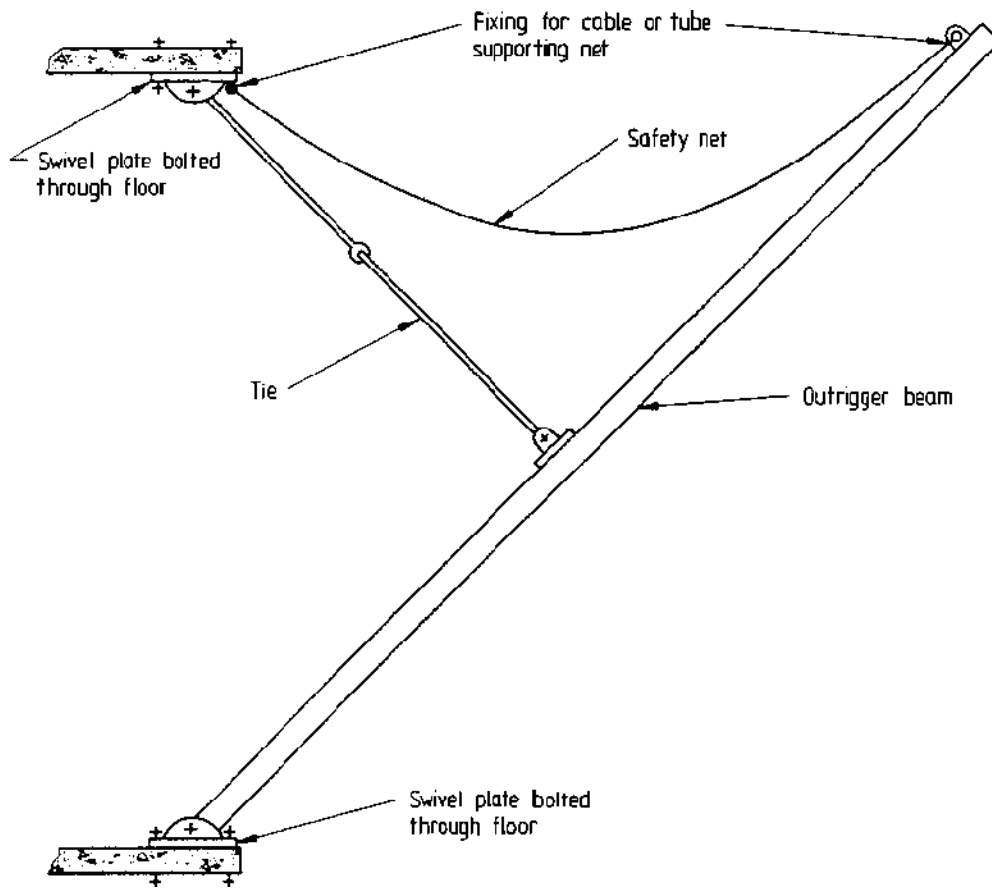
Both these British Standards are available from Standards Australia and Standards New Zealand.



(a) Attached to scaffolding

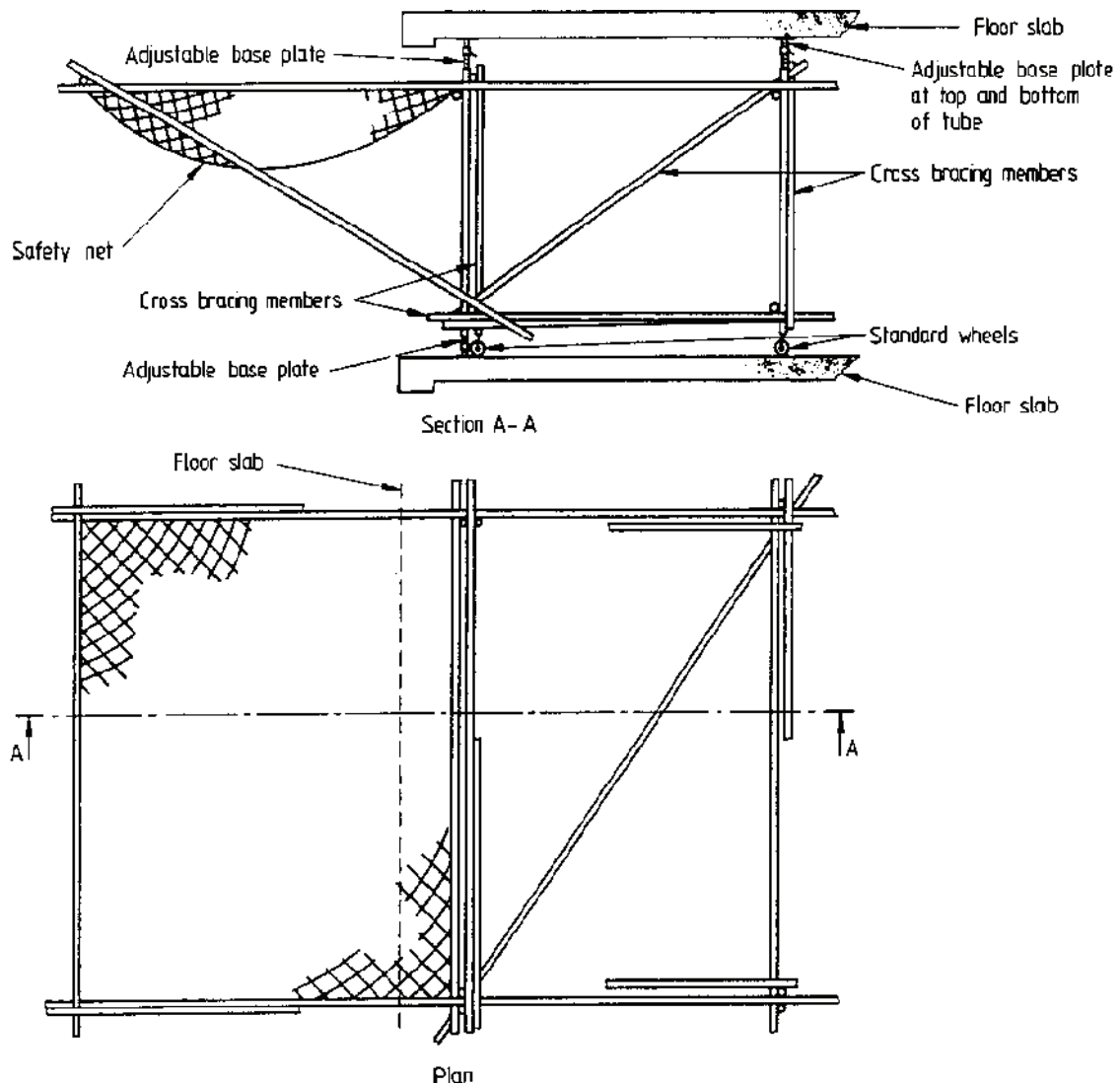
NOTE: The net and its supporting framework are attached to the scaffolding structure which should be securely tied back to the building. This arrangement may also be suitable for attachment direct to the building.

FIGURE F3(A) (in part) TYPICAL ARRANGEMENT OF
OUTRIGGER OR PERIMETER NETS



(b) Attached to the structure

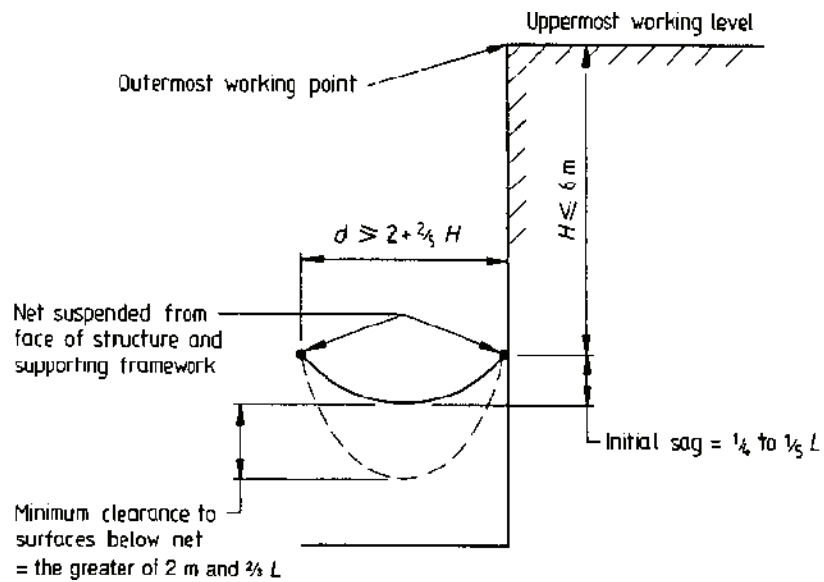
FIGURE F3(A) (in part) TYPICAL ARRANGEMENT OF OUTRIGGERED OR PERIMETER NETS



(c) Mobile net assembly

NOTE: Extreme care should be taken when securing the net assembly to the structure, to ensure that it cannot be displaced during use. Due regard should be given to the need to take additional precautions, to prevent the net assembly from running over the edge of the floor when moving from one position to another.

FIGURE F3(A) (in part) TYPICAL ARRANGEMENT OF
OUTRIGGERED OR PERIMETER NETS

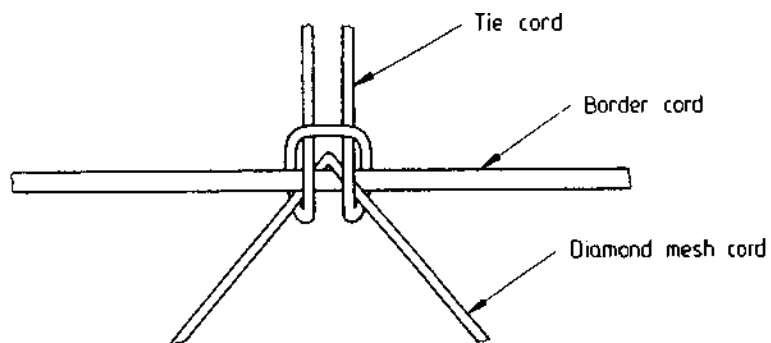


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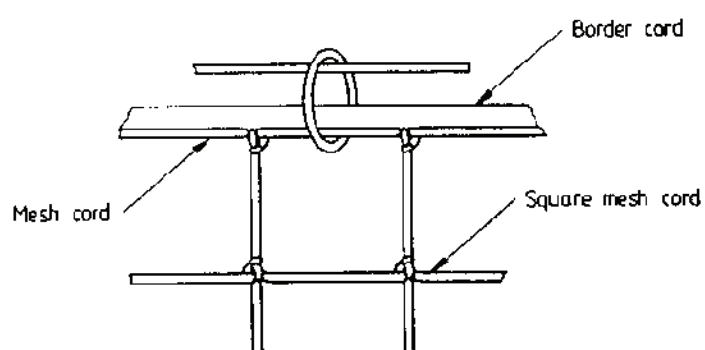
- H is the vertical distance in metres between net and uppermost working level above net for which the net has been designed (see Paragraph F3).
- L is the length of shortest side of net in metres, which should be not less than 2 m.
- d is the horizontal projection from outermost working point in metres.

NOTE: Supporting framework not shown.

FIGURE F3(B) SITING OF OUTRIGGERED NETS



(a) Diamond mesh cord



(b) Square mesh cord

FIGURE F3(C) ATTACHMENT OF SAFETY NET USING A TIE CORD

APPENDIX G

INSPECTION OF BELTS AND HARNESSES—CHECK LIST

(Informative)

This Appendix is copied from Appendix A of AS 2626—1983 and lists items of safety belts and harnesses to be inspected and those conditions/faults to be checked.

Component	Condition/fault to be checked
Webbing	Cuts or tears Abrasion damage, especially where there is contact with hardware Excessive stretching Damage due to contact with heat, corrosives or solvents Deterioration due to rotting, mildew or ultraviolet exposure
Snap hooks	Distortion of hook or latch Cracks or forging folds Wear at swivels and latch pivot pin Open rollers Free movement of the latch over its full travel Broken, weak or misplaced latch springs (compare if possible with a new snap hook) Free from dirt or other obstructions (e.g. rust)
D-rings	Excessive 'vertical' movement of the straight portion of the D-ring at its attachment point on to the belt, so that the corners between the straight and curved sections of the D become completely exposed NOTE: Excessive vertical movement of the D-ring in its mounting can allow the nose of larger snap hooks to become lodged behind the straight portion of the D, in which position the snap hook can often accidentally 'roll out' of the D under load. Cracks, especially at the intersection of the straight and curved portions Distortion or other physical damage of the D-ring Excessive loss of cross-section due to wear
Buckles and adjusters	Distortion or other physical damage Cracks and forging laps where applicable Bent tongues Open rollers
Sewing	Broken, cut or worn threads Damage or weakening of threads due to contact with heat, corrosives, solvents or mildew
Ropes	Cuts Abrasion or fraying Stretching Damage due to contact with heat, corrosives, solvents, etc Deterioration due to ultraviolet light or mildew Note: See also Appendix C
Chains	Physical damage Security of attachments to snap hooks, rings and similar components

APPENDIX H

INSPECTION OF SELF-LOCKING ANCHORAGES—CHECK LIST

(Informative)

This Appendix is copied from Appendix B of AS 2626—1983 and lists items of self-locking anchorages to be inspected and particular conditions/faults to be checked.

Component	Condition/fault to be checked
Rope (fully extend rewind drum anchorage)	Cuts Abrasion or fraying Stretching Damage due to contact with heat, corrosives or solvents Excessive dirt or grease impregnation NOTE: See also Appendix C With rewind anchorages, give a firm pull with the rope fully extended to check that the rope end is securely anchored to the drum
Anchorage body	• <i>Mounting ring:</i> Physical damage or wear, especially at any pivot points Cracks, especially in corners Mounting security • <i>Anchorage body proper:</i> Physical damage such as significant dents, distortion or corrosion As far as possible, but without dismantling, check for the entry of foreign bodies such as small stones Loose or missing screws, nuts or similar objects (external check only) Position of the clutch compression indicator button (fitted only to rewind drums with steel rope)
Locking mechanism and rope guides	Check externally visible rope guides for excessive wear or ridging Check that the rope-locking mechanism locks and holds securely when the rope is given a sharp tug Ensure that the rope runs freely through the anchorage with no tendency to stick or bind, and that on rewind drum anchorages the rope rewinds completely without loss of tension
Hardware	Examine the condition and locking action of any associated snap hooks or links

APPENDIX I

CANTILEVER BUILDERS' HOISTS

(Informative)

The erection and use of movable cantilever builders' hoists complying with AS 1418.7 (see Figure I1) should comply with the following:

- Ensure the hoist is set on an adequate foundation and maintained vertically.
- Needles and screw jacks should be suitably packed with timbers and tightened to maintain the tower in a vertical position.
- Towers should not freestand more than 3 m above the last tie.
- Towers should be braced irrespective of their height.
- Towers should be supported with a lateral brace at a height of not more than 6 m above the base and thereafter at levels of not more than 6 m apart.
- It may be necessary to brace towers at levels of less than 6 m apart, for rigidity or as specified.
- The horizontal clearance between the edge of the moving platform and any landing or floor should not be less than 25 mm nor more than 100 mm.
- The hoist and tower should be constructed, secured and maintained in position so that the platform does not foul any landing or other obstruction.
- The overrun clearance between the attachment of the hoist rope to the platform and the head sheave on the tower should be not less than 1.5 m, while the platform is at its highest working level. A top limit hoisting device that disengages the drive mechanism should be fitted at this level.
- Ensure the area at the base of the tower is effectively guarded with suitable material secured in position at a height of 1 m above the working platform and set back 800 mm from the edge of the platform.
- Effective overhead protection should be provided for the hoist driver.
- The working load limit should be clearly marked on the platform.
- A notice stating that persons are prohibited from riding on the platform should be clearly marked on the platform.

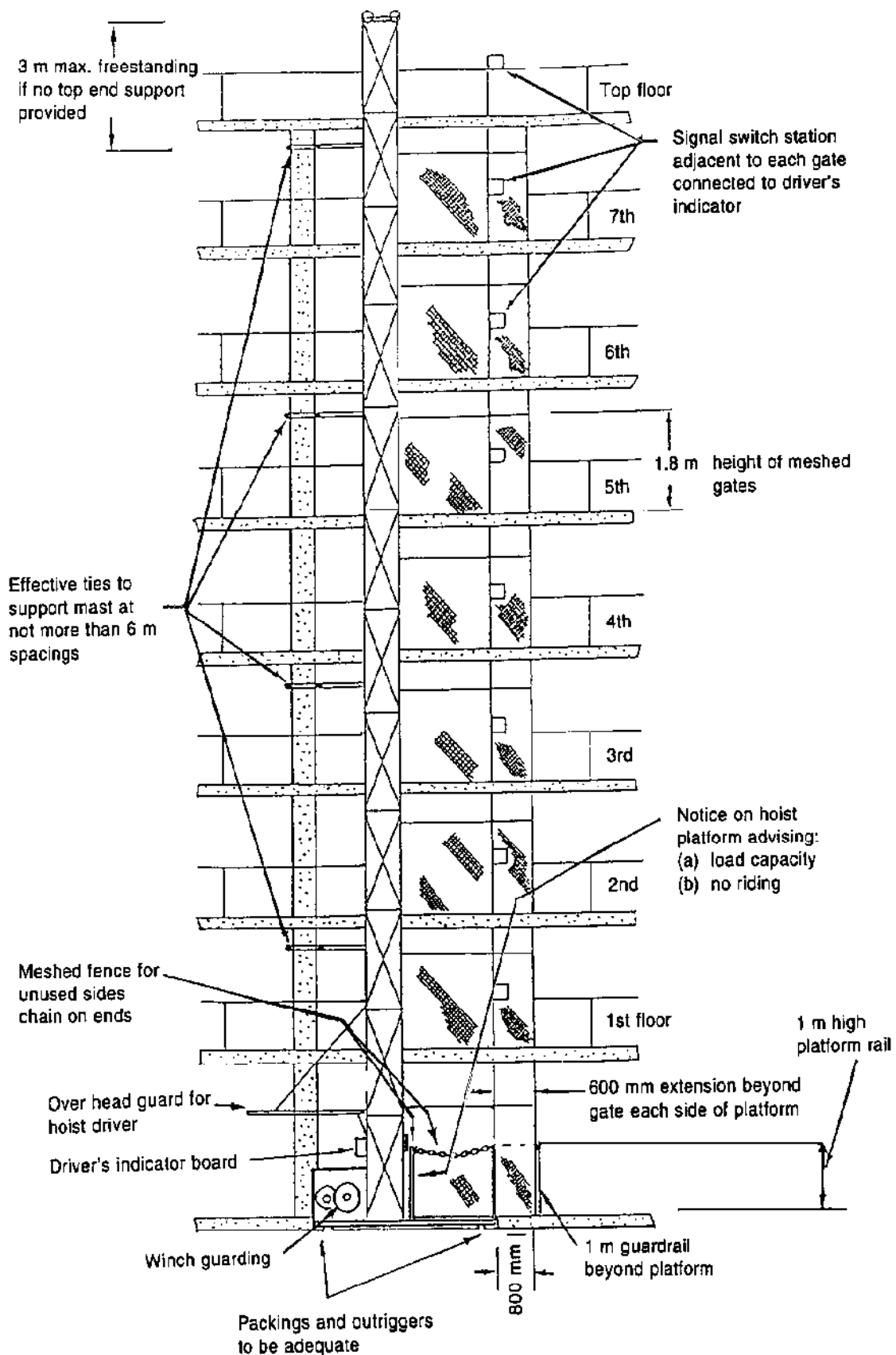


FIGURE 11 CANTILEVER BUILDERS' HOIST

APPENDIX J

INSPECTION CHECKLIST

(Informative)

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Question	Answer	Comments	Explanatory notes
1 Scaffold vicinity			
1.1 Has sufficient public protection been provided?			Where the scaffold is to be erected adjacent to or over public space or adjoining property, there may be a need to provide specific controls (e.g. hoardings, catch platforms, barricades, warning lights, illumination, shade cloth).
1.2 Have sufficient safeguards against electric powerlines been provided?			Power lines are a major hazard. Unless there is positive confirmation that de-energizing has occurred, scaffolds constructed of timber or other non-conductive materials should be not closer than 1.5 m to powerlines. Scaffolds with metallic components should be not closer than 4.0 m to powerlines. Refer to the supply authority for precise information.
1.3 Is there sufficient control over vehicle movement?			Uncontrolled vehicle movement in close proximity to a scaffold is a hazard that may lead to the scaffold or its supporting structure collapsing or becoming unstable. It may be necessary to provide protective buffers, re-route traffic or otherwise control the movement of vehicles.
1.4 Is there sufficient control over crane operation?			Where cranes operate in close proximity to a scaffold, there is a risk that loads may snag on the scaffold or endanger persons using the scaffold. Specific site procedures may need to be developed to minimize this risk.
1.5 Are there sufficient controls over the storage, handling and use of hazardous substances?			Where corrosive substances are to be used on the scaffold or in its vicinity, it may be necessary to develop specific procedures to minimize the risk.
2 Supporting structure			
2.1 Is the supporting structure in good condition?			Soil under the scaffold should be well consolidated and drained to prevent waterlogging. Floors, walls and other structures that provide support or transmit loads from the scaffold should be checked for soundness.
2.2 Does the supporting structure have adequate strength?			Some structures may require an engineer's certificate. Point loads transmitted by standards and other members such as ties should be estimated, taking into account dead loads, live loads and environmental loads. The supporting structure may need to be strengthened by back-propping or other suitable means.

(continued)

Question	Answer	Comments	Explanatory notes
2.3 Are there sufficient controls to prevent adverse deterioration of the supporting structure?			Where a scaffold is founded on soil, trenching or other excavation work in the vicinity may be a hazard unless control measures are implemented. Partial demolition of supporting structures also needs close control.
2.4 Are all measures to strengthen the supporting structure adequate?			Back-propping should comply with AS 3610. Controls should be implemented to ensure that any material or equipment used to strengthen the supporting structure is not inadvertently removed or damaged.
2.5 Is the risk of the supporting structure being overloaded from other sources adequately controlled?			The supporting structure may be subject to various loads from vehicles, stored materials, impact forces and build-up of debris. Where this could result in the overloading of the supporting structure, control measures should be implemented.
3 Soleplates and baseplates			
3.1 Are there sufficient soleplates?			Soleplates should be provided to distribute the load transmitted by standards unless the strength and rigidity of the supporting structure is sufficient to prevent any subsidence or failure under full load.
3.2 Are the soleplates of suitable material and in a serviceable condition?			Soleplates should be of robust material such as timber scaffold planks or plate steel. They should be able to distribute the loads transmitted by standards when the scaffold is under full load without damage or failure.
3.3 Are the soleplates secure?			Soleplates should be positioned to prevent them being dislodged or undermined.
3.4 Are there sufficient baseplates?			Unless standards are bearing on steel surfaces at least 6 mm thick or are mounted on castors, they should be fitted with metal baseplates to distribute the load.
3.5 Are the baseplates of the appropriate type?			Prefabricated scaffolds generally require adjustable baseplates to provide levelling. Standards founded on beams generally require u-heads to prevent dislodgment.
3.6 Are the baseplates serviceable and of suitable dimensions?			Baseplates that are bowed or bent will not provide an even bearing surface. To comply with AS 1576.2, baseplates need to be at least 6 mm thick and have a minimum plan area equivalent to 150 mm × 150 mm.
3.7 Are the baseplates secure?			Where there is a possibility of slip or dislodgement, baseplates should be fixed with nails, screws or welding, as appropriate.
4 Scaffold Structure			
4.1 Are the standards bearing firmly?			A 'floating' standard will cause additional loading on ledgers and adjacent standards, which may lead to failure.

(continued)

Question	Answer	Comments	Explanatory notes
4.2 Are the standards plumb (or as designed)?			Standards that are not vertical will take less load than standards which are vertical. Unless specifically allowed for in the design, this may lead to failure of the scaffold.
4.3 Are the longitudinal standard spacings correct?			If the longitudinal standard spacing exceeds that allowed for by the design, there may be excessive loads placed on ledgers and standards.
4.4 Are the transverse standard spacings correct?			If the transverse standard spacing is wider than designed, excessive loads may be placed on putlogs or transoms. If it is narrower than designed, the stability of the scaffold may be adversely affected and it may not be possible to provide platforms that are wide enough for the intended work.
4.5 Are the joints in standards correctly positioned?			Wrongly positioned joints in standards may lead to failure. Joints should be located in accordance with the design specification and supplier's information.
4.6 Are the joints in standards correctly secured?			Unless otherwise allowed for in the supplier's information or design specification, joints should be positively fixed to preclude uplift or dislodgment.
4.7 Are the ledgers level (or as designed)?			Ledgers that are not level may place additional loads on standards and other members. Unless specifically allowed for in the design, this may lead to structural failure and prevent the provision of adequate platforms.
4.8 Are the ledgers continuous (or as designed)?			Unless specifically designed, ledgers should be continuous for the full length of the scaffold.
4.9 Are the lift heights correct?			Where the vertical distance between ledgers exceeds the design specifications or supplier's information, the load-bearing capacity of the standards will be reduced and the rigidity of the scaffold may be adversely affected.
4.10 Are the horizontal ledger spacings correct?			Except where provided in the supplier's information or specifically designed, at each lift there must be a ledger for each longitudinal row of standards. At any lift, all ledgers should be in the same plane.
4.11 Are the ledgers correctly secured?			Ledgers that are not fixed in accordance with the supplier's information or the design specification may not be capable of sustaining the intended loads.
4.12 Are ledger joints correctly positioned?			The incorrect position of ledger joints may lead to failure under load. Joints should be staggered and should not occur in end bays.
4.13 Are the joints in ledgers correctly secured?			Unless otherwise designed, joints in ledgers should be fixed with sleeve-type end-to-end couplers or should be scarfed to prevent separation under load.

(continued)

Question	Answer	Comments	Explanatory notes
4.14 Are there sufficient transoms/putlogs			Unless otherwise designed, in each lift there should be a transom or putlogs at each standard/ledger node point. Missing transoms or putlog will reduce the load capacity of standards, decrease rigidity and create traps in the scaffold.
4.15 Are the transoms/putlogs correctly positioned and secured?			Transoms should be positioned as closely as possible to the standard/ledger node points. Putlogs should be positioned on the upper surface of ledgers. The fixing method should be in accordance with the supplier's information or the design specification.
4.16 Is the bracing adequate?			A scaffold generally requires longitudinal and transverse bracing systems to achieve the required rigidity. Plan bracing may also be required. The minimum amount, maximum spacing, positioning and fixing of braces should comply with the supplier's information or design specification.
4.17 Is the scaffold sufficiently stable?			Unless it is designed to be freestanding, the stability of a scaffold will depend upon the ties by which it is fixed to the supporting structure. Needles or counterweights that provide free standing stability should be positively secured against displacement or removal.
4.18 Are the ties correctly positioned and correctly fixed?			The spacing and fixing methods must comply with the supplier's information or design specification. Ties should rigidly fix the scaffold to the supporting structure but should not interfere with access along working platforms, access platforms or landings.
5 Platforms			
5.1 Does the scaffold have the required number of working platforms?			The number of working platforms should be sufficient for the intended work, but should be not more than the number stated in the design specification or the supplier's information.
5.2 Are the working platforms at the required locations?			Working platforms should be positioned so that the intended task can be carried out without unnecessary restriction or over-reaching. The height of the platform and the amount of clearance from the working face are critical factors.
5.3 Are catch platforms correctly positioned?			Where catch platforms are required they should be positioned so as to minimize the distance that debris can fall and should be positioned to comply with the design specification or supplier's information. The provision of cantilevered catch platforms (fans) may require the scaffold to be provided with additional ties.
5.4 Are the platforms and supporting scaffold constructed for the appropriate duty live loads?			The nature of the work to be performed from the scaffold and the weight of materials to be deposited on the platforms will determine whether the platforms should be light duty, medium duty, heavy duty or special duty.

(continued)

Question	Answer	Comments	Explanatory notes
5.5 Are the platform dimensions suitable for the intended work?			Working platforms should be wide enough to allow for the storage of any material that will need to be placed on the platform while maintaining clear access along their entire length and allowing sufficient space to perform the intended work tasks without unnecessary obstruction. Catch platforms need to be large enough to trap any falling debris.
5.6 Is there adequate edge protection?			Edge protection is required at the open sides and ends of all working platforms and access platforms from which a person or object could fall 2 m or more. Additional infill may be required to adequately contain materials and equipment.
5.7 Are the platforms correctly constructed?			Scaffold planks and prefabricated platform units should be checked for damage or deterioration. Working platforms are required to have a slip-resistant surface, be closely decked, be incapable of uplift under working conditions, be level and free of trip hazards. Where specifically designed, a working platform may have a slope of up to 7° from the horizontal. This is equal to a rise of 1 in 8. dislodge planks.
5.8 Are planks secured against wind?			Some locations (e.g. highrise city buildings, cyclone prone areas, alpine regions) are prone to excessive wind gusts, which may dislodge planks.
6 Access and Egress			
6.1 Is there access and egress to all working platforms?			Means of safe access and egress to working platforms should be provided. This can be achieved with stairways, access ways, ladders or other suitable means.
6.2 Are temporary stairways correctly installed?			
6.3 Are portable ladders of an industrial grade, serviceable and correctly installed?			Portable ladders used for access and egress should be industrial-grade ladders. They should be checked for defects and deterioration.
6.4 Are access ways correctly installed?			Access ways need to be 450 mm wide for persons or 675 mm wide for persons and materials. Slope must not exceed 20° (a rise of 1 in 3) to the horizontal and if over 7° must be cleated.
7 Containment sheeting			
7.1 Has the scaffold been designed for wind loading on any containment sheeting?			Additional ties and bracing may be required to ensure stability.
7.2 Are the fixing ties secure?			

(continued)

Question	Answer	Comments	Explanatory notes
7.3 Are there any rips or tears?			
7.4 Are the overlap joints satisfactory?			
8 General fitness for purpose			
8.1 Is there adequate provision for material handling?			Where material or equipment is intended to be placed on platforms or removed from the scaffold, suitable means or devices should be incorporated or provided. Depending upon the circumstances, they may include items such as gin wheels, builders' hoists, tile or brick conveyors, rubbish chutes, personnel and materials hoists, crane loading bays and winches. The design specification should allow for any significant loads that may be transmitted to the scaffold through such items.
8.2 Are the clearances between the scaffold and adjacent structures correct?			Specific clearances may be critical, particularly where cladding or similar work is to be done from the scaffold. Boilers and similar plant may be subject to significant expansion and contraction due to temperature variations.
8.3 Is there adequate protection from falling debris?			Some work tasks, such as demolition and abrasive blasting may require the scaffold to be sheeted. Precautions may need to be taken to protect people from sparks and molten metal caused by hot work such as welding or oxy cutting.
8.4 Has the scaffold been adequately designed to support all attachments?			Where lifting equipment such as chain blocks are supported from the scaffold, significant loads can be applied to structural members. Concrete pumpjacks can transmit impact forces. The sheeting of scaffolds increases the impact of wind loads.
8.5 Are all approaches and platforms effectively lit?			Where there is insufficient natural lighting, artificial lighting should be provided and set up so as to avoid glare and deep shadows.
9 Mobile scaffolds			
9.1 Is the supporting surface hard and flat?			A soft or uneven supporting surface will cause instability and may lead to the collapse of the scaffold.
9.2 Is the area of operation free of floor penetrations, powerlines and other hazards?			It may be necessary to clearly limit the operational area of a mobile scaffold by erecting barricades or implementing other forms of control in order to isolate the scaffold from hazards.
9.3 Are the castor wheel locks in working order?			A mobile scaffold should not be left unattended or worked from while the castors are in a free-running condition. Castors with inoperative or missing wheel locks should be replaced.

(Informative)

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RECORD SHEET FOR SCAFFOLDING HOISTS AND PROTECTIVE DEVICES

Record sheet for *scaffolding hoist/*protective device

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RECORD SHEET FOR LOAD LIMITING DEVICES

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Record sheet for wire rope

Rope identification No.

Rope diametermm

Original lengthm

For use with *scaffold hoist/*protective device

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Unit No.

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Model No.

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APPENDIX L

AUSTRALIAN TEST FOR TIMBER SCAFFOLD PLANKS

(Informative)

L1 SIMPLY SUPPORTED IMPACT TEST

The simply supported impact test sets out a field method for determining a working impact (or a brittle resistance) value for a timber scaffold plank in Australia.

The plank must be supported at the maximum intended design span (i.e. similar to that in Figure L1), to a maximum height of 150 mm above the supporting surface measured to the top of the plank. A stop must be placed at midspan under the plank to prevent the deflection or displacement from exceeding 50 mm, measured from the underside of the plank.

Planks with a design length of less than 2.5 m must be supported at their respective spans accordingly, and a stop must be placed at midspan under the plank to prevent the deflection or displacement from exceeding a value equal to 2 percent of the length of the span, measured from the underside of the plank.

The worst visible defect or the worst suspected defective area must be positioned at midspan and on the tension side (underside) of the plank. The planks should be in a dry condition.

A bag of sand or similar weighing approximately 80 kg must be dropped from a height of 600 mm onto the centre of the plank with the intention of imparting the greatest load possible. This test can be repeated to both sides of the plank as desired.

During this test, there must be no visual or audible sign of distress. It is of no consequence should the plank touch the stop in the course of the test. The stop is intended to reduce the severity of the test, so as not to damage planks that would otherwise be useable for less exacting applications.

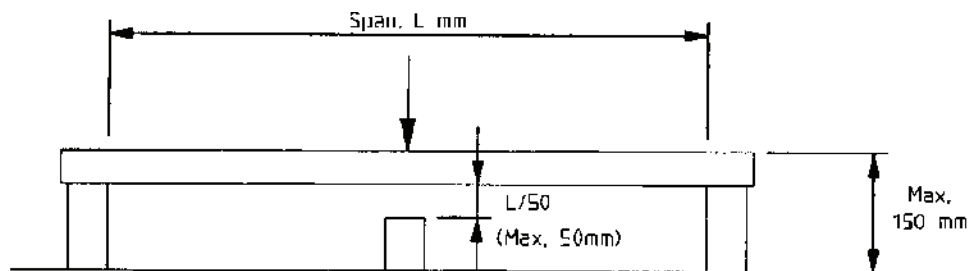


FIGURE L1 SIMPLY SUPPORTED PLANK FOR JUMP TEST

L2 CANTILEVERED IMPACT TEST

The cantilevered impact test sets out a field method for determining a working impact (or a brittle resistance) value for a timber scaffold plank.

The plank must be supported at half the maximum intended design span (i.e. similar to that in Figure L2), to a maximum height of 150 mm above the supporting surface measured to the top of the plank. A stop must be placed at the free end under the plank to prevent the deflection or displacement from exceeding 25 mm, measured from the underside of the plank.

Planks with a design length of less than 2.5 m must be supported at half their respective spans accordingly, and a stop must be placed at the free end under the plank to prevent the deflection or displacement from exceeding a value equal to 1 percent of the length of span, measured from the underside of the plank.

The worst visible defect or the worst suspected defective areas must be positioned at midspan and on the tension side (i.e. the topside) of the plank. The planks should be in a dry condition. A bag of sand or similar, weighing approximately 80 kg must be dropped from a height of 600 mm onto the end of the plank with the intention of imparting the greatest load possible. This test can be repeated to both sides of the plank as desired.

During this test there must be no visual or audible sign of distress. It is of no consequence should the plank touch the stop in the course of the test. The stop is intended to reduce the severity of the test, so as not to damage planks that would otherwise be usable for less exacting applications.

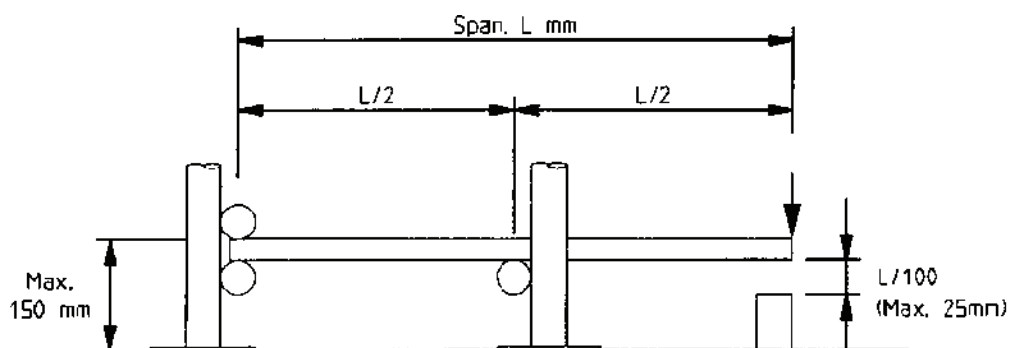


FIGURE L2 CANTILEVERED PLANK FOR JUMP TEST